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(54) **ORGANIC LIGHT EMITTING DIODE AND ORGANIC LIGHT EMITTING DEVICE HAVING THEREOF**

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H10K 85/30 (2023.01)
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H10K 50/11 (2023.01)
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H10K 50/18 (2023.01)
H10K 50/19 (2023.01)
H10K 101/00 (2023.01)
H10K 101/10 (2023.01)

(52) **U.S. Cl.**

CPC **C09K 11/06** (2013.01); **H10K 59/12** (2023.02); **H10K 85/322** (2023.02); **H10K 85/615** (2023.02); **H10K 85/624** (2023.02);

H10K 85/626 (2023.02); **H10K 85/633** (2023.02); **H10K 85/636** (2023.02); **H10K 85/6572** (2023.02); **H10K 85/6574** (2023.02); **H10K 85/6576** (2023.02); **H10K 85/658** (2023.02); **H10K 50/11** (2023.02); **H10K 50/15** (2023.02); **H10K 50/181** (2023.02); **H10K 50/19** (2023.02); **H10K 2101/10** (2023.02); **H10K 2101/90** (2023.02)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

An organic light emitting diode and an organic light emitting device containing the same are provided. The organic light emitting diode includes at least two emitting parts each of which includes independently an emitting material layer. Each of the emitting material layers includes independently an anthracene-based host having different deuterium substitution rates. The organic light emitting device includes the organic light emitting diode. Hosts with different deuterium substitution rates are introduced into each organic emitting material layer to improve luminous efficiency and luminous lifespan of the organic light emitting diode and the organic light emitting device.

21 Claims, 5 Drawing Sheets

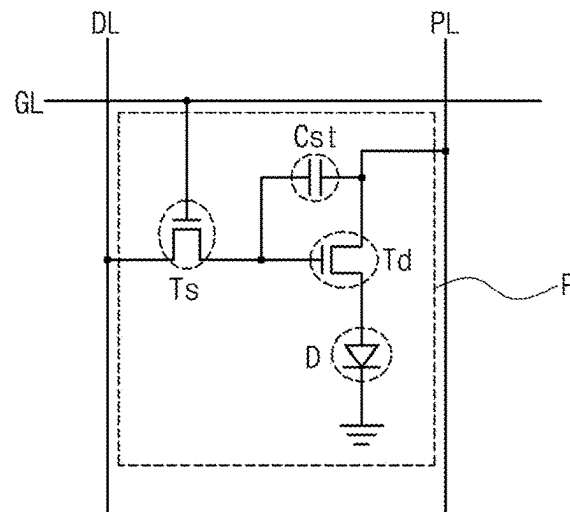


Fig. 1

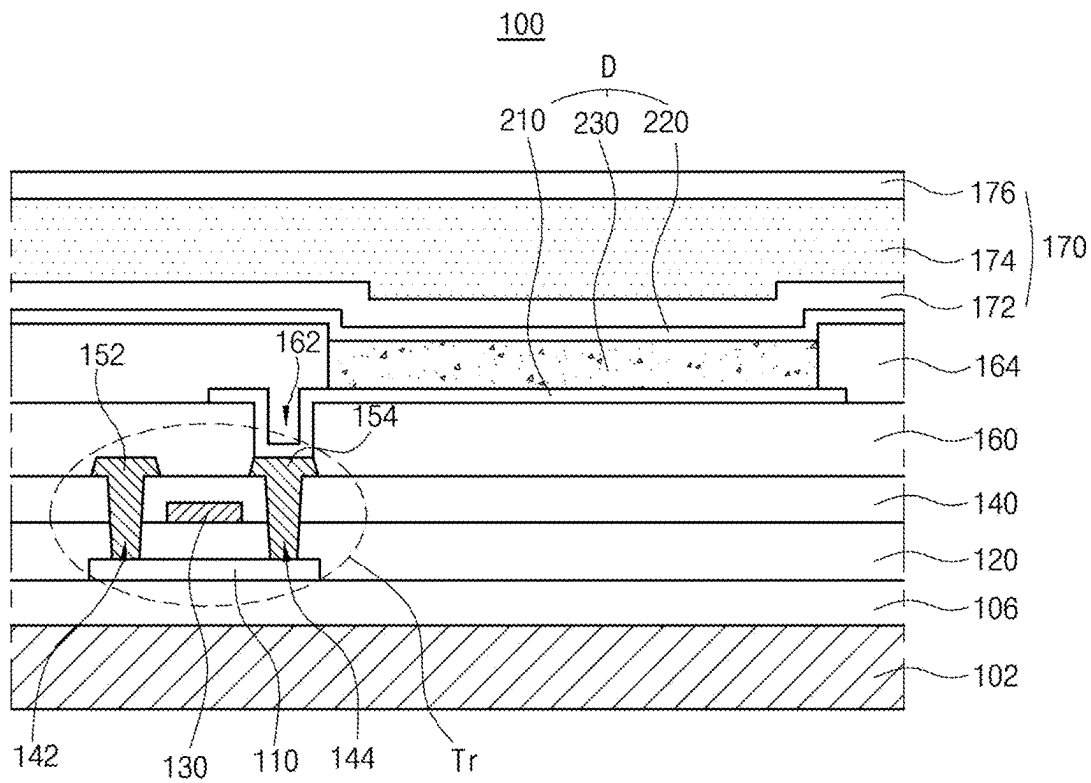


Fig. 2

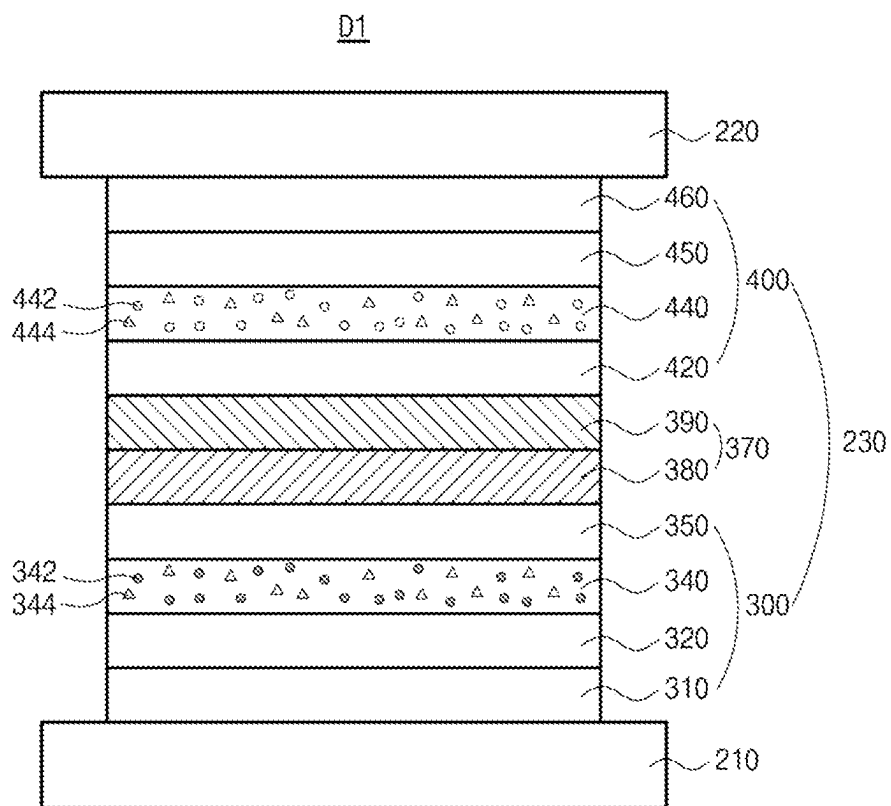


Fig. 3

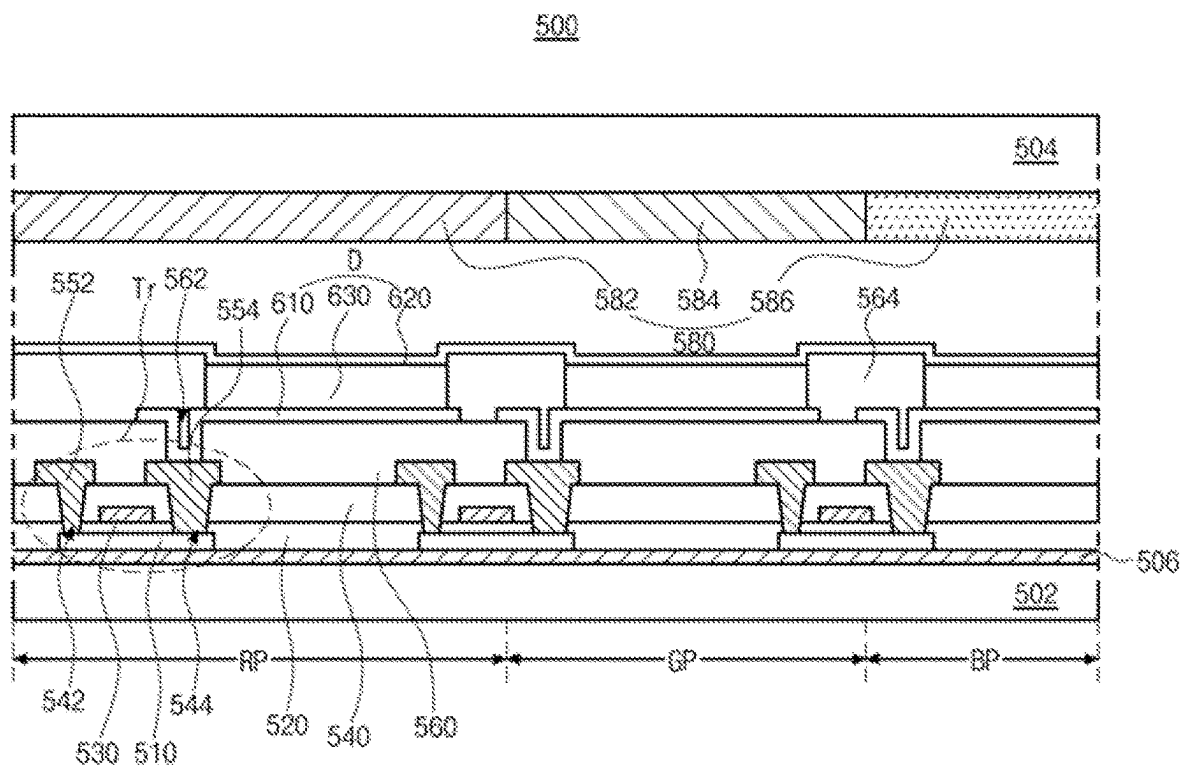


Fig. 4

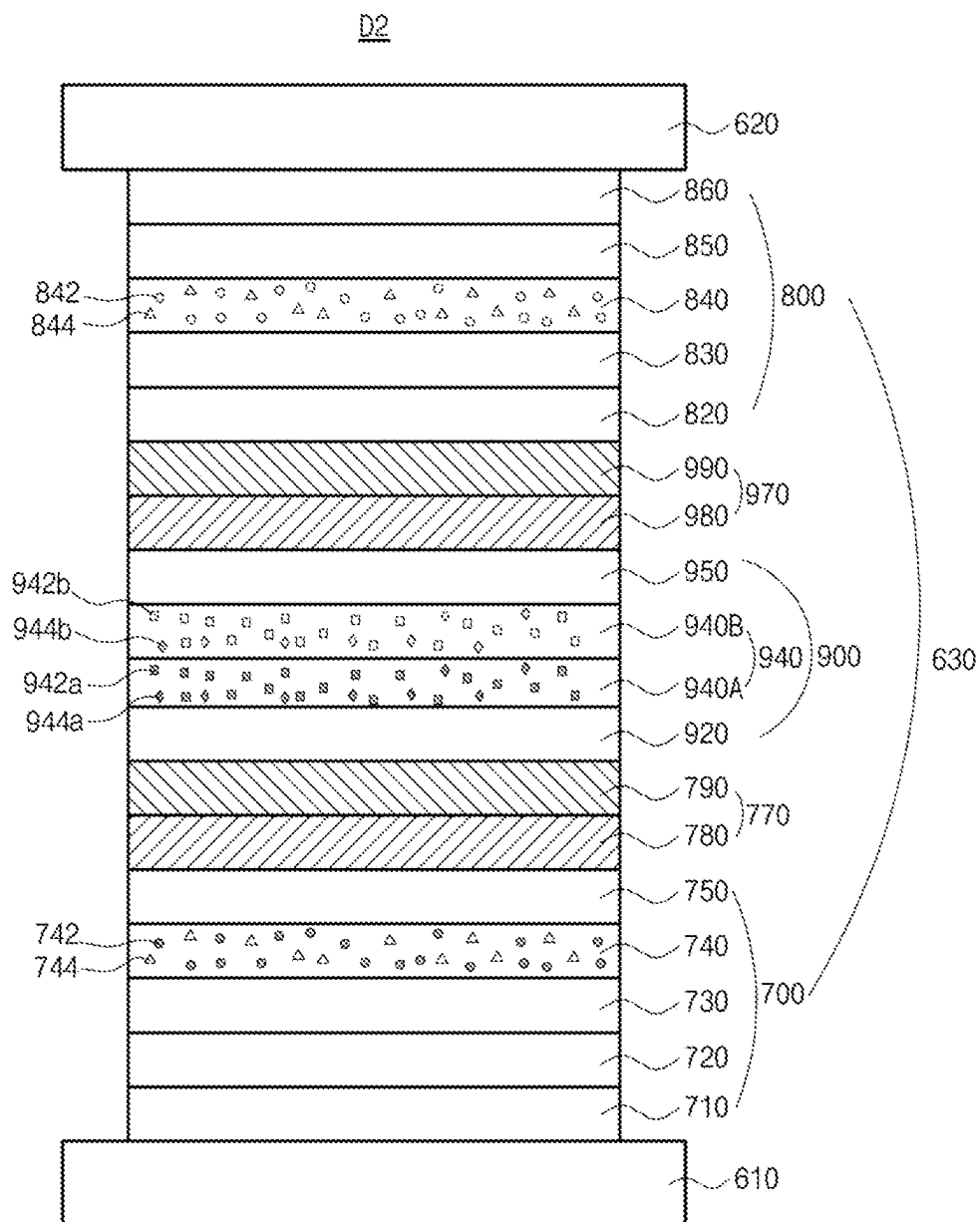


Fig. 5

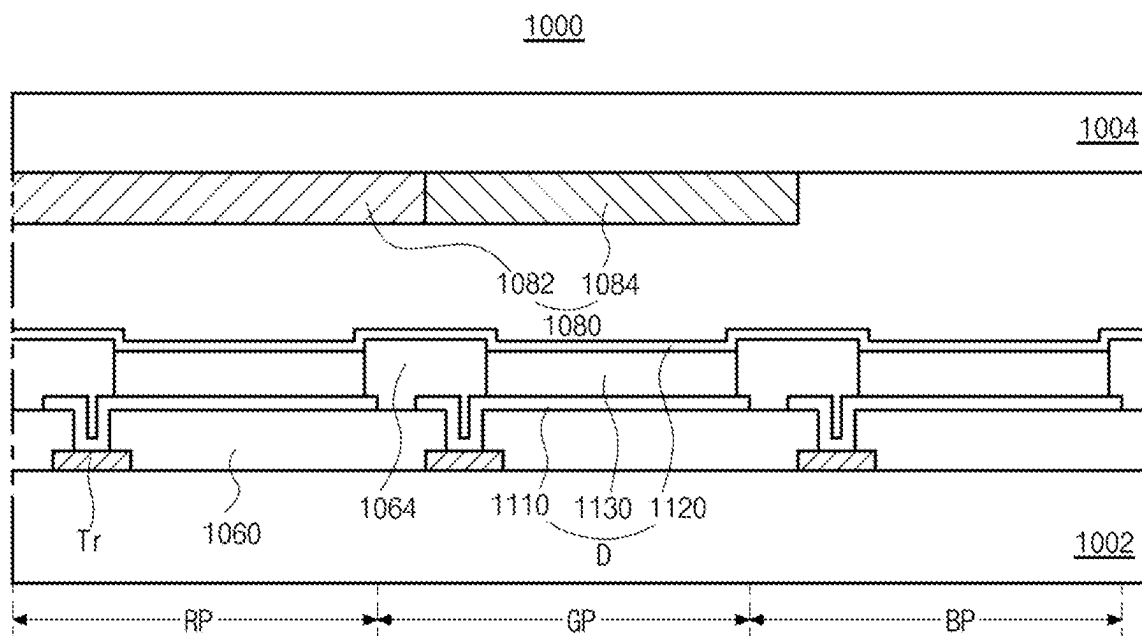


Fig. 6

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ORGANIC LIGHT EMITTING DIODE AND ORGANIC LIGHT EMITTING DEVICE HAVING THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the priority of Korean Patent Application No. 10-2020-0165607, filed in the Korean Intellectual Property Office on Dec. 1, 2020, the entire content of which is expressly incorporated herein by reference in its entirety.

BACKGROUND

Technical Field

The present disclosure relates to an organic light emitting diode, and more specifically, to an organic light emitting diode having excellent luminous efficiency and luminous lifespan and an organic light emitting device including the diode.

Discussion of the Related Art

An organic light emitting diode (OLED) among a flat display device used widely has come into the spotlight as a display device replacing rapidly a liquid crystal display device (LCD). The OLED can be formed as a thin organic film less than 2000 Å and can implement unidirectional or bidirectional images by electrode configurations. Also, the OLED can be formed even on a flexible transparent substrate such as a plastic substrate so that a flexible or a foldable display device can be realized with ease using the OLED. In addition, the OLED can be driven at a lower voltage and the OLED has excellent high color purity compared to the LCD.

Since fluorescent material uses only singlet exciton energy in the luminous process, the related art fluorescent material shows low luminous efficiency. On the contrary, phosphorescent material can show high luminous efficiency since it uses triplet exciton energy as well as singlet exciton energy in the luminous process. However, metal complex, representative phosphorescent material, has short luminous lifespan for commercial use. Particularly, blue luminous materials have not showed satisfactory luminous efficiency and luminous lifespan compared to other color luminous materials. Therefore, there remains a need to develop a new compound or a device structure that can enhance luminous efficiency and luminous lifespan of the organic light emitting diode.

BRIEF SUMMARY

Accordingly, embodiments of the present disclosure are directed to an organic light emitting device that substantially obviates one or more of the problems due to the limitations and disadvantages of the related art.

An aspect of the present disclosure is to provide an organic light emitting diode with improved luminous efficiency and luminous lifespan and an organic light emitting device including the diode.

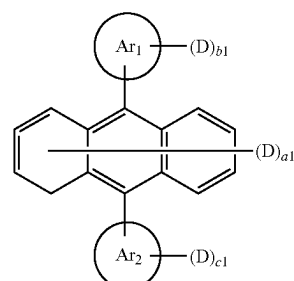
Additional features and aspects will be set forth in the description that follows, and in part will be apparent from the description, or can be learned by practice of the inventive concepts provided herein. Other features and aspects of the inventive concept can be realized and attained by the struc-

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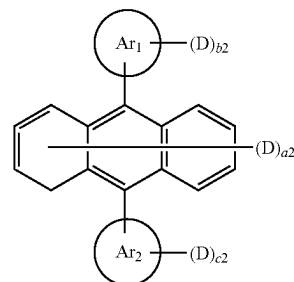
ture particularly pointed out in the written description, or derivable therefrom, and the claims hereof as well as the appended drawings.

To achieve these and other aspects of the inventive concepts, as embodied and broadly described, the present disclosure provides an organic light emitting diode comprising a first electrode; a second electrode facing the first electrode; a first emitting part disposed between the first and second electrode and comprising a first emitting material layer; a second emitting part disposed between the first emitting part and the second electrode and comprising a second emitting material layer; and a first charge generation layer disposed between the first and second emitting parts, wherein the first emitting material layer includes a first host having the following structure of Formula 1; and wherein the second emitting material layer includes a second host having the following structure of Formula 3:

Formula 1



Formula 3



wherein each of Ar₁ and Ar₂ is independently a C₆-C₂₀ aryl group; D represents deuterium; each of a1 and a2 is independently an integer of 0 to 8; each of b1, b2, c1 and c2 is independently a number of deuterium substituted carbon atoms in Ar₁ and Ar₂ not linked to the anthracene core; and a sum of a1, b1 and c1 is different from a sum of a2, b2 and c2.

For example, the organic light emitting diode may further comprise a third emitting part disposed between the first charge generation layer and the second emitting part and including a third emitting material layer, and a second charge generation layer disposed between the second emitting part and the third emitting part.

The organic light emitting diode having a tandem structure may emit a blue (B) light or a white (W) light.

In another aspect, the present disclosure provides an organic light emitting device comprising a substrate and the organic light emitting diode over the substrate.

The substrate may define a red pixel region, a green pixel region and a blue pixel region and the organic light emitting diode may be located correspondingly to the red pixel region, the green pixel region and the blue pixel region, and

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the organic light emitting device may further comprise a color conversion layer disposed between the substrate and the organic light emitting diode or over the organic light emitting diode correspondingly to the red pixel region and the green pixel region.

The substrate may define a red pixel region, a green pixel region and a blue pixel region and the organic light emitting diode may be located correspondingly to the red pixel region, the green pixel region and the blue pixel region, and the organic light emitting device may further comprise a color filter layer disposed between the substrate and the organic light emitting diode or over the organic light emitting diode correspondingly to the red pixel region, the green pixel region and the blue pixel region.

It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the inventive concepts as claimed.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the disclosure, are incorporated in and constitute a part of this application, illustrate embodiments of the disclosure and together with the description serve to explain principles of the disclosure.

FIG. 1 is a schematic circuit diagram illustrating an organic light emitting display device in accordance with the present disclosure.

FIG. 2 is a schematic cross-sectional view illustrating an organic light emitting display device as an example of an organic light emitting device in accordance with one exemplary aspect of the present disclosure.

FIG. 3 is a schematic cross-sectional view illustrating an organic light emitting diode having two emitting part in accordance with an exemplary aspect of the present disclosure.

FIG. 4 is a schematic cross-sectional view illustrating an organic light emitting display device in accordance with another exemplary aspect of the present disclosure.

FIG. 5 is a cross-sectional view illustrating an organic light emitting diode having three emitting parts in accordance with another exemplary aspect of the present disclosure.

FIG. 6 is a schematic cross-section view illustrating an organic light emitting display device in accordance with still another exemplary aspect of the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to aspects of the disclosure, examples of which are illustrated in the accompanying drawings.

The present disclosure relates to an organic light emitting diode including multiple emitting parts where an anthracene-based host having a different deuterium substitution rate is introduced into an emitting material layer constituting each emitting part, and an organic light emitting device including the organic light emitting diode. The configuration of the present disclosure enables the organic light emitting diode and the organic light emitting device to maximize their luminous efficiency and luminous lifespan. The organic light emitting diode can be applied into an organic light emitting device such as an organic light emitting display device or an organic light emitting illumination device.

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FIG. 1 is a schematic circuit diagram illustrating an organic light emitting display device of the present disclosure. As illustrated in FIG. 1, a gate line GL, a data line DL and power line PL, each of which cross each other to define a pixel region P, in the organic light emitting display device. A switching thin film transistor Ts, a driving thin film transistor Td, a storage capacitor Cst and an organic light emitting diode D are formed within the pixel region P. The pixel region P may include a red (R) pixel region, a green (G) pixel region and a blue (B) pixel region.

The switching thin film transistor Ts is connected to the gate line GL and the data line DL, and the driving thin film transistor Td and the storage capacitor Cst are connected between the switching thin film transistor Ts and the power line PL. The organic light emitting diode D is connected to the driving thin film transistor Td. When the switching thin film transistor Ts is turned on by a gate signal applied into the gate line GL, a data signal applied into the data line DL is applied into a gate electrode of the driving thin film transistor Td and one electrode of the storage capacitor Cst through the switching thin film transistor Ts.

The driving thin film transistor Td is turned on by the data signal applied into the gate electrode so that a current proportional to the data signal is supplied from the power line PL to the organic light emitting diode D through the driving thin film transistor Td. And then, the organic light emitting diode D emits light having a luminance proportional to the current flowing through the driving thin film transistor Td. In this case, the storage capacitor Cst is charge with a voltage proportional to the data signal so that the voltage of the gate electrode in the driving thin film transistor Td is kept constant during one frame. Therefore, the organic light emitting display device can display a desired image.

FIG. 2 is a schematic cross-sectional view illustrating an organic light emitting display device in accordance with an exemplary aspect of the present disclosure. As illustrated in FIG. 2, the organic light emitting display device 100 comprises a substrate 102, a thin-film transistor Tr over the substrate 102, and an organic light emitting diode D connected to the thin film transistor Tr. As an example, the substrate 102 defines a red pixel region, a green pixel region and a blue pixel region and the organic light emitting diode D is located in each pixel region. In other words, the organic light emitting diode D, each of which emits red, green or blue light, is located correspondingly in the red pixel region, the green pixel region and the blue pixel region.

The substrate 102 may include, but is not limited to, glass, thin flexible material and/or polymer plastics. For example, the flexible material may be selected from the group, but is not limited to, polyimide (PI), polyethersulfone (PES), polyethylenenaphthalate (PEN), polyethylene terephthalate (PET), polycarbonate (PC) and combination thereof. The substrate 102, over which the thin film transistor Tr and the organic light emitting diode D are arranged, forms an array substrate.

A buffer layer 106 may be disposed over the substrate 102, and the thin film transistor Tr is disposed over the buffer layer 106. The buffer layer 106 may be omitted.

A semiconductor layer 110 is disposed over the buffer layer 106. In one exemplary aspect, the semiconductor layer 110 may include, but is not limited to, oxide semiconductor materials. In this case, a light-shield pattern may be disposed under the semiconductor layer 110, and the light-shield pattern can prevent light from being incident toward the semiconductor layer 110, and thereby, preventing the semiconductor layer 110 from being deteriorated by the light.

Alternatively, the semiconductor layer **110** may include polycrystalline silicon. In this case, opposite edges of the semiconductor layer **110** may be doped with impurities.

A gate insulating layer **120** including an insulating material is disposed on the semiconductor layer **110**. The gate insulating layer **120** may include, but is not limited to, an inorganic insulating material such as silicon oxide (SiO_x) or silicon nitride (SiN_x).

A gate electrode **130** made of a conductive material such as a metal is disposed over the gate insulating layer **120** so as to correspond to a center of the semiconductor layer **110**. While the gate insulating layer **120** is disposed over a whole area of the substrate **102** in FIG. 2, the gate insulating layer **120** may be patterned identically as the gate electrode **130**.

An interlayer insulating layer **140** including an insulating material is disposed on the gate electrode **130** with covering over an entire surface of the substrate **102**. The interlayer insulating layer **140** may include an inorganic insulating material such as silicon oxide (SiO_x) or silicon nitride (SiN_x), or an organic insulating material such as benzocyclobutene or photo-acryl.

The interlayer insulating layer **140** has first and second semiconductor layer contact holes **142** and **144** that expose both sides of the semiconductor layer **110**. The first and second semiconductor layer contact holes **142** and **144** are disposed over opposite sides of the gate electrode **130** with spacing apart from the gate electrode **130**. The first and second semiconductor layer contact holes **142** and **144** are formed within the gate insulating layer **120** in FIG. 2. Alternatively, the first and second semiconductor layer contact holes **142** and **144** are formed only within the interlayer insulating layer **140** when the gate insulating layer **120** is patterned identically as the gate electrode **130**.

A source electrode **152** and a drain electrode **154**, which are made of conductive material such as a metal, are disposed on the interlayer insulating layer **140**. The source electrode **152** and the drain electrode **154** are spaced apart from each other with respect to the gate electrode **130**, and contact both sides of the semiconductor layer **110** through the first and second semiconductor layer contact holes **142** and **144**, respectively.

The semiconductor layer **110**, the gate electrode **130**, the source electrode **152** and the drain electrode **154** constitute the thin film transistor Tr, which acts as a driving element. Namely, the thin film transistor T4 may correspond to the driving thin film transistor Td (of FIG. 1). The thin film transistor Tr in FIG. 2 has a coplanar structure in which the gate electrode **130**, the source electrode **152** and the drain electrode **154** are disposed over the semiconductor layer **110**. Alternatively, the thin film transistor Tr may have an inverted staggered structure in which a gate electrode is disposed under a semiconductor layer and a source and drain electrodes are disposed over the semiconductor layer. In this case, the semiconductor layer may include amorphous silicon.

Although not shown in FIG. 2, a gate line and a data line, which cross each other to define a pixel region, and a switching element, which is connected to the gate line and the data line, may be further formed in the pixel region. The switching element is connected to the thin film transistor Tr, which is a driving element. In addition, a power line is spaced apart in parallel from the gate line or the data line, and the thin film transistor Tr may further include a storage capacitor configured to constantly keep a voltage of the gate electrode for one frame.

A passivation layer **160** is disposed on the source and drain electrodes **152** and **154** with covering the thin film

transistor Tr over the whole substrate **102**. The passivation layer **160** has a flat top surface and a drain contact hole **162** that exposes the drain electrode **154** of the thin film transistor Tr. While the drain contact hole **162** is disposed on the second semiconductor layer contact hole **144**, it may be spaced apart from the second semiconductor layer contact hole **144**.

The organic light emitting diode (OLED) D includes a first electrode **210** that is disposed on the passivation layer **160** and connected to the drain electrode **154** of the thin film transistor Tr. The OLED D further includes an emissive layer **230** and a second electrode **220** each of which is disposed sequentially on the first electrode **210**.

The first electrode **210** is disposed in each pixel region. The first electrode **210** may be an anode and include conductive material having relatively high work function value. For example, the first electrode **210** may include, but is not limited to, a transparent conductive oxide (TCO) such as indium tin oxide (ITO), indium zinc oxide (IZO), indium tin zinc oxide (ITZO), SnO, ZnO, indium cerium oxide (ICO), aluminum doped zinc oxide (AZO), and the like.

In one exemplary aspect, when the organic light emitting display device **100** is a bottom-emission type, the first electrode **210** may have a single-layered structure of TCO. Alternatively, when the organic light emitting display device **100** is a top-emission type, a reflective electrode or a reflective layer may be disposed under the first electrode **210**. For example, the reflective electrode or the reflective layer may include, but is not limited to, Ag or aluminum-palladium-copper (APC) alloy. In the OLED of the top-emission type, the first electrode **210** may have a triple-layered structure of ITO/Ag/ITO or ITO/APC/ITO.

In addition, a bank layer **164** is disposed on the passivation layer **160** in order to cover edges of the first electrode **210**. The bank layer **164** exposes a center of the first electrode **210**. The bank layer **164** may be omitted.

An emissive layer **230** is disposed on the first electrode **210**. In one exemplary embodiment, the emissive layer **230** may include a hole injection layer (HIL), a hole transport layer (HTL), an electron blocking layer (EBL), an emitting material layer (EML), a hole blocking layer (HBL), an electron transport layer (ETL), an electron injection layer (EIL) and/or a charge generation layer (CGL), as illustrated in FIG. 3. The emissive layer **230** may have multiple emitting parts to form a tandem structure.

The emissive layer **230** may include at least one emitting material layer including an anthracene-based host and a boron-based dopant. Such an emissive layer **230** enables the OLED D and the organic light emitting display device **100** to improve their luminous efficiency and luminous lifespan. We will describe the OLED in more detail below.

The second electrode **220** is disposed over the substrate **102** above which the emissive layer **230** is disposed. The second electrode **220** may be disposed over a whole display area, and may include a conductive material with a relatively low work function value compared to the first electrode **210**, and may be a cathode. For example, the second electrode **220** may include, but is not limited to, aluminum (Al), magnesium (Mg), calcium (Ca), silver (Ag), alloy thereof such as aluminum-magnesium alloy (Al—Mg) and combination thereof such as Ag:Mg. For example, when the second electrode **220** comprises Ag:Mg, Ag and Mg may be mixed with, but is not limited to, a weight ratio between about 5:1 and about 10:1, for example, about 8:1 and 10:1. When the organic light emitting display device **100** is a top-emission type, the second electrode **220** is thin so as to have light-transmissive (or semi-transmissive) property.

In addition, an encapsulation film **170** may be disposed over the second electrode **220** in order to prevent outer moisture from penetrating into the organic light emitting diode **D**. The encapsulation film **170** may have, but is not limited to, a laminated structure of a first inorganic insulating film **172**, an organic insulating film **174** and a second inorganic insulating film **176**. The encapsulation film **170** may be omitted.

The organic light emitting display device **100** may further include a polarizing plate to reduce reflection of external light. For example, the polarizing plate may be a circular polarizing plate. When the organic light emitting display device **100** is a bottom-emission type, the polarizing plate may be disposed under the substrate **102**. Alternatively, when the organic light emitting display device **100** is a top-emission type, the polarizing plate may be attached onto the encapsulation film. Further, a cover window may be attached onto the encapsulation film **170** or the polarizing plate in the organic light emitting display device **100** of a top-emission type. In this case, the substrate **102** and the cover window have flexible properties so that a flexible display device can be constructed.

As described above, introducing a separate host having a different deuterium substitution rate into an emitting material layer constituting each emitting part enables the OLED to maximize its luminous efficiency and luminous lifespan. FIG. **3** is a schematic cross-sectional view illustrating an organic light emitting diode having two emitting parts in accordance with an exemplary embodiment of the present disclosure.

As illustrated in FIG. **3**, the organic light emitting diode (OLED) **D1** in accordance with the first aspect of the present disclosure includes first and second electrodes **210** and **220** facing each other and an emissive layer **230** disposed between the first and second electrodes **210** and **220**. The organic light emitting display device **100** (FIG. **2**) includes a red pixel region, a green pixel region and a blue pixel region, and the OLED **D1** may be disposed in the blue pixel region.

One of the first and second electrodes **210** and **220** is an anode and the other of the first and second electrodes **210** and **220** is a cathode. For example, the first electrode **210** may be an anode injecting holes and the second electrode **220** may be a cathode injecting electrons. In addition, one of the first and second electrodes **210** and **220** is a reflective electrode and the other of the first and second electrodes **210** and **220** is a transmissive (semi-transmissive) electrode. For example, each of the first and second electrodes **210** and **220** may have a thickness, but is not limited to, between about 100 Å about 2000 Å, for example, about 100 Å and about 1000 Å.

The emissive layer **230** includes a first emitting part **300** and a second emitting part **400**. In addition, the emissive layer **230** may further include a charge generation layer (CGL) **370** disposed between the first emitting part **300** and the second emitting part **400**, thus, the first emitting part **300**, the CGL **370** and the second emitting part **400** are disposed sequentially on the first electrode **210**. In other words, the first emitting part **300** is disposed between the first electrode

210 and the CGL **370** and the second emitting part **400** is disposed between the second electrode **220** and the CGL **370**.

The first emitting part **300** includes a first emitting material layer (lower emitting material layer, EML1) **340**. The first emitting part **300** may further include at least one of an HIL **310** disposed between the first electrode **210** and the EML1 **340**, a first hole transport layer (lower hole transport layer, HTL1) **320** disposed between the HIL **310** and the EML1 **340**, and a first electron transport layer (lower electron transport layer, ETL1) **350** disposed between the EML1 **340** and the CGL **370**. Alternatively, the first emitting part **300** may further include first electron blocking layer (lower electron blocking layer, EBL1, not shown) disposed between the HTL1 **320** and the EML1 **340**.

The second emitting part **400** includes a second emitting material layer (upper emitting material layer, EML2) **440**. The second emitting part **400** may further include at least one of a second hole transport layer (upper hole transport layer, HTL2) **420** disposed between the CGL **370** and the EML2 **440**, a second electron transport layer (upper electron transport layer, ETL2) **450** disposed between the second electrode **220** and the EML2 **440** and an EIL **460** disposed between the second electrode **220** and the ETL2 **450**. Alternatively, the second emitting part **400** may further include a second electron blocking layer (upper electron blocking layer, EBL2, not shown) disposed between the HTL2 **420** and the EML2 **440**.

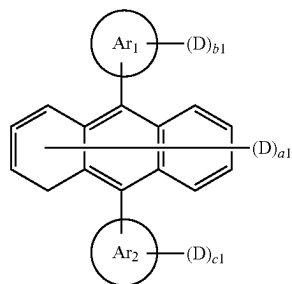
The EML1 **340** may include a first host **342** and a first dopant **344** and the EML2 **440** may include a second host **442** and a second dopant **444**. In this case, each of the first host **342** and the second host **442** forms a medium or a matrix in the EML1 **340** and the EML2 **440**, respectively. In the drawing, for convenience of description, some of the first and second hosts **342** and **442** are shown in the form of particles.

The EML1 **340** includes the first host **342** of an anthracene-based organic compound and the first dopant **344** of a boron-based organic compound so as to emit blue light. The EML2 **440** includes the second host **442** of an anthracene-based organic compound and the second dopant **444** of a boron-based organic compound so as to emit blue light. The first host **342** is different from the second host **442** in the substitution rate of deuterium introduced into the molecule.

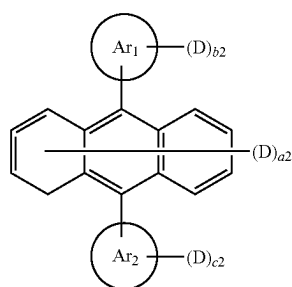
For example, the first host **342** may be an anthracene-based organic compound where all hydrogen (protium) atoms linked to a carbon atom of the molecule, for example, carbon atoms of an aromatic ring in the molecule, are substituted with deuterium atoms. The first host **342** may have the following structure of Formula 1. In contrast, the second host **442** may be an anthracene-based organic compound where hydrogen (protium) atoms linked to a carbon atom of the molecule, for example, carbon atoms of an aromatic ring in the molecule, are not substituted with deuterium atoms or the a portion of the protium atoms are substituted with deuterium atoms. The second host **442** may have the following structure of Formula 3:

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Formula 1



Formula 3



wherein each of Ar_1 and Ar_2 is independently a C_6 - C_{20} aryl group; D represents deuterium; each of a1 and a2 is independently a number of deuterium substituted carbons in the anthracene core and each of a1 and a2 is independently an integer of 0 to 8; each of b1, b2, c1 and c2 is independently a number of deuterium substituted carbon atoms in Ar_1 and Ar_2 not linked to the anthracene core; a sum of a1, b1 and c1 is different from a sum of a2, b2 and c2.

For example, a1 in Formula 1 may differ from a2 in Formula 3, b1 in Formula 1 may differ from b2 in Formula 3 and/or c1 in Formula 1 may differ from c2 in Formula 3. As an example, each of b1, c1, b2 and c2 in Formulae 1 and 3 may be independently an integer of 0 to 19. Both the first host **342** and the second host **442** are anthracene-based derivatives having substantially the same chemical structure except the deuteration rate (deuterium substitution rate). In other words, the first host **342** has a first deuteration rate and the second host **442** has a second deuteration rate different from the first deuteration rate.

In one exemplary aspect, each of Ar_1 and Ar_2 in Formulae 1 and 3 may be independently selected from a phenyl group, a naphthyl group and an anthracenyl group. As an example, each of Ar_1 and Ar_2 in Formulae 1 and 3 may be independently 1-naphthyl or 2-naphthyl. For example, one of Ar_1 and Ar_2 in Formulae 1 and 3 may be 1-naphthyl and the other Ar_1 and Ar_2 in Formulae 1 and 3 may be 2-naphthyl. In this case, each of b1, b2, c1 and c2 in Formulae 1 and 3 may be independently an integer of 0 to 7.

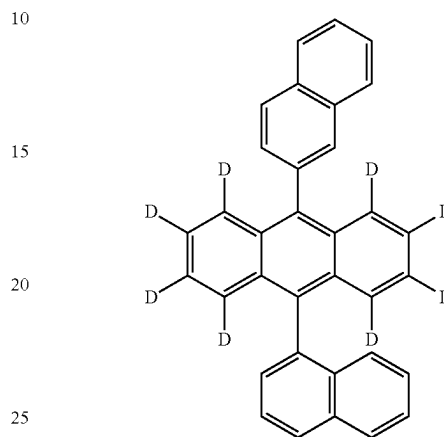
As an example, the first host **342** having the structure of Formula 1 may comprises an anthracene-based organic compound where at least one of carbon atoms in the anthracene core is substituted with deuterium, and the carbon atoms in both Ar_1 and Ar_2 are not substituted with deuterium (i.e., a1 is an integer of 1 to 8, each of b1 and c1 is 0). For example, the first host **342** may have a deuteration rate of about 27% to about 37%. Alternatively, the first host **342** of the anthracene-based compound having the structure of

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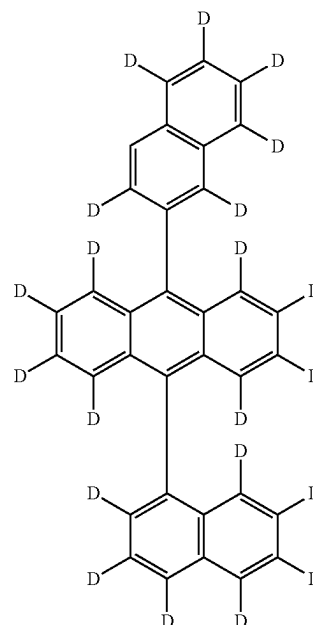
Formula 1 may have a deuteration rate of equal to or more than about 90%, for example, 100%. In one exemplary aspect, the first host **342** may be selected from, but is not limited to, the following anthracene-based organic compounds having the structure of Formula 2:

Formula 2

Host 1-2



Host 1-5

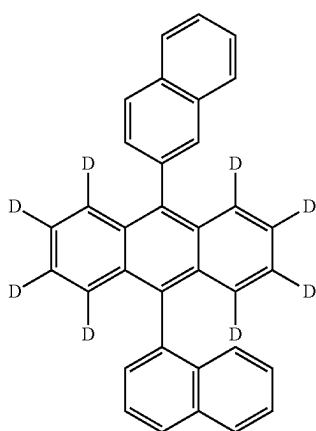
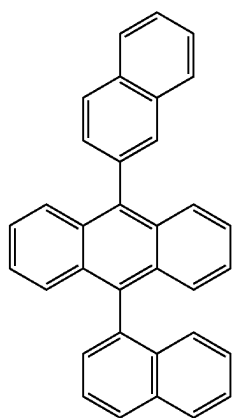


In an exemplary aspect, the second host **442** having the structure of Formula 3 may comprises an anthracene-based organic compound where none of the carbon atoms in the aromatic rings is substituted with deuterium (0% of deuteration rate). Alternatively, the second host **442** having the structure of Formula 3 may comprise an anthracene-based organic compound where at least one of carbon atoms in the anthracene core is substituted with deuterium, and none of the carbon atoms in both Ar_1 and Ar_2 is substituted with deuterium (i.e., a2 is an integer of 1 to 8, each of b2 and c2 is 0). For example, the second host **442** may have a deuteration rate of about 27% to about 37%. In another exem-

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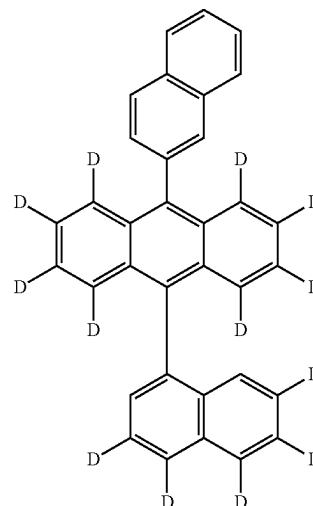
plary aspect, the second host **442** having the structure of Formula 3 may comprise an anthracene-based organic compound where at least one of carbon atoms in the anthracene core is substituted with deuterium, and at least one of the carbon atom(s) in one of Ar₁ and Ar₂ is substituted with deuterium, and the carbon atoms in the other of Ar₁ and Ar₂ is not substitute with deuterium (i.e., a2 is an integer of 1 to 8, one of b2 and c2 is 0 and the other of b2 and c2 is not 0). For example, the second host **442** may have a deuteration rate of about 59% to about 68%. In still another exemplary aspect, the second host **442** of the anthracene-based organic compound having the structure of Formula 3 may have a deuteration rate of equal to or more than about 90%, for example, 100%. For example, the second host **442** may be selected from, but is not limited to, the following anthracene-based organic compounds having the structure of Formula 4:

Formula 4

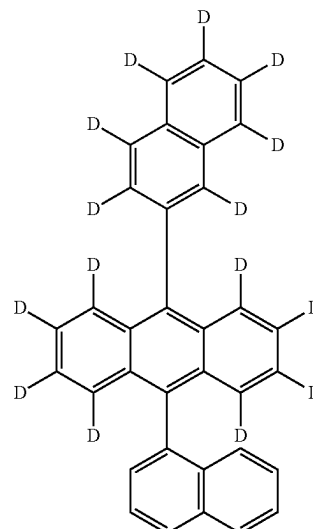


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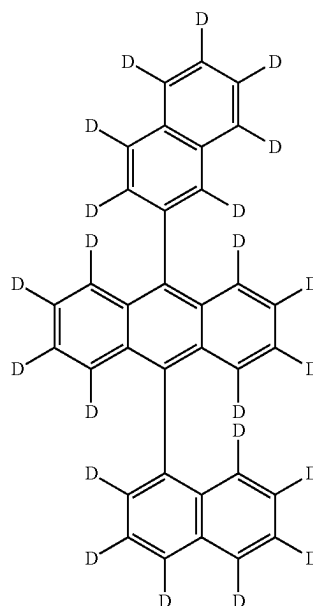
-continued



Host 1-3



Host 1-4



Host 1-5

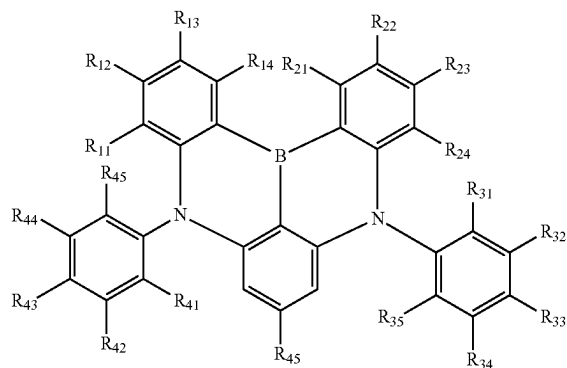
Host 1-1

Host 1-2

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Each of the first dopant **344** and the second dopant **444** may comprise independently a boron-based organic compound so that both the EML1 **340** and the EML2 **440** may emit blue light. The first dopant **344** may be identical to or different from the second dopant **444**. For example, each of the first dopant **344** and the second dopant **444** may comprise a boron-based organic compound having the following structure of Formula 5:

Formula 5



wherein each of R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ is independently selected from the group consisting of protium, deuterium, a C₁-C₁₀ alkyl group, a C₆-C₃₀ aryl group, a C₆-C₃₀ aryl amino group and a C₅-C₃₀ hetero aryl group, each of R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may be identical to or different from each other, or adjacent two of R₃₁, R₃₂, R₃₃, R₃₄ and R₃₅ together with the carbon atoms to which they are attached or adjacent two of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ together with the carbon atoms to which they are attached forms independently an unsubstituted or substituted C₆-C₁₀ aromatic ring or an unsubstituted or substituted C₅-C₁₀ hetero aromatic ring; R₅₁ is selected from the group consisting of protium, deuterium, a C₁-C₁₀ alkyl group, a C₃-C₁₅ cyclo alkyl group, a C₆-C₃₀ aryl group, a C₅-C₃₀ hetero aryl group and a C₆-C₃₀ aryl amino group, wherein each of the C₆-C₃₀ aryl group and the C₅-C₃₀ hetero aryl group may be independently unsubstituted or substituted with deuterium or a C₁-C₁₀ alkyl group, wherein the aryl group of the C₆-C₃₀ aryl amino group may be unsubstituted or substituted with deuterium or at least one of a C₁-C₁₀ alkyl group and a C₆-C₂₀ aryl group.

Each of the C₆-C₃₀ aryl group, the C₅-C₃₀ hetero aryl group and the C₆-C₃₀ aryl amino group of R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄, R₄₅ and R₅₁ may be independently unsubstituted or substituted with a C₁-C₁₀ alkyl group, for example, a C₁-C₅ alkyl group such as tert-butyl and/or tert-pentyl.

More particularly, the C₆-C₃₀ aryl amino group of R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄, R₄₅ and R₅₁ may comprise independently, but is not limited to, a diphenyl amino group and a phenyl-naphthyl amino group. The C₆-C₃₀ aryl group of R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄, R₄₅ and R₅₁ may comprise independently, but is not limited to, a phenyl group and a naphthyl group each of which is independently unsubstituted or substituted with at least one, for example 1-2, a C₁-C₁₀ alkyl

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group. The C₅-C₃₀ aryl hetero aryl group of R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄, R₄₅ and R₅₁ may comprise independently, but is not limited to, a carbazolyl group. The C₁-C₁₀ alkyl group of R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄, R₄₅ and R₅₁ may comprise independently, but is not limited to, a C₁-C₅ alkyl group such as methyl, ethyl, propyl, butyl (e.g., tert-butyl), pentyl (e.g., tert-pentyl). In this case, each of the C₆-C₃₀ aryl amino group, the C₆-C₃₀ aryl group, the C₅-C₃₀ hetero aryl group and the C₁-C₁₀ alkyl group as well as the substituent linked to those groups may be further substituted with deuterium.

The C₆-C₃₀ aromatic ring and the C₅-C₃₀ hetero aromatic ring formed by adjacent two of R₃₁ to R₃₅ and/or adjacent two of R₄₁ to R₄₅ may comprise, but is not limited to, a benzofuran ring and a benzothiophene ring each of which may be unsubstituted or substituted with one to three C₁-C₅ alkyl groups.

In one exemplary aspect, one of R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise independently, but is not limited to, methyl, tert-butyl or tert-pentyl, and the rest of R₁₁, R₁₂, R₁₃ and R₁₄, the rest of R₂₁, R₂₂, R₂₃ and R₂₄, the rest of R₃₁, R₃₂, R₃₃, R₃₄ and R₃₅ and the rest of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise independently, but is not limited to, protium or deuterium, and R₅₁ may comprise protium, deuterium, methyl, a diphenyl amino group, a phenyl-naphthyl amino group or a carbazolyl group. In another exemplary aspect, R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄, R₃₅, R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise hydrogen and R₅₁ may be a diphenyl amino group.

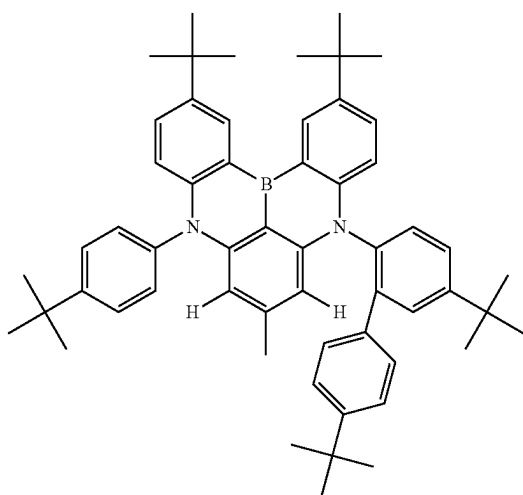
In still another exemplary aspect, R₁₁, R₁₂, R₁₃, R₁₄, R₂₁, R₂₂, R₂₃, R₂₄, R₃₁, R₃₂, R₃₃, R₃₄ and R₃₅ may comprise hydrogen, one of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise a phenyl group, the rest of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise hydrogen, and R₅₁ may comprise methyl. In further still another exemplary aspect, one of R₁₁, R₁₂, R₁₃ and R₁₄, one R₂₁, R₂₂, R₂₃ and R₂₄, one of R₃₁, R₃₂, R₃₃, R₃₄ and R₃₅ and one of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise independently, but is not limited to, methyl, tert-butyl or tert-pentyl, the other of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise a phenyl group and a naphthyl group each of which may be substituted with tert-butyl or tert-pentyl, and the rest of R₁₁, R₁₂, R₁₃ and R₁₄, the rest of R₂₁, R₂₂, R₂₃ and R₂₄, the rest of R₃₁, R₃₂, R₃₃, R₃₄ and R₃₅ and the rest of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise independently, but is not limited to, protium or deuterium, and R₅₁ may comprise methyl.

In further still another exemplary aspect, one of R₁₁, R₁₂, R₁₃ and R₁₄, one of R₂₁, R₂₂, R₂₃ and R₂₄, one of R₃₁, R₃₂, R₃₃, R₃₄ and R₃₅ and one of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise independently tert-butyl and tert-pentyl, other two of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may form a benzofuran ring or a benzothiophene ring that may be substituted with tert-butyl, the rest of R₁₁, R₁₂, R₁₃ and R₁₄, the rest of R₂₁, R₂₂, R₂₃ and R₂₄, the rest of R₃₁, R₃₂, R₃₃, R₃₄ and R₃₅ and the rest of R₄₁, R₄₂, R₄₃, R₄₄ and R₄₅ may comprise protium and deuterium, and R₅₁ may comprise methyl.

As an example, each of the first dopant **344** and the second dopant **444** may be independently selected from, but is not limited to, the following boron-based organic compounds having the structure of Formula 6:

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Formula 6



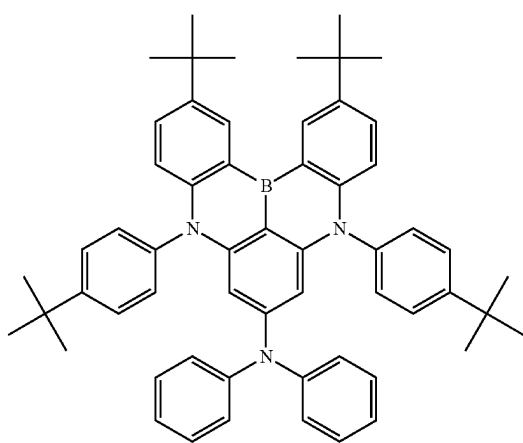
Dopant 1

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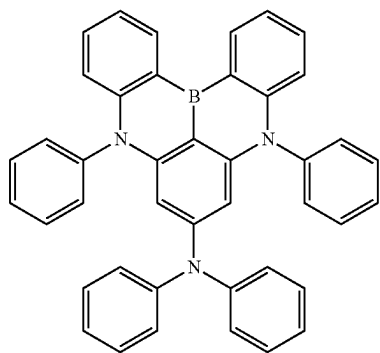
Dopant 2

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Dopant 3

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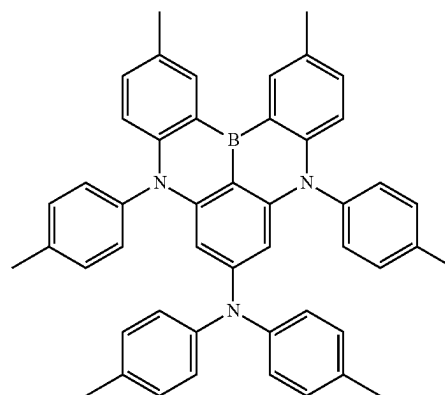
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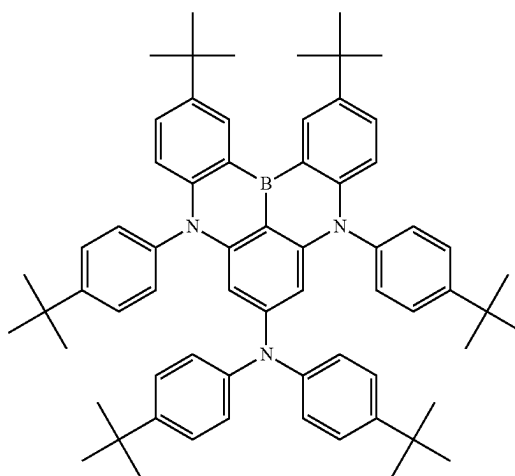
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Dopant 4



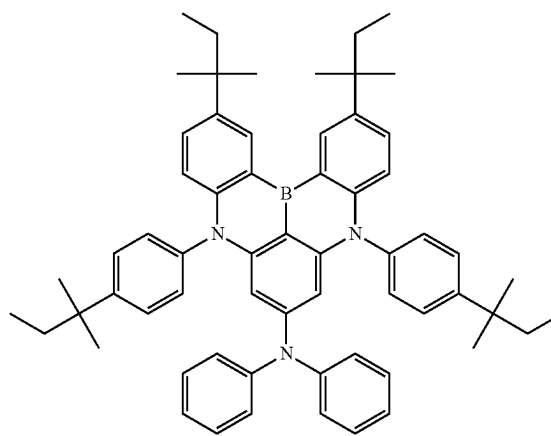
Dopant 5

25



Dopant 6

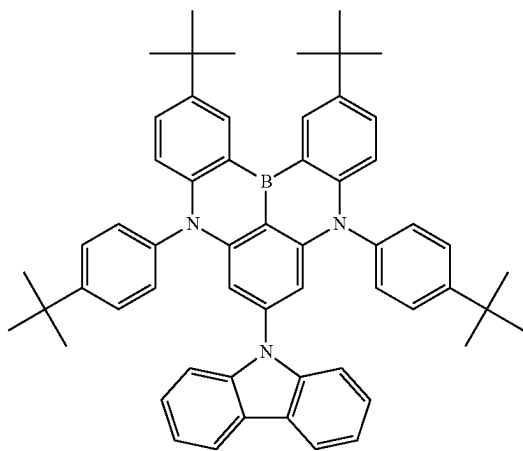
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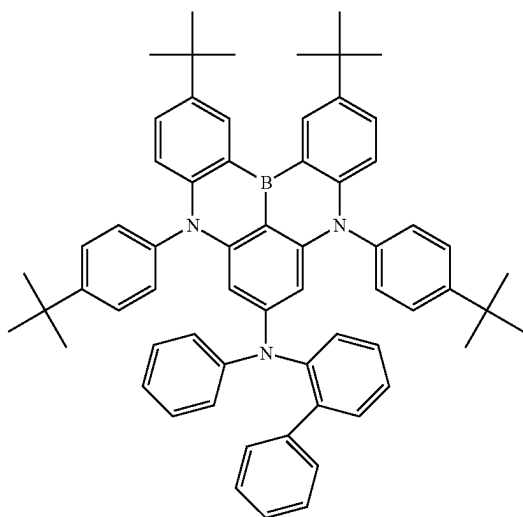
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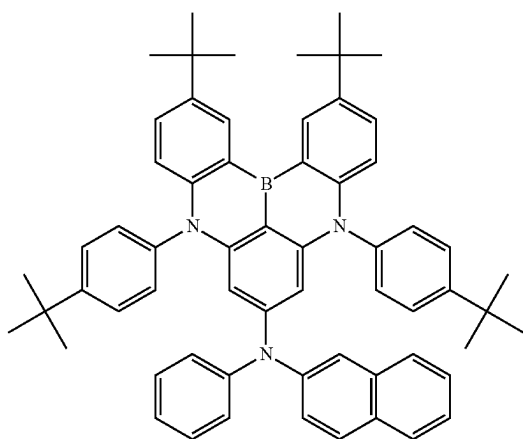
Dopant 7



Dopant 8



Dopant 9

**18**

-continued

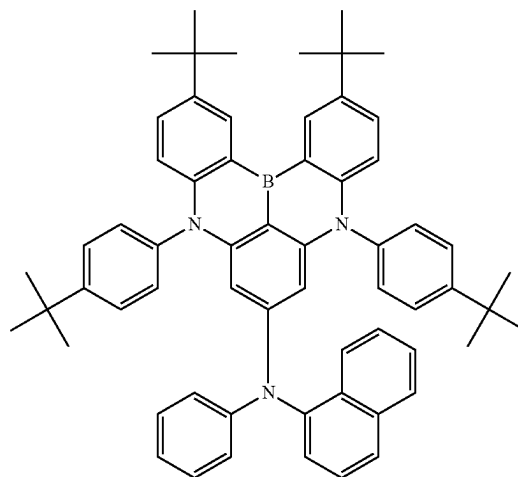
Dopant 10

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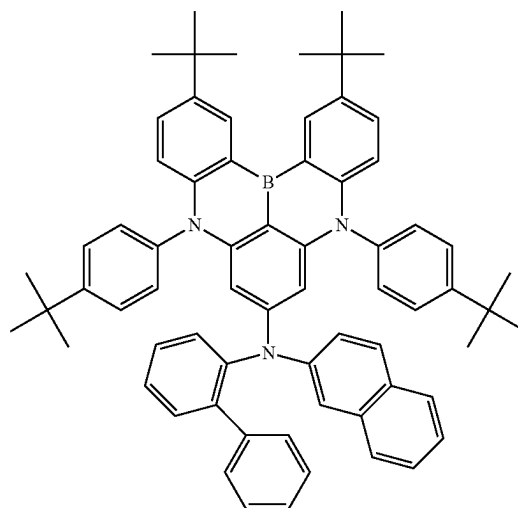
Dopant 11

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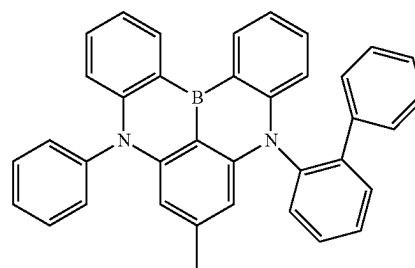
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Dopant 12

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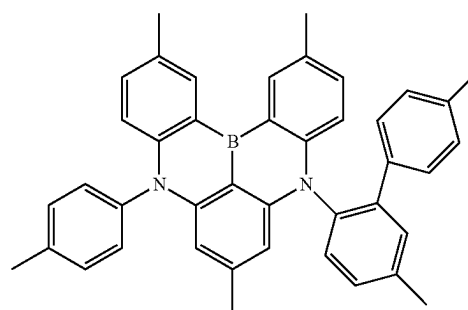


Dopant 13

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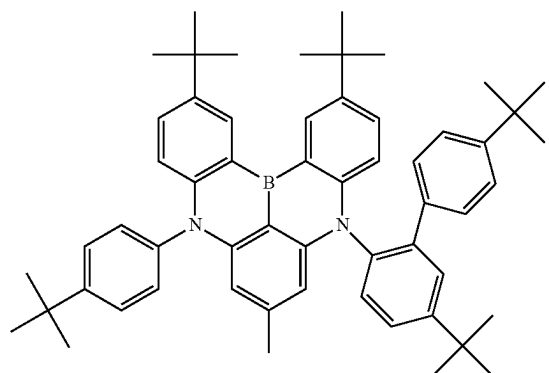
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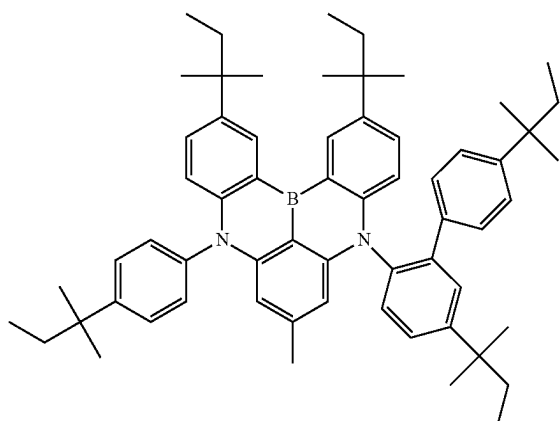
19

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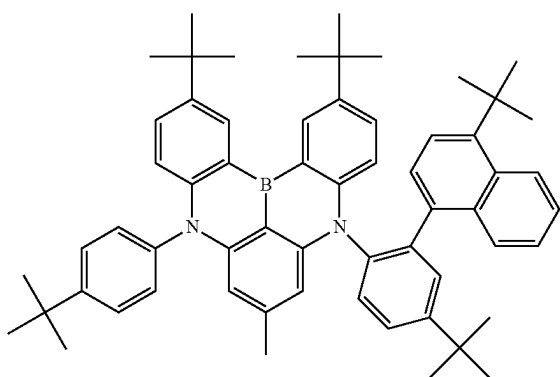
Dopant 14



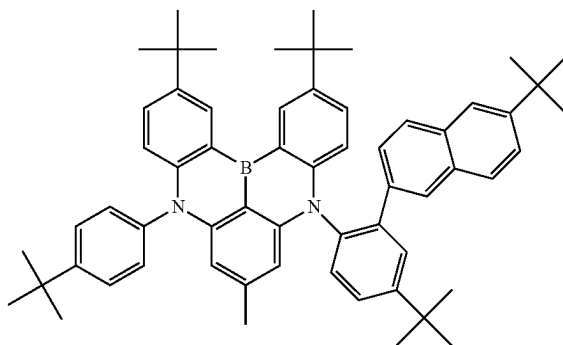
Dopant 15



Dopant 16

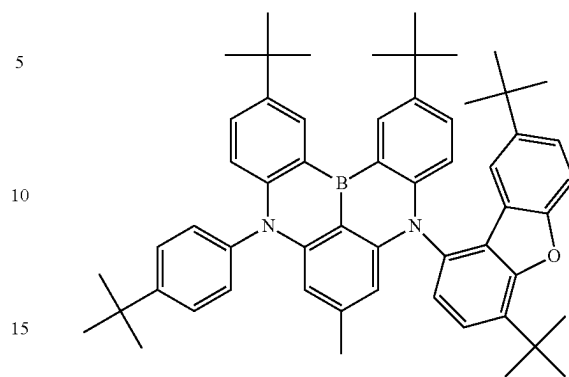


Dopant 17

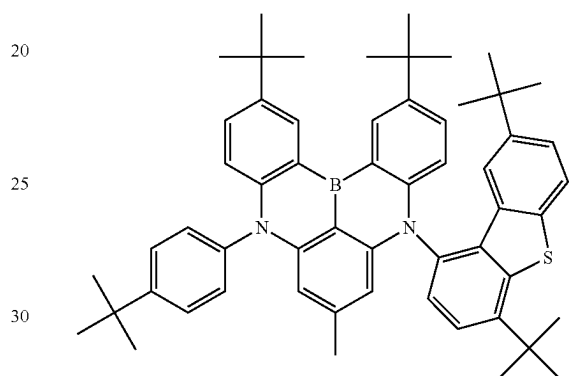
**20**

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Dopant 18



Dopant 19

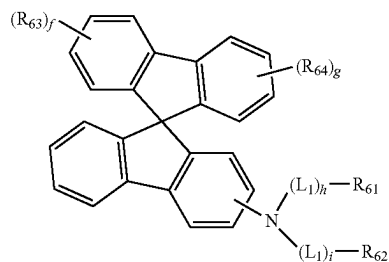


The contents of the first dopant **344** and the second dopant **444** in each of the EML1 **340** and the EML2 **440** may be, but is not limited to, about 1 wt % to about 10 wt %, for example, about 1 wt % to about 5 wt %. Each of the EML1 **340** and the EML2 **440** may have a thickness of, but is not limited to, about 100 Å to about 1000 Å, for example, about 200 Å to about 500 Å.

The HIL **310** is disposed between the first electrode **210** and the HTL1 **320** and improves an interface property between the inorganic first electrode **210** and the organic HTL1 **320**. In one exemplary embodiment, the HIL **310** may include hole injection host and hole injection dopant.

In one exemplary aspect, the hole injection host may include, but is not limited to, a spirofluorene-based organic compound having the following structure of Formula 7:

Formula 7



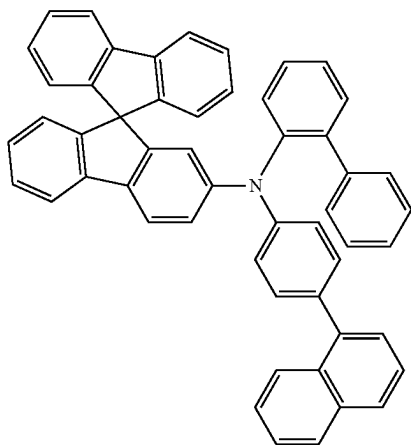
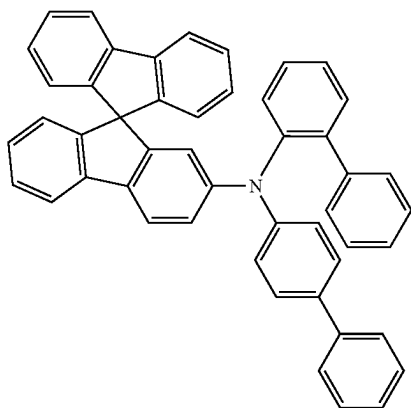
wherein each of R_{61} and R_{62} is independently a C_6 - C_{30} aryl group or a C_3 - C_{30} hetero aryl group, wherein each of the C_6 - C_{30} aryl group and the C_3 - C_{30} hetero aryl group may be

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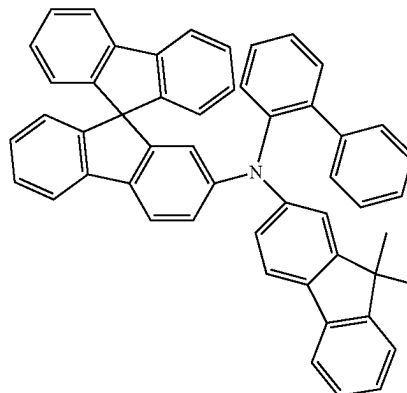
independently unsubstituted or substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{30} aryl group; each of R_{63} and R_{64} is independently protium, deuterium or a C_1 - C_{20} alkyl group; each of f and g is a number of substituents and is independently an integer of 0 to 4; each of L_1 and L_2 is independently a C_6 - C_{30} arylene group, wherein the C_6 - C_{30} arylene group may be unsubstituted or substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{30} aryl group; and each of h and i is an integer of 0 or 1.

For example, each of L_1 and L_2 in Formula 7 may comprise independently a phenylene group unsubstituted or substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{20} aryl group (e.g., phenyl group), and each of R_{61} and R_{62} may comprise independently a phenyl group, a naphthyl group, a fluorenyl group and a carbazolyl group each of which may be independently unsubstituted or substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{30} aryl group (e.g., phenyl group). As an example, the hole injection host may be selected from, but is not limited to, the following spirofluorene-based organic compound having the structure of Formula 8:

Formula 8

**22**

-continued



E3

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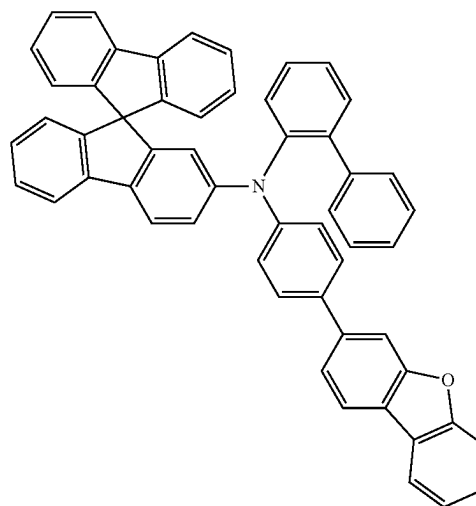
E1

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E4

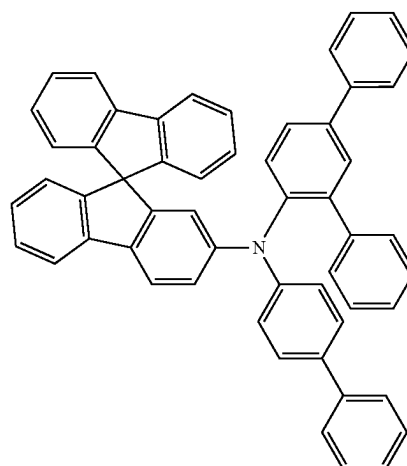
E2

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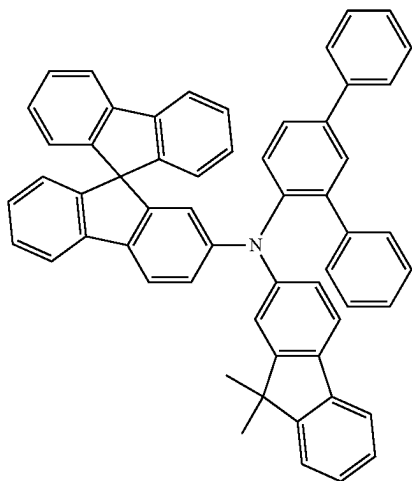
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E5

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E6

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E7

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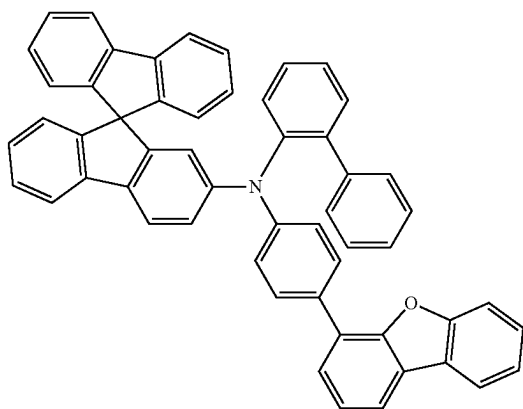
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E8

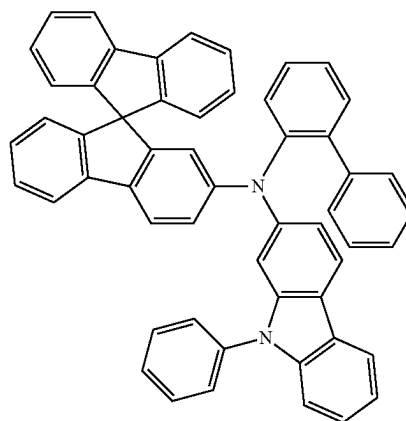
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**24**

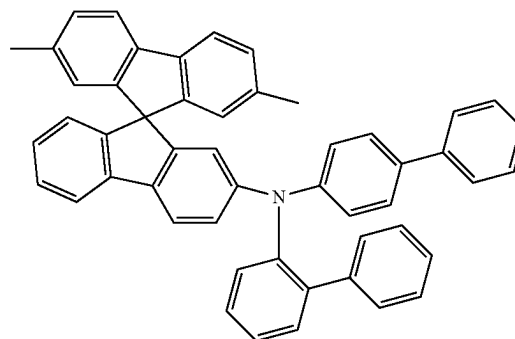
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E9

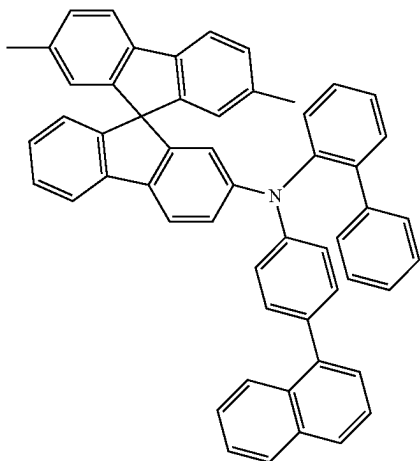
E10

E11



25

-continued



E12

26

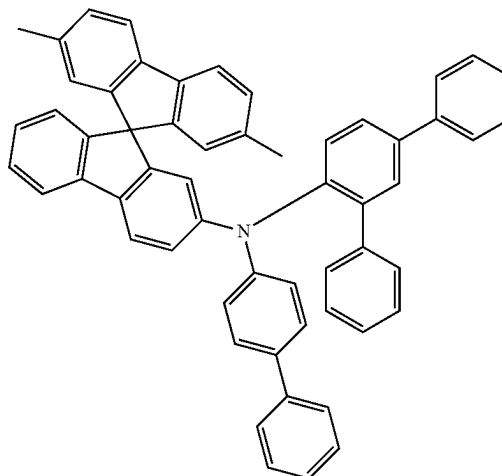
-continued

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E15

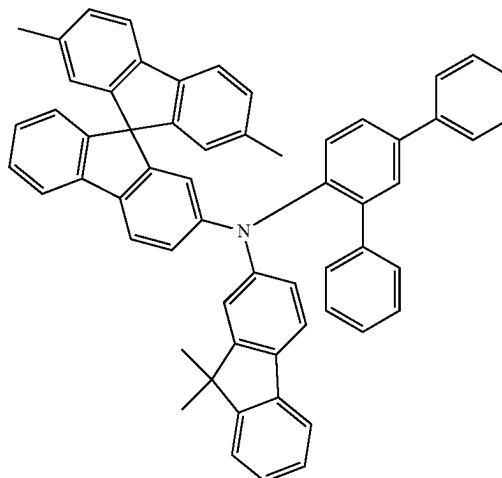
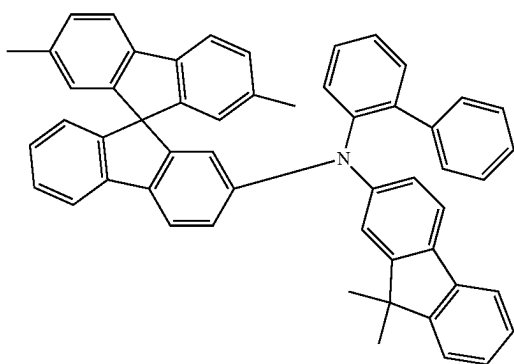
25

E13

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E16

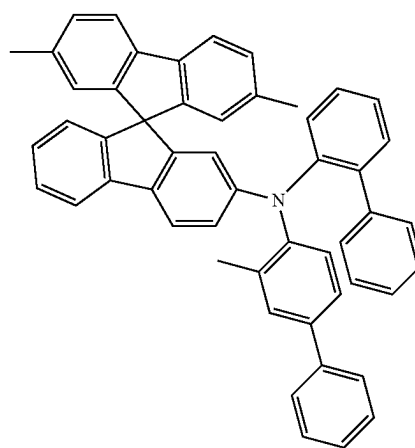
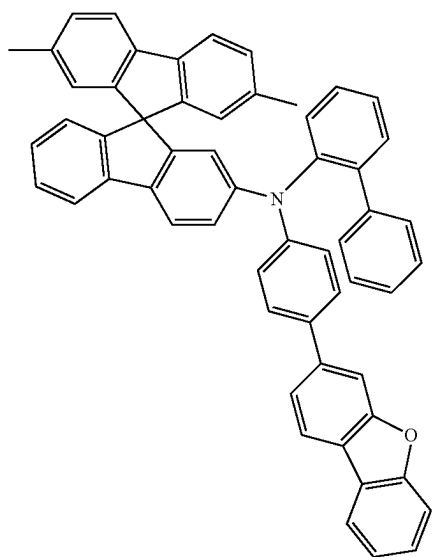
E14 45

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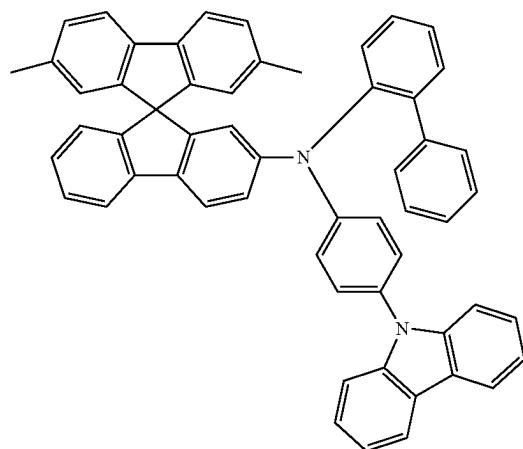
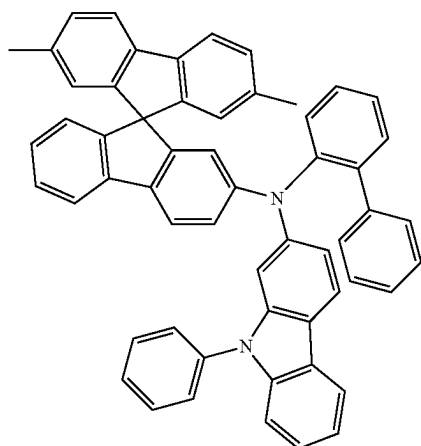
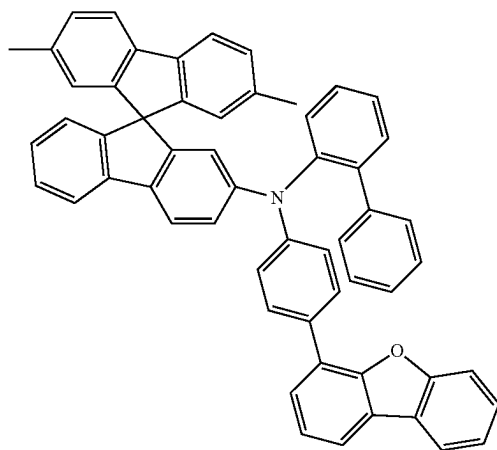
65



E17

27

-continued



In an exemplary aspect, the hole injection dopant may have a radialene structure. The hole injection dopant having the radialene structure may be selected from, but is not limited to, the following organic compounds having the structure of Formula 9:

28

E18 Formula 9

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E19 20

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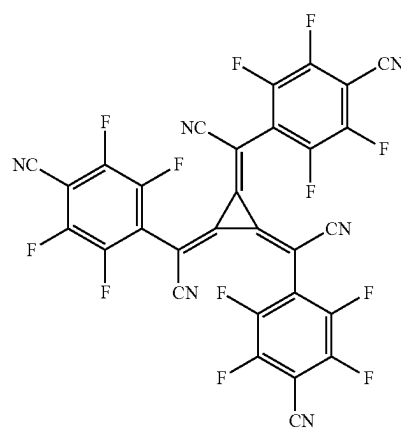
E20

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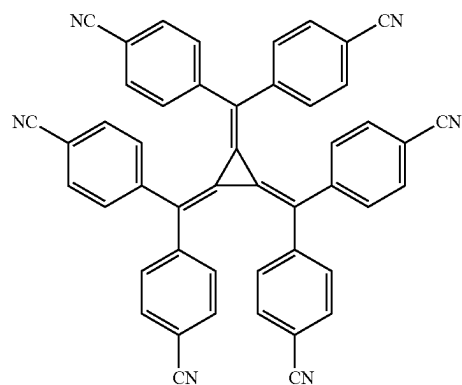
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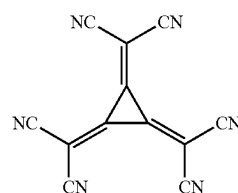
60



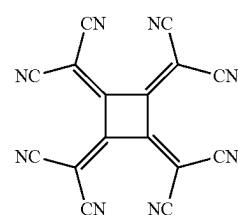
I1



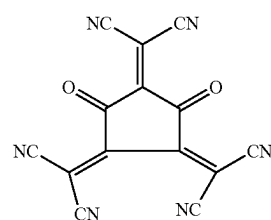
I2



I3



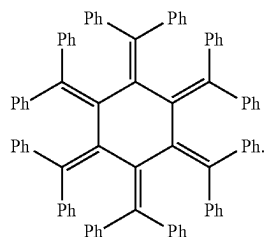
I4



I5

29

-continued



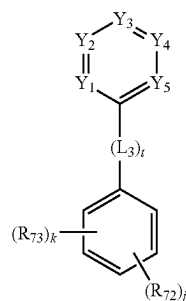
The contents of the hole injection dopant in the HIL **310** may be, but is not limited to, about 0.1 wt % to about 10 wt %, for example, about 0.1 wt % to about 5 wt %.

The HIL **310** may have a thickness of, but is not limited to, about 10 Å to about 300 Å, for example, about 50 Å to about 150 Å.

Each of the HTL1 **320** and the HTL2 **420** transports holes to the EML1 **340** and the EML2 **440**, respectively. In one exemplary aspect, each of the HTL1 **320** and the HTL2 **420** may include independently, but is not limited to, the spiro-fluorene-based organic compound having the structure of Formulae 7 and 8. Each of the HTL1 **320** and the HTL2 **420** may have a thickness of, but is not limited to, about 200 Å to about 1000 Å, for example, about 200 Å to about 500 Å.

The ETL1 **350** transports electrons into the EML1 **340**. In an exemplary aspect, the ETL1 **350** may include an organic compound having triplet energy level T^1 of equal to or more than about 2.6 eV. As an example, the ETL1 **350** may include an azine-based organic compound having the following structure of Formula 10:

Formula 10

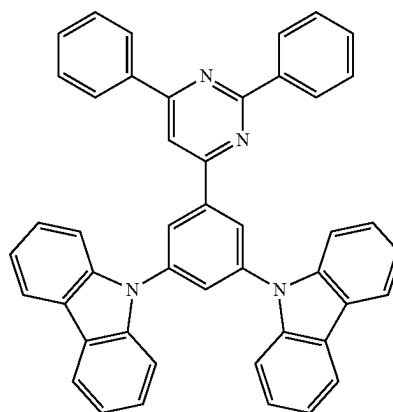
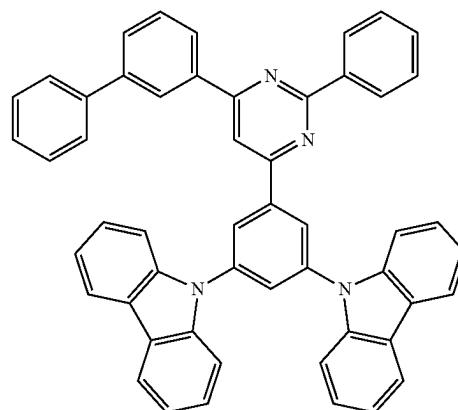
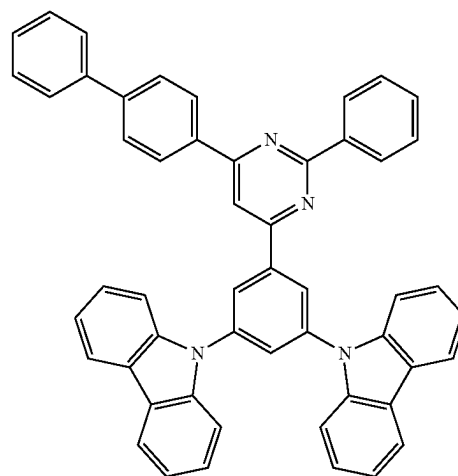


wherein each of Y_1 , Y_2 , Y_3 , Y_4 and Y_5 is independently CR₇₁ or nitrogen (N), provided that one to three of Y_1 , Y_2 , Y_3 , Y_4 and Y_5 is nitrogen; R₇₁ is hydrogen or a C₆-C₃₀ aryl group; L₃ is a C₆-C₃₀ arylene group; R₇₂ is a C₆-C₃₀ aryl group or a C₅-C₃₀ hetero aryl group, wherein each of the C₆-C₃₀ aryl group and the C₅-C₃₀ hetero aryl group may be independently unsubstituted or substituted with a C₆-C₃₀ aryl group or a C₃-C₃₀ hetero aryl group; R₇₃ is hydrogen or adjacent two of R₇₃ form a C₆-C₂₀ aromatic ring when k is 2 or more; j is 1 or 2; k is an integer of 0 to 4; and l is 0 or 1.

For example, the C₆-C₃₀ aryl group or C₃-C₃₀ hetero aryl group substituted to R₇₂ may comprise a C₁₀-C₃₀ fused aryl group and a C₁₀-C₃₀ fused hetero aryl group. R₇₃ in Formula 10 may be fused to form a naphthyl group. As an example, the ETL1 **350** may be selected from, but is not limited to, the following azine-based organic compounds having the structure of Formula 11:

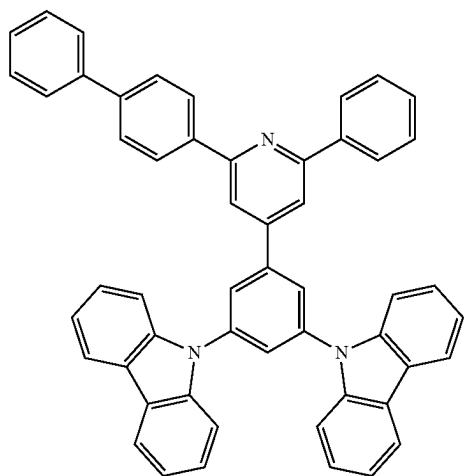
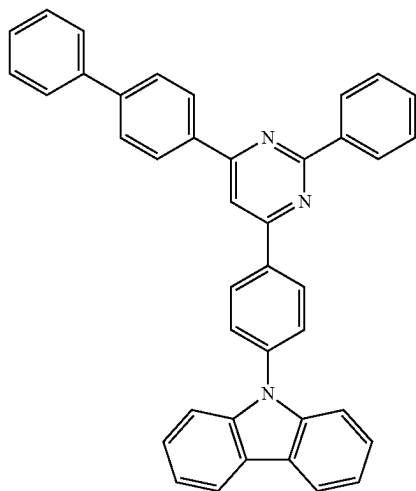
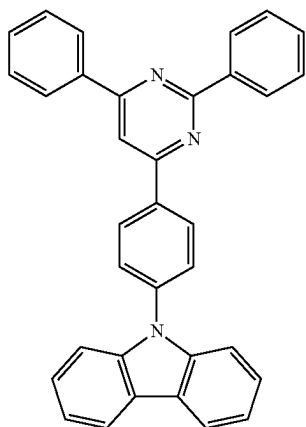
30

Formula 11

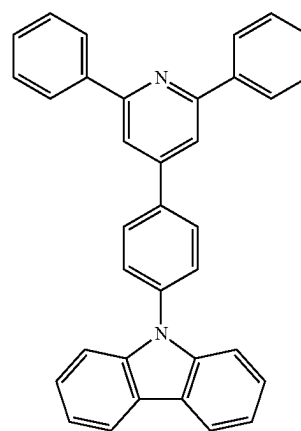
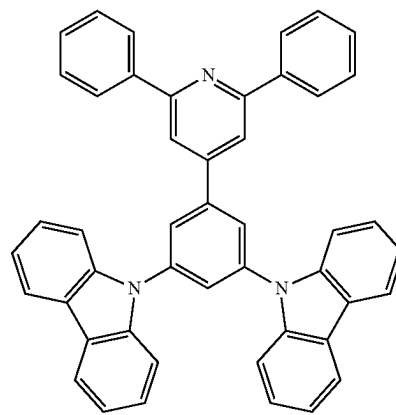
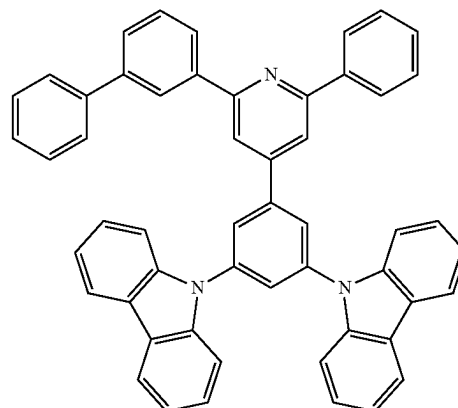


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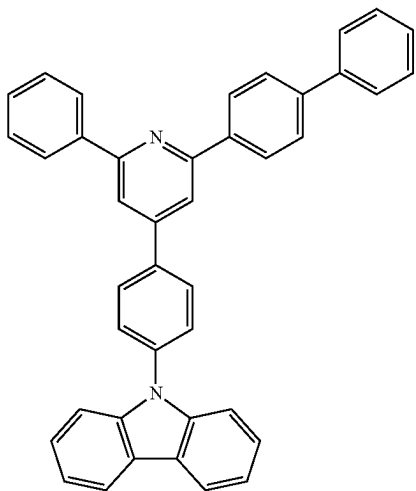
**32**

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33

-continued



F10

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F11

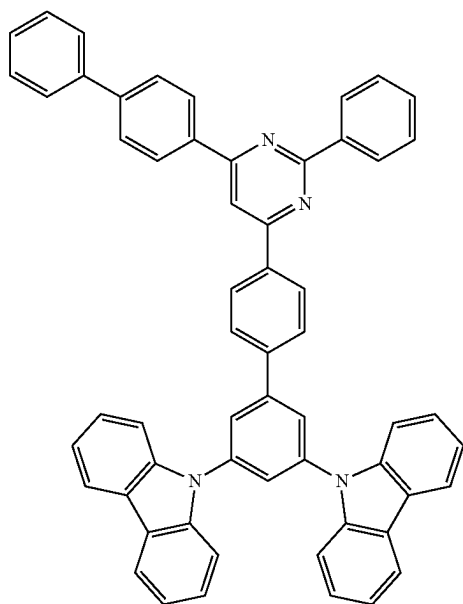
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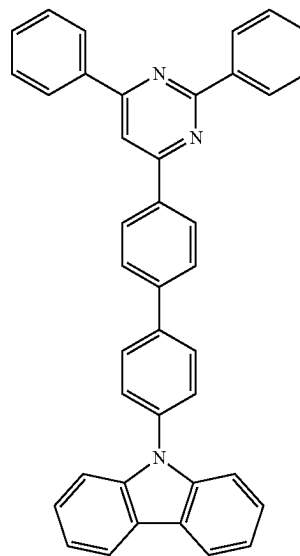
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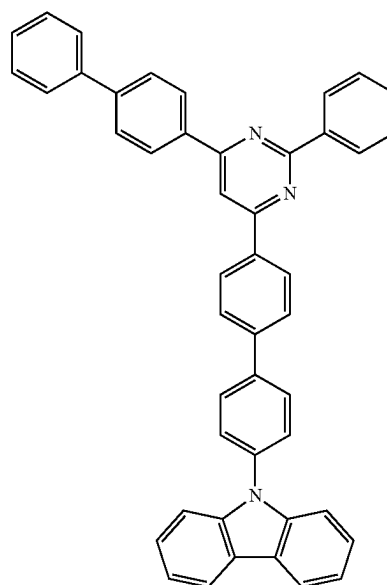
65

**34**

-continued



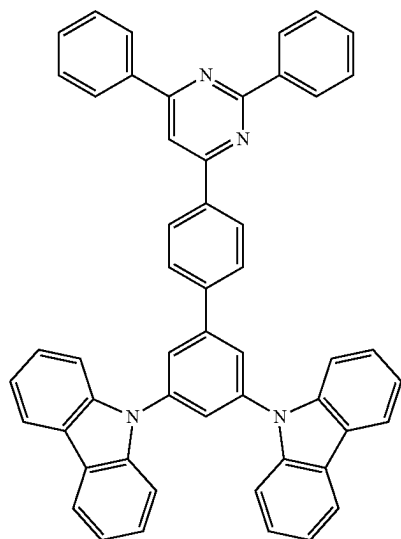
F12



F13

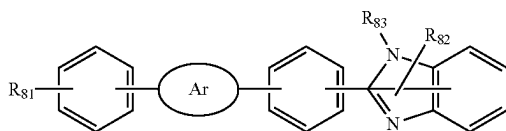
35

-continued



The ETL2 450 transports electrons into the EML2 440. The ETL2 450 may include an organic compound having excellent electron injection property and charge mobility. As an example, the ETL2 450 may include a benzimidazole-based organic compound having the following structure of Formula 12:

Formula 12

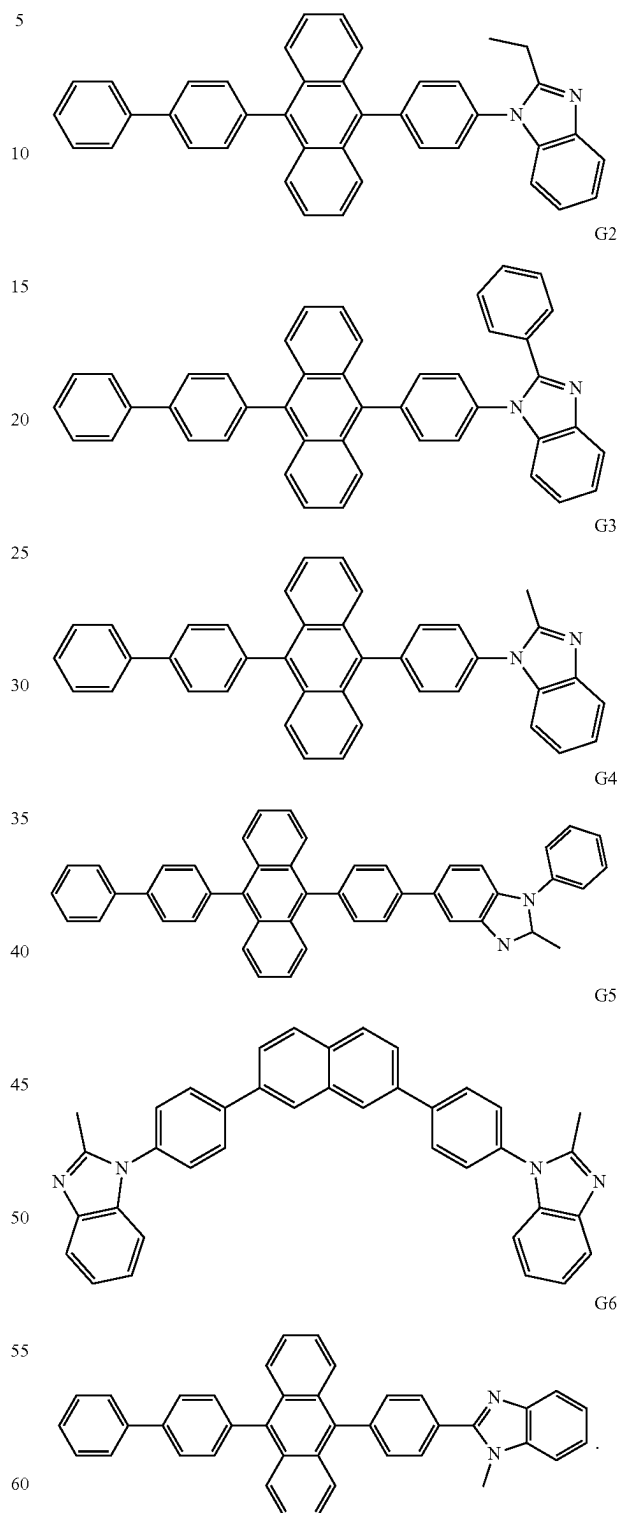


wherein Ar is a C₁₀-C₃₀ arylene group; R₈₁ is a C₆-C₃₀ aryl group or a C₅-C₃₀ hetero aryl group, wherein each of the C₆-C₃₀ aryl group and the C₅-C₃₀ hetero aryl group is unsubstituted or substituted with a C₁-C₁₀ alkyl group; each of R₈₂ and R₈₃ is independently hydrogen, a C₁-C₁₀ alkyl group or a C₆-C₃₀ aryl group.

For example, Ar may comprise a naphthylene group and an anthracenylene group, R₈₁ may comprise a phenyl group and a benzimidazole group each of which is independently unsubstituted or substituted with a C₁-C₁₀ alkyl group and each of R₈₂ and R₈₃ may comprise independently hydrogen, methyl, ethyl and a phenyl group. As an example, the benzimidazole-based organic compound in the ETL2 450 may be selected from, but is not limited to, the following benzimidazole-based organic compounds having the structure of Formula 13:

36

F14 Formula 13



In one exemplary aspect, the ETL2 450 may include the benzimidazole-based organic compound having the structure of Formulae 12 to 13 and other electron transport material being an organometallic compound. The other

37

electron transport material being an organometallic compound in the ETL2 450 may comprise, but is not limited to, lithium quinolate (Liq), tris(8-hydroxyquinoline aluminum (Alq₃) and bis(2-methyl-8-quinolinolato-N1,O8)-(1,1'-biphenyl-4-olato)aluminum (BALq). The contents of the other electron transport material being the organometallic compound in the ETL2 450 may be, but is not limited to, equal to or less than about 50 wt %, for example, about 2 wt % to about 50 wt %. When the ETL2 450 includes the benzimidazole-based organic compound and the other electron transport material being the organometal compound, the ETL2 450 may have a thickness of, but is not limited to, about 50 Å to about 500 Å, for example, about 50 Å to about 350 Å.

In another exemplary aspect, the ETL2 450 may include the benzimidazole-based organic compound having the structure of Formulae 12 to 13, the azine-based organic compound having the structure of Formulae 10 to 11, and other electron transport material of the organometallic compound. In this case, the benzimidazole-based organic compound having the structure of Formulae 12 to 13 and the azine-based organic compound having the structure of Formulae 10 to 11 may be mixed with a weight ratio of about 1:4 to about 4:1, for example, about 1:2 to about 2:1.

The contents of the other electron transport material being the organometallic compound in the ETL2 450 may be, but is not limited to, equal to or less than about 50 wt %, for example, about 2 wt % to about 50 wt %. When the ETL2 450 includes the benzimidazole-based organic compound, the azine-based organic compound and the other electron transport material being the organometal compound, the ETL2 450 may have a thickness of, but is not limited to, about 100 Å to about 500 Å, for example, about 100 Å to about 350 Å.

The EIL 460 is disposed between the second electrode 220 and the ETL2 450, and can improve physical properties of the second electrode 220 and therefore, can enhance the lifespan of the OLED D1. In one exemplary embodiment, the EIL 460 may include, but is not limited to, an alkali metal halide or alkaline earth metal halide such as LiF, CsF, NaF, BaF₂ and the like, and/or an organic metal compound such as lithium benzoate, sodium stearate, and the like.

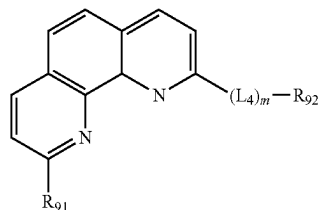
Alternatively, the EIL 460 may be doped with an alkali metal such as Li, Na, K and Cs, an alkaline earth metal such as Mg, Sr, Ba and Ra and/or a lanthanide metal such as Yb in addition to the alkali metal halide, the alkaline earth metal halide and the organic metal compound. In this case, the alkali metal halide/alkaline earth metal halide/organic metal compound and the alkali metal/alkaline earth metal/lanthanide metal in the EIL 460 may be mixed with, but is not limited to, a weight ratio of about 4:1 to about 1:4, for example, about 2:1 to about 1:2. The EIL 460 may have a thickness of, but is not limited to, about 5 Å to about 100 Å, for example, about 10 Å to about 50 Å.

The CGL 370 is disposed between the first emitting part 300 and the second emitting part 400. The CGL 370 includes an N-type CGL (N-CGL) 380 disposed between the ETL1 350 and the HTL2 420 and a P-type CGL (P-CGL) 390 disposed between the N-CGL 380 and the HTL2 420. The N-CGL 380 provides electrons with the EML1 340 of the first emitting part 300 and the P-CGL 390 provides holes with the EML2 440 of the second emitting part 400.

The N-CGL 380 may be an organic layer including an N-type host and an N-type dopant. For example, the N-type host may include a phenanthroline-based organic compound having the following structure of Formula 14:

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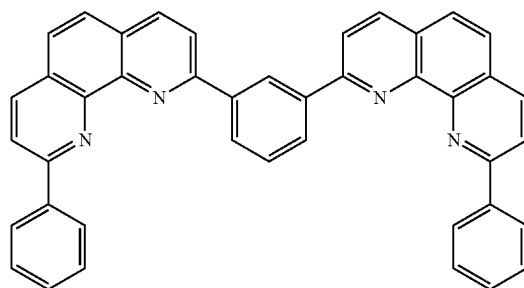
Formula 14



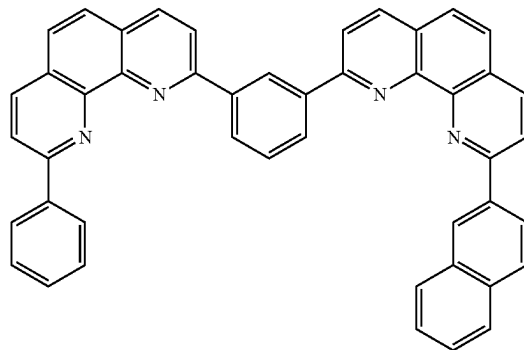
wherein R₉₁ is hydrogen or a C₆-C₃₀ aryl group, wherein the C₆-C₃₀ aryl group may be unsubstituted or substituted with a C₁-C₁₀ alkyl group; R₉₂ is a C₆-C₃₀ aryl group, wherein the C₆-C₃₀ aryl group may be unsubstituted or substituted with a C₁-C₁₀ alkyl group; each of L₄ is independently a C₆-C₃₀ arylene group or a C₅-C₃₀ hetero arylene group; and m is 1 or 2.

For example, R₉₁ in Formula 14 may comprise hydrogen and an unsubstituted or methyl-substituted phenyl group, an unsubstituted or methyl-substituted naphthyl group, R₉₂ may comprise a phenyl group, a naphthyl group and a phenanthrenyl group each of which is independently unsubstituted or substituted with methyl, and L₄ may comprise a phenylene group, a naphthylene group, an anthracenylene group and a phenanthrolylene group. As an example, the N-type host may be selected from, but is not limited to, the following phenanthroline-based organic compounds having the structure of Formula 15:

Formula 15



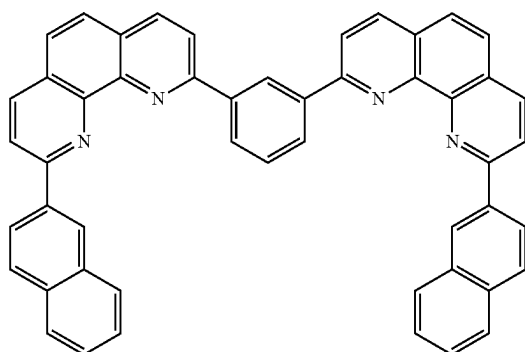
H1



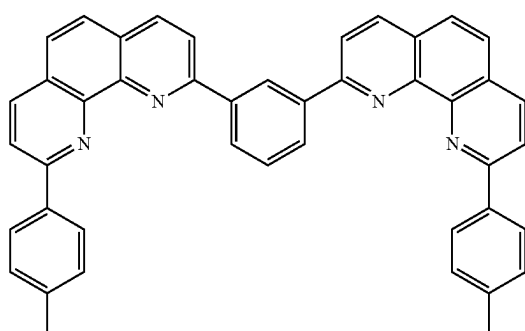
H2

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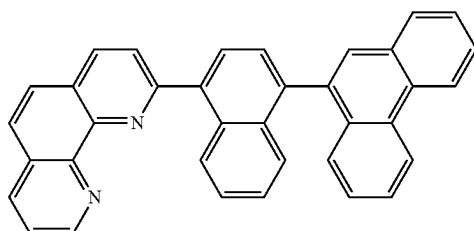
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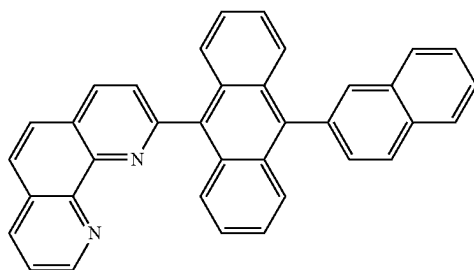
H3



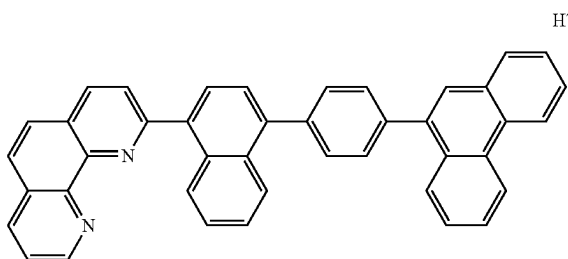
H4



H5



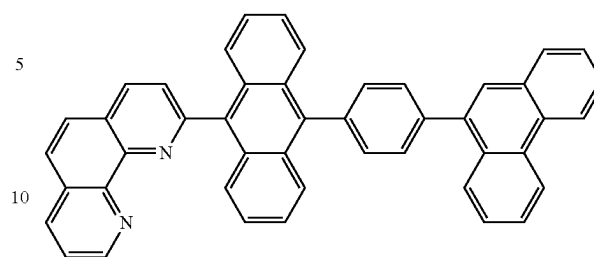
H6



H7

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-continued



H8

The N-type dopant may include an alkali metal such as Li, Na, K and Cs and/or an alkaline earth metal such as Mg, Sr, Ba and Ra. The N-type dopant allows the N-CGL **380** to have improved electron generations and electron injection. For example, the contents of the N-type dopant in the N-CGL **380** may be, but is not limited to, about 1 wt % to about 10 wt %. The N-CGL **380** may have a thickness of, but is not limited to, about 30 Å to about 500 Å, for example, about 50 Å to about 300 Å.

The P-CGL **390** may be an organic layer including a P-type host and a P-type dopant. For example, the P-type host may comprise the spirofluorene-based organic compound having the structure of Formulae 7 and 8, and the P-type dopant may comprise anyone selected from the radialene-based organic compound having the structure of Formula 9, but is not limited thereto. The contents of the P-type dopant in the P-CGL **390** may be, but is not limited to, about 1 wt % to about 40 wt %, for example, about 3 wt % to about 30 wt %. The P-CGL **390** may have a thickness of, but is not limited to, about 30 Å to about 500 Å, for example, about 50 Å to about 200 Å.

In accordance with this aspect, the EML **1 340** includes an anthracene-based organic compound having the structure of Formulae 1 to 2 where at least the anthracene-core is fully deuterated so as to have excellent hole injection property as the first host **342**, and the EML **2 440** includes an anthracene-based organic compound having the structure of Formulae 3 to 4 where all the carbon atoms are not deuterated or a part of carbon atom is deuterated so as to have excellent electron injection property as the second host **442**. Accordingly, the OLED **D1** can lower its driving voltage and maximize its luminous efficiency and luminous lifespan.

In the above embodiment, the OLED has two emitting parts. Unlikely, an OLED can emit white light. FIG. **4** is a schematic cross-sectional view illustrating an organic light emitting display device as an organic light emitting device in accordance with another exemplary aspect of the present disclosure.

As illustrated in FIG. **4**, the organic light emitting display device **500** comprises a first substrate **502** that defines each of a red pixel region RP, a green pixel region GP and a blue pixel region BP, a second substrate **504** facing the first substrate **502**, a thin film transistor Tr over the first substrate **502**, an organic light emitting diode D disposed between the first and second substrates **502** and **504** and emitting white (W) light and a color filter layer **580** disposed between the organic light emitting diode D and the second substrate **504**.

Each of the first and second substrates **502** and **504** may include, but is not limited to, glass, flexible material and/or polymer plastics. For example, each of the first and second substrates **502** and **504** may be made of PI, PES, PEN, PET, PC or combination thereof. The first substrate **502**, over

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which a thin film transistor Tr and an organic light emitting diode D are arranged, forms an array substrate.

A buffer layer **506** may be disposed over the first substrate **502**, and the thin film transistor Tr is disposed over the buffer layer **506** correspondingly to each of the red pixel region RP, the green pixel region GP and the blue pixel region BP. The buffer layer **506** may be omitted.

A semiconductor layer **510** is disposed over the buffer layer **506**. The semiconductor layer **510** may be made of oxide semiconductor material or polycrystalline silicon.

A gate insulating layer **520** including an insulating material, for example, inorganic insulating material such as silicon oxide (SiO_x) or silicon nitride (SiN_x) is disposed on the semiconductor layer **510**.

A gate electrode **530** made of a conductive material such as a metal is disposed over the gate insulating layer **520** so as to correspond to a center of the semiconductor layer **510**. An interlayer insulating layer **540** including an insulating material, for example, inorganic insulating material such as silicon oxide (SiO_x) or silicon nitride (SiN_x), or an organic insulating material such as benzocyclobutene or photo-acryl, is disposed on the gate electrode **530**.

The interlayer insulating layer **540** has first and second semiconductor layer contact holes **542** and **544** that expose both sides of the semiconductor layer **510**. The first and second semiconductor layer contact holes **542** and **544** are disposed over opposite sides of the gate electrode **530** with spacing apart from the gate electrode **530**.

A source electrode **552** and a drain electrode **554**, which are made of a conductive material such as a metal, are disposed on the interlayer insulating layer **540**. The source electrode **552** and the drain electrode **554** are spaced apart from each other with respect to the gate electrode **530**, and contact both sides of the semiconductor layer **510** through the first and second semiconductor layer contact holes **542** and **544**, respectively.

The semiconductor layer **510**, the gate electrode **530**, the source electrode **552** and the drain electrode **554** constitute the thin film transistor Tr, which acts as a driving element.

Although not shown in FIG. 4, a gate line and a data line, which cross each other to define a pixel region, and a switching element, which is connected to the gate line and the data line, may be further formed in the pixel region. The switching element is connected to the thin film transistor Tr, which is a driving element. In addition, a power line is spaced apart in parallel from the gate line or the data line, and the thin film transistor Tr may further include a storage capacitor configured to constantly keep a voltage of the gate electrode for one frame.

A passivation layer **560** is disposed on the source and drain electrodes **552** and **554** with covering the thin film transistor Tr over the whole first substrate **502**. The passivation layer **560** has a drain contact hole **562** that exposes the drain electrode **554** of the thin film transistor Tr.

The organic light emitting diode (OLED) D is located over the passivation layer **560**. The OLED D includes a first electrode **610** that is connected to the drain electrode **554** of the thin film transistor Tr, a second electrode **620** facing from the first electrode **610** and an emissive layer **630** disposed between the first and second electrodes **610** and **620**.

One of the first electrode **610** formed for each pixel region and the second electrode **620** formed integrally may be an anode and the other of the first electrode **610** and the second electrode **620** may be a cathode. One of the first and second electrodes **610** and **620** may be a transmissive (or semi-

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transmissive) electrode and the other of the first and second electrodes **610** and **620** may be reflective electrode.

For example, the first electrode **610** may be an anode and may include a conductive material having relatively high work function value. For example, the first electrode **610** may include, ITO, IZO, ITZO, SnO, ZnO, ICO, AZO, and the like.

The second electrode **620** is disposed over the first substrate **502** above which the emissive layer **630** is disposed. The second electrode **620** may be disposed over a whole display area. The second electrode **620** may be a cathode and may include a conductive material with a relatively low work function value, for example, a low-resistant metal. As an example, the second electrode **620** may include, but is not limited to, Al, Mg, Ca, Ag, alloy thereof or combination thereof such as Al—Mg or Ag:Mg.

When the organic light emitting display device **500** is a bottom-emission type, the first electrode may have a single-layered structure of transparent conductive oxide. Alternatively, when the organic light emitting display device **500** is a bottom-emission type, a reflective electrode or a reflective layer may be disposed under the first electrode. For example, the reflective electrode or the reflective layer may include, but is not limited to, Ag or APC alloy. In the OLED of a top-emission type, the first electrode **610** may have a triple-layered structure of ITO/Ag/ITO or ITO/APC/ITO. The second electrode **620** is thin so as to have light-transmissive (or semi-transmissive) property.

A bank layer **564** is disposed on the passivation layer **560** in order to cover edges of the first electrode **610**. The bank layer **564** exposes a center of the first electrode **610** corresponding to each of the red pixel region RP, the green pixel region GP and the blue pixel region BP. The bank layer **564** may be omitted.

An emissive layer **630** including multiple emitting parts is disposed on the first electrode **610**. Each emitting part may include an emitting material layer, respectively. Each emitting part may further include at least one of an HIL, an HTL, an EBL, an ETL and an EIL.

The color filter layer **580** is disposed over the OLED D and includes a red color filter **582**, a green color filter **584** and a blue color filter **586** each of which is disposed correspondingly to the red pixel region RP, the green pixel region GP and the blue pixel region BP, respectively. Although not shown in FIG. 4, the color filter layer **580** may be attached to the OLED D through an adhesive layer. Alternatively, the color filter layer **580** may be disposed directly on the OLED D.

In addition, an encapsulation film may be disposed over the second electrode **620** in order to prevent outer moisture from penetrating into the OLED D. The encapsulation film may have, but is not limited to, a laminated structure of a first inorganic insulating film, an organic insulating film and a second inorganic insulating film (**170** in FIG. 2).

The organic light emitting display device **500** may further comprise a polarizing plate in order to reduce reflection of external light. As an example, the polarizing plate may be a circular polarizing plate. When the organic light emitting display device is a bottom-emission type, the polarizing plate may be disposed under the first substrate **502**. Alternatively, when the organic light emitting display device **500** is a top-emission type, the polarizing plate may be attached onto the encapsulation film, for example, on the second substrate **504**.

In FIG. 4, the light emitted from the OLED D is transmitted through the second electrode **620** and the color filter layer **580** is disposed over the OLED D. Alternatively, the

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light emitted from the OLED D is transmitted through the first electrode **610** and the color filter layer **580** may be disposed between the OLED D and the first substrate **502**. In addition, a color conversion layer may be formed between the OLED D and the color filter layer **580**. The color conversion layer may include a red color conversion layer, a green color conversion layer and a blue color conversion layer each of which is disposed correspondingly to each pixel region (RP, GP and BP), respectively, so as to convert the white (W) color light to each of a red, green and blue color lights, respectively.

As described above, the white (W) color light emitted from the OLED D is transmitted through the red color filter **582**, the green color filter **584** and the blue color filter **586** each of which is disposed correspondingly to the red pixel region RP, the green pixel region GP and the blue pixel region BP, respectively, so that red, green and blue color lights are displayed in the red pixel region RP, the green pixel region GP and the blue pixel region BP.

FIG. 5 is a schematic cross-sectional view illustrating an organic light emitting diode having three emitting parts in accordance with another exemplary aspect of the present disclosure. As illustrated in FIG. 5, the organic light emitting diode (OLED) D2 includes first and second electrodes **610** and **620** and an emissive layer **630** disposed between the first and second electrodes **610** and **620**.

One of the first and second electrodes **610** and **620** may be an anode and the other of the first and second electrodes **610** and **620** may be a cathode. As an example, the first electrode **610** may be an anode injecting holes and the second electrode **620** may be a cathode injecting electrons. One of the first and second electrodes **610** and **620** may be a reflective electrode and the other of the first and second electrodes **610** and **620** may be a transmissive (or semi-transmissive) electrode. As an example, each of the first and second electrodes **610** and **620** may have a thickness of, but is not limited to, about 100 Å to about 2000 Å, for example, about 100 Å to about 1000 Å.

The emissive layer **630** includes a first emitting part **700**, a second emitting part **800** and a third emitting part **900**. The emissive layer **630** further includes a first charge generation layer (CGL1) **770** disposed between the first emitting part **700** and the third emitting part **900** and a second charge generation layer (CGL2) **970** disposed between the second emitting part **800** and the third emitting part **900**. As the CGL1 **770** is disposed between the first and third emitting parts **700** and **900**, and the CGL2 **970** is disposed between the second and third emitting parts **800** and **900**, the first emitting part **700**, the CGL1 **770**, the third emitting part **900**, the CGL2 **970** and the second emitting part **800** are disposed sequentially over the first electrode **610**. In other words, the first emitting part **700** is disposed between the first electrode **610** and the CGL1 **770**, the second emitting part **800** is disposed between the second electrode **620** and the CGL2 **970**, and the third emitting part **900** is disposed between the CGL1 **770** and the CGL2 **970**.

The first emitting part **700** includes a first emitting material layer (lower EML, EML1) **740**. The first emitting part **700** may further comprise at least one of an HIL **710** disposed between the first electrode **610** and the EML1 **740**, a HTL1 (lower HTL) **720** disposed between the HIL **710** and the EML1 **740** and an ETL1 (lower ETL) **750** disposed between the EML1 **740** and the CGL1 **770**. Alternatively, the first emitting part **700** may further comprise an EBL1 (lower EBL) **730** disposed between the HTL1 **720** and the EML1 **740**.

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The second emitting part **800** includes a second emitting material layer (upper EML, EML2) **840**. The second emitting part **800** may further comprise at least one of a HTL2 (upper HTL) **820** disposed between the CGL2 **970** and the EML2 **840**, an ETL2 (upper ETL) **850** disposed between the second electrode **620** and the EML2 **840** and an EIL **860** disposed between the second electrode **620** and the ETL2 **850**. Alternatively, the second emitting part **800** may further comprise an EBL2 (upper EBL) **830** disposed between the HTL2 **820** and the EML2 **840**.

The third emitting part **900** includes a third emitting material layer (middle EML, EML3) **940**. The third emitting part **900** may further include at least one of a third hole transport layer (middle HTL, HTL3) **920** disposed between the CGL1 **770** and the EML3 **940** and a third electron transport layer (middle ETL, ETL3) **950** disposed between the EML3 **940** and the CGL2 **970**.

The EML1 **740** includes a first host **742** of an anthracene-based organic compound and a first dopant **744** of a boron-based organic compound, so that the EML1 **740** emits blue light. The EML2 **840** includes a second host **842** of an anthracene-based organic compound and a second dopant **844** of a boron-based organic compound, so that the EML2 **840** emits blue light.

As an example, the first host **742** may be an anthracene-based organic compound where at least the carbon atoms in the anthracene core are deuterated and may have the structure of Formulae 1 and 2. The second host **842** may be an anthracene-based organic compound where none of the carbon atoms of the aromatic rings in the molecules is deuterated or a part of the carbon atoms is deuterated, and may have the structure of Formulae 3 and 4.

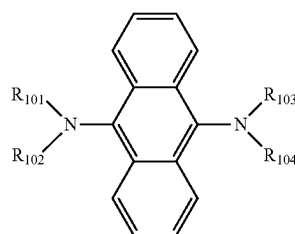
Each of the first and second dopants **744** and **844** includes a boron-based organic compound, so that both the EML1 **740** and the EML2 **840** emit blue light. The first dopant **744** may be identical to or different from the second dopant **844**. Each of the first and second dopants **744** and **844** may have independently the structure of Formulae 5 and 6.

The contents of the first dopant **744** and the second dopant **844** in each of the EML1 **740** and the EML2 **840** may be, but is not limited to, about 1 wt % to about 10 wt %, for example, about 1 wt % to about 5 wt %. As an example, each of the EML1 **740** and the EML2 **840** may have a thickness of, but is not limited to, about 100 Å to about 1000 Å, for example, about 200 Å to about 500 Å.

The HIL **710** is disposed between the first electrode **610** and the HTL1 **720** and improves an interface property between the inorganic first electrode **610** and the organic HTL1 **720**. The HIL **710** may include a hole injection host and a hole injection dopant.

In one exemplary aspect, the hole injection host may include, but is not limited to, an anthracene-based organic compound having the following structure of Formula 16:

Formula 16

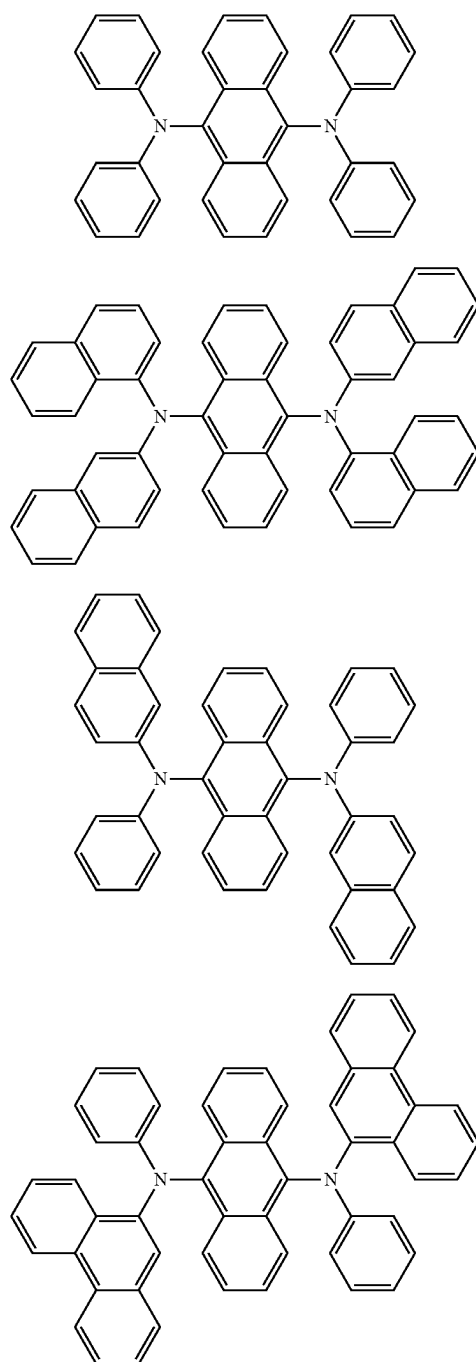


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wherein each of R_{101} , R_{102} , R_{103} and R_{104} is independently a C_6 - C_{30} aryl group, wherein the C_6 - C_{30} aryl group may be unsubstituted or substituted with a C_1 - C_{10} alkyl group.

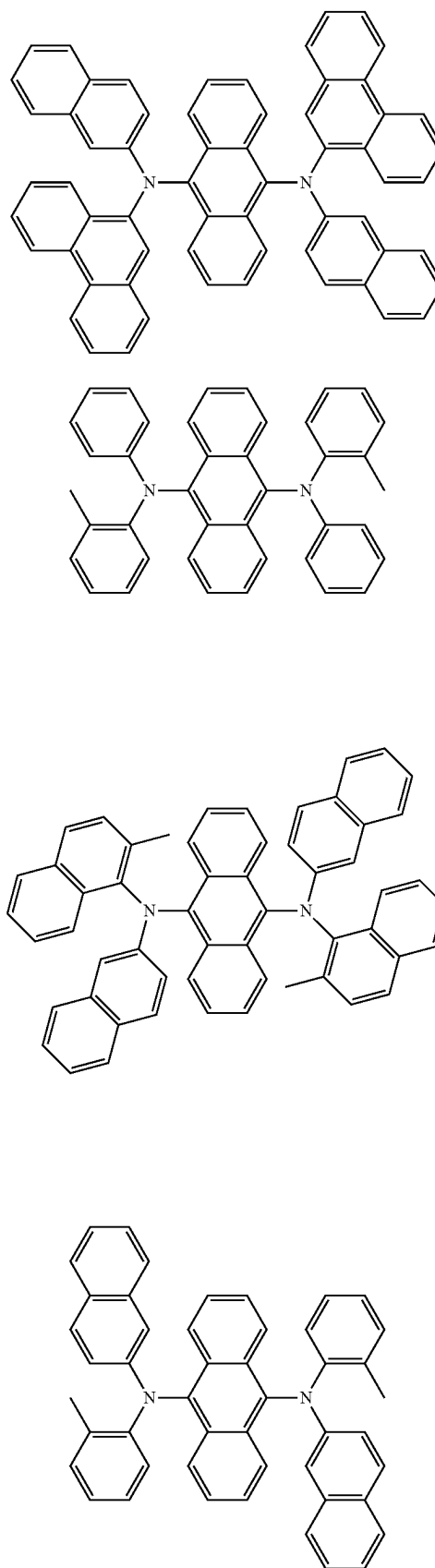
For example, each of R_{101} , R_{102} , R_{103} and R_{104} in Formula 16 may comprise independently a phenyl group, a naphthyl group (e.g., 1-naphthyl group and 2-naphthyl group) and a phenanthrenyl group each of which is unsubstituted or substituted with a C_1 - C_{10} alkyl group. As an example, the hole injection host may be selected from, but is not limited to, the following anthracene-based compounds having the structure of Formula 17:

Formula 17



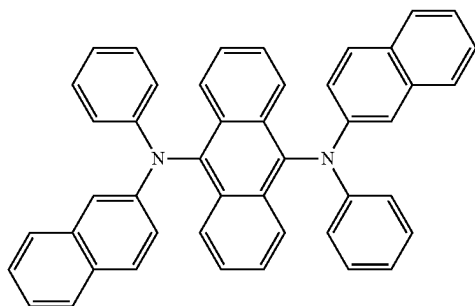
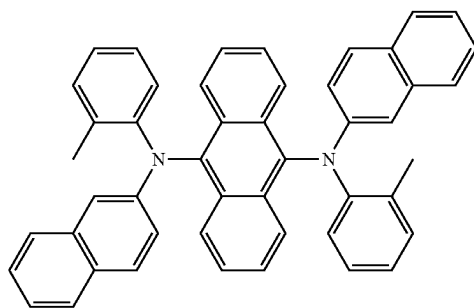
46

-continued



47

-continued



The hole injection dopant may comprise, but is not limited to, an alkali metal halide such as LiF, NaF and CsF and/or an alkaline earth metal halide such as BaF₂ and MgF₂.

The hole injection host having the structure of Formulae 16 and 17 and the hole injection dopant of the alkali metal halide and/or the alkaline earth metal halide in the HIL 710 may be mixed with a weight ratio of, but is not limited to, about 9:1 to about 5:5, for example, about 8:2 to about 5:5. As an example, the HIL 710 may have a thickness of, but is not limited to, about 5 Å to about 200 Å, for example, about 10 Å to about 100 Å.

Each of the HTL1 720, the HTL2 820 and the HTL3 920 transports holes into the EML1 740, the EML2 840 and the EML3 940, respectively. In one exemplary aspect, each of the HTL1 720, the HTL2 820 and the HTL3 920 may include independently, but is not limited to, the spirofluorene-based organic compound having the structure of Formulae 7 and 8. The HTL1 720 may have a thickness of about 300 Å to about 2000 Å, for example, about 500 Å to about 1000 Å, the HTL2 820 may have a thickness of about 300 Å to about 2000 Å, for example, about 500 Å to about 900 Å and the HTL3 920 may have a thickness of about 10 Å to about 300 Å, for example, about 10 Å to about 150 Å, but is not limited thereto.

The ETL1 750 transports electrons into the EML1 740. In one exemplary aspect, the ETL1 750 may include an organic compound having the triplet energy level of equal to or more than about 2.6 eV. As an example, the ETL1 750 may include the azine-based organic compound having the structure of Formulae 10 and 11. The ETL1 750 may have a thickness of, but is not limited to, about 30 Å to about 300 Å, for example, about 50 Å to about 150 Å.

The ETL2 850 transports electrons into the EML2 840 and may include an organic compound having excellent

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electron injection property. As an example, the ETL2 850 may include the benzimidazole-based organic compound having the structure of Formulae 12 to 13.

In one exemplary aspect, the ETL2 850 may consist of the benzimidazole-based organic compound. Since the benzimidazole-based organic compound having the structure of Formulae 12 to 13 has excellent electron injection property and charge mobility, the ETL2 850 including the benzimidazole-based organic compound and disposed adjacently to the second electrode 620 providing electrons can maximize its electron injection property and charge mobility.

In another exemplary aspect, the ETL2 850 may include the benzimidazole-based organic compound having the structure of Formulae 12 to 13 and the azine-based organic compound having the structure of Formulae 10 to 11. In this case, the benzimidazole-based organic compound and the azine-based organic compound in the ETL2 850 may be mixed with a weight ratio of, but is not limited to, about 1:4 to about 4:1, for example, about 1:2 to about 2:1. The ETL2 850 may have a thickness of, but is not limited to, about 30 Å to about 500 Å, for example, about 50 Å to about 350 Å.

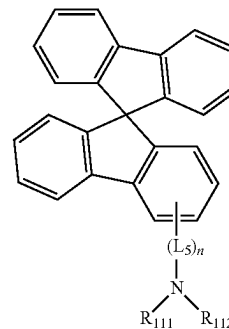
The ETL3 950 transports electrons into the EML3 840. In an exemplary aspect, the ETL3 950 may include the benzimidazole-based organic compound having the structure of Formulae 12 to 13. The ETL3 950 may have a thickness of, but is not limited to, about 30 Å to about 500 Å, for example, about 50 Å to about 350 Å.

The EIL 860 is disposed between the second electrode 620 and the ETL2 850, and can improve physical properties of the second electrode 620, and therefore, can enhance the lifespan of the OLED D2. In one exemplary embodiment, the EIL 860 may include, but is not limited to, an alkali metal halide or alkaline earth metal halide such as LiF, CsF, NaF, BaF₂ and the like, and/or an organic metal compound such as lithium benzoate, sodium stearate, and the like.

Alternatively, the EIL 860 may be doped with an alkali metal such as Li, Na, K and Cs, an alkaline earth metal such as Mg, Sr, Ba and Ra and/or a lanthanide metal such as Yb in addition to the alkali metal halide, the alkaline earth metal halide and the organic metal compound. The EIL 860 may have a thickness of, but is not limited to, about 5 Å to about 100 Å, for example, about 10 Å to about 50 Å.

The EBL1 730 prevents electrons from transporting to the first electrode 610 via the EML1 740, and the EBL2 830 prevents electrons from transporting to the CGL2 970 via the EML2 840. In one exemplary aspect, each of the EBL1 730 and the EBL2 830 may include independently, but is not limited to, an spirofluorene-based organic compound having the following structure of Formula 18:

Formula 18



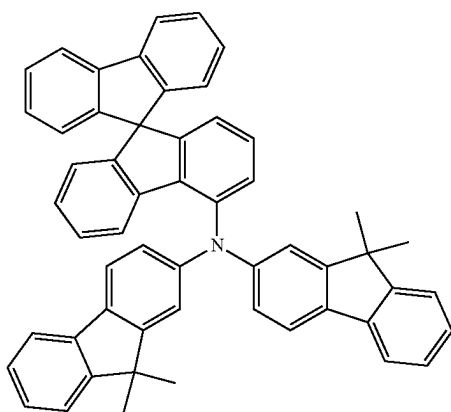
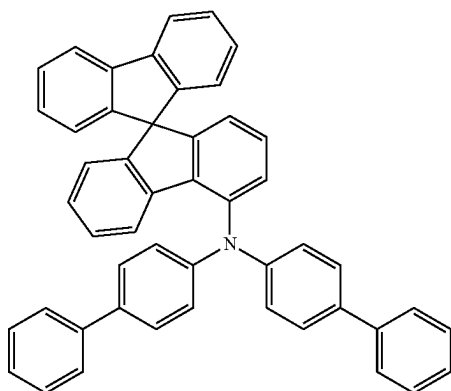
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wherein L_5 is a C_6 - C_{30} arylene group; n is 0 or 1; each of R_{11} and R_{112} is independently a C_6 - C_{30} aryl group or a C_5 - C_{30} hetero aryl group, wherein each of the C_6 - C_{30} aryl group and the C_5 - C_{30} hetero aryl group may be unsubstituted or substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{20} aryl group.

For example, L_5 in Formula 18 comprise a phenylene group and each of R_{111} and R_{112} in Formula 18 may comprise independently a phenyl group, a biphenyl group, a fluorenyl group, a carbazolyl group, a phenyl-carbazolyl group, a carbazolyl-phenyl group, a dibenzofuranyl group and a dibenzothiophenyl group each of which is independently unsubstituted or substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{20} aryl group (e.g., phenyl group).

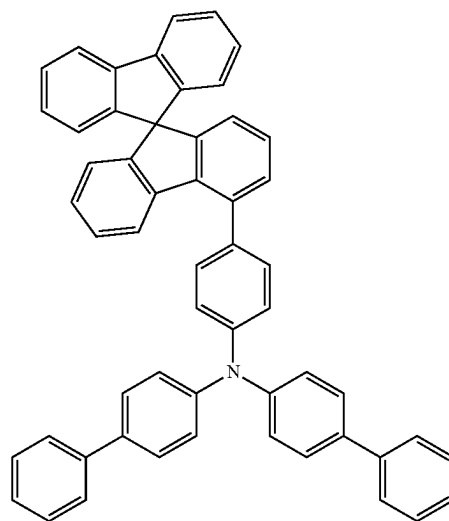
As an example, each of the EBL1 730 and the EBL2 830 may be independently selected from, but is not limited to, the following spirofluorene-based organic compounds having the structure of Formula 19:

Formula 19

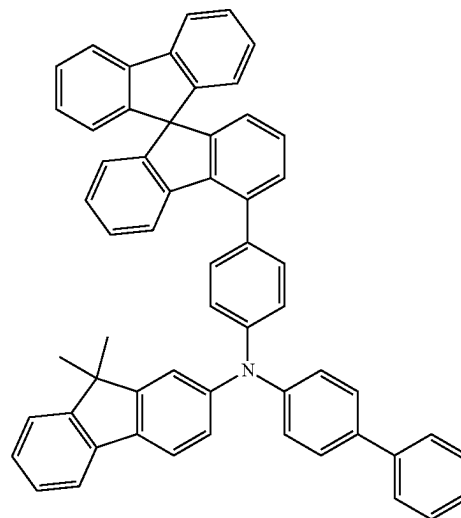


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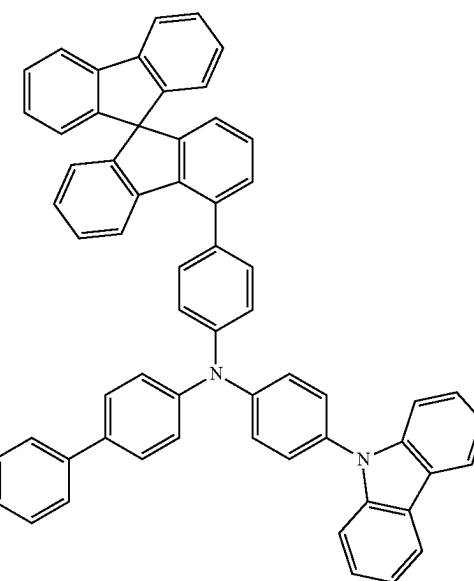
-continued



K3



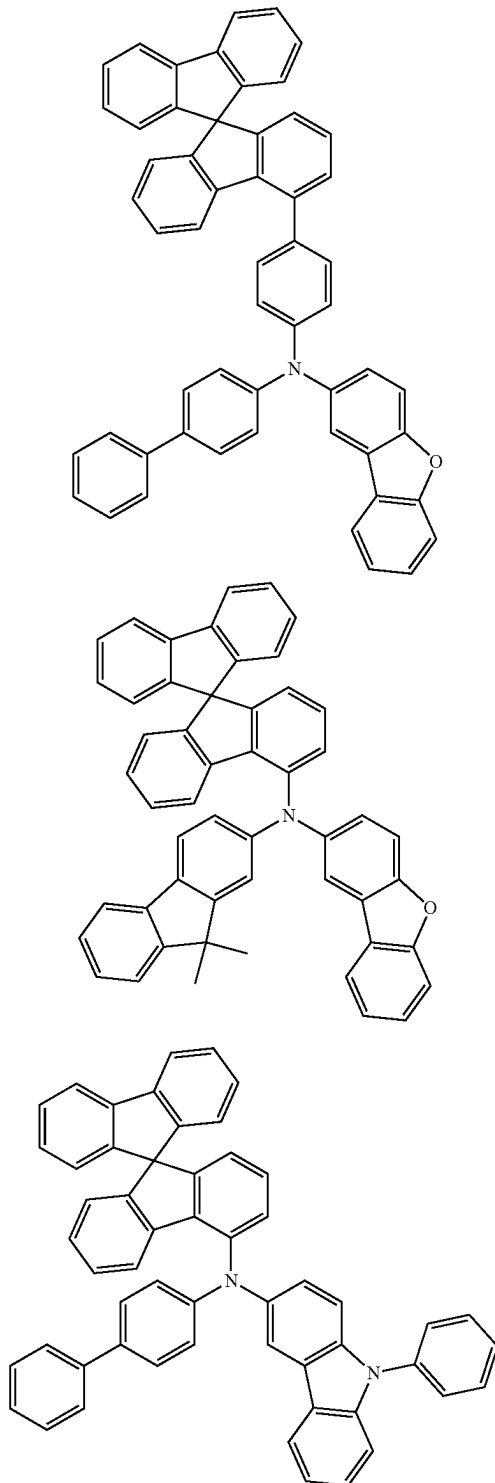
K4



K5

51

-continued



Each of the EBL1 730 and the EBL2 830 may have independently a thickness of, but is not limited to, about 30 Å to about 500 Å, for example, about 50 Å to about 250 Å.

The CGL1 770 is disposed between the first emitting part 700 and the second emitting part 800, particularly the third emitting part 900. The CGL1 770 includes a first N-type CGL (N-CGL1) 780 disposed between the ETL1 750 and the HTL3 920 and a first P-type CGL (P-CGL1) 790

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disposed between the N-CGL1 780 and the HTL3 920. The N-CGL1 780 provides electrons with the EML1 740 of the first emitting part 700 and the P-CGL1 790 provides holes with the EML3 940 of the third emitting part 900.

The CGL2 970 is disposed between the second emitting part 800 and the third emitting part 900. The CGL2 970 includes a second N-CGL (N-CGL2) 980 disposed between the ETL3 950 and the HTL2 820 and a second P-type CGL (P-CGL2) 990 disposed between the N-CGL2 980 and the HTL2 820. The N-CGL2 980 provides electrons with the EML3 940 of the third emitting part and the P-CGL2 990 provides holes with the EML2 840 of the second emitting part.

Each of the N-CGL1 780 and the N-CGL2 980 may be an organic layer including an N-type host and an N-type dopant, respectively. In one exemplary aspect, the N-type host may include the phenanthroline-based organic compound having the structure of Formulae 14 to 15. The N-type dopant may include an alkali metal such as Li, Na, K and Cs and/or an alkaline earth metal such as Mg, Sr, Ba and Ra. For example, the contents of the N-type dopant in each of the N-CGL1 780 and the N-CGL2 980 may be, but is not limited to, about 1 wt % to about 10 wt %. The N-CGL1 780 may have a thickness of, but is not limited to, about 30 Å to about 500 Å, for example, about 50 Å to about 250 Å, and the N-CGL2 980 may have a thickness of, but is not limited to, about 30 Å to about 600 Å, for example, about 50 Å to about 400 Å.

Each of the P-CGL1 790 and the P-CGL2 990 may be an organic layer including a P-type host and a P-type dopant, respectively. In one exemplary aspect, the P-type host may include the spirofluorene-based organic compound having the structure of Formulae 7 and 8, and the P-type dopant may include anyone selected from the radialene-based organic compound having the structure of Formula 9, but is not limited thereto. The contents of the P-type dopant in each of the P-CGL1 790 and the P-CGL2 990 may be, but is not limited to, about 1 wt % to about 40 wt %, for example, about 3 wt % to about 30 wt %. Each of the P-CGL1 790 and the P-CGL2 990 may have independently a thickness of, but is not limited to, about 30 Å to about 500 Å, for example, about 50 Å to about 200 Å.

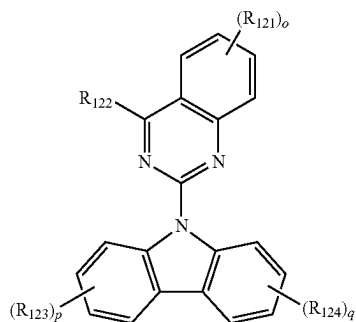
The EML3 940 includes a lower middle EML (first layer) 940A and an upper middle EML (second layer) 940B. The lower middle EML 940A is disposed adjacently to the first electrode 610 and the upper middle EML 940B is disposed adjacently to the second electrode 620. One of the lower middle EML 940A and the upper middle EML 940B may be a green emitting material layer and the other of the lower middle EML 940A and the upper middle EML 940B may be a red emitting material layer. The green emitting material layer and the red emitting material layers are disposed sequentially so as to form the EML3 940.

For example, the lower middle EML 940A may be a red emitting material layer. In this case, the lower middle EML 940A may include red host (third host) 942a and red dopant (third dopant) 944a. In an exemplary aspect, the red host 942a may include a P-type red host (hole-type red host) and an N-type red host (electron-type red host).

As an example, the P-type red host may include, but is not limited to, the spirofluorene-based organic compound having the structure of Formulae 7 to 8. The N-type red host may include, but is not limited to, a quinazoline-carbazole-based organic compound having the following structure of Formula 20:

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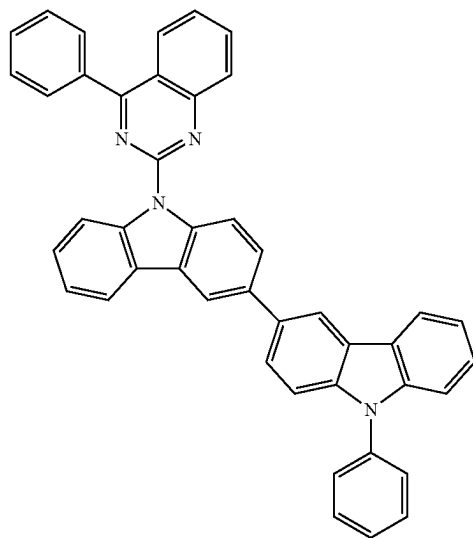
Formula 20



wherein R₁₂₁ is protium, deuterium, a C₁-C₂₀ alkyl group or a C₆-C₃₀ aryl group; R₁₂₂ is a C₆-C₃₀ aryl group; each of R₁₂₃ and R₁₂₄ is independently protium, deuterium, a C₅-C₃₀ hetero aryl group, wherein the C₅-C₃₀ hetero aryl group is unsubstituted or substituted with a C₆-C₃₀ aryl group which may be unsubstituted or further substituted with other C₆-C₃₀ aryl group, or each of adjacent two of R₁₂₃ and adjacent two of R₁₂₄ form a C₆-C₁₀ aromatic ring in case of each of p and q is independently two or more, wherein at least one of R₁₂₃ and R₁₂₄ may be a C₅-C₃₀ hetero aryl group or at least one of adjacent two of R₁₂₃ and adjacent two of R₁₂₄ may form a C₆-C₁₀ aromatic ring; each of o, p and q is a number of substituents and is independently an integer of 0 to 4.

For example, R₁₂₁ may comprise protium and deuterium, R₁₂₂ may comprise a phenyl group, R₁₂₃ may comprise protium and deuterium or adjacent two of R₁₂₃ may form a phenyl ring, and R₁₂₄ may comprise a carbazolyl group and a benzocarbazolyl group each of which may be substituted with independently a phenyl group and/or a naphthyl group each of which may be independently unsubstituted or further substituted with other phenyl group and/or other naphthyl group in Formula 20. As an example, the N-type red host may be selected from, but is not limited to, the following organic compounds having the structure of Formula 21:

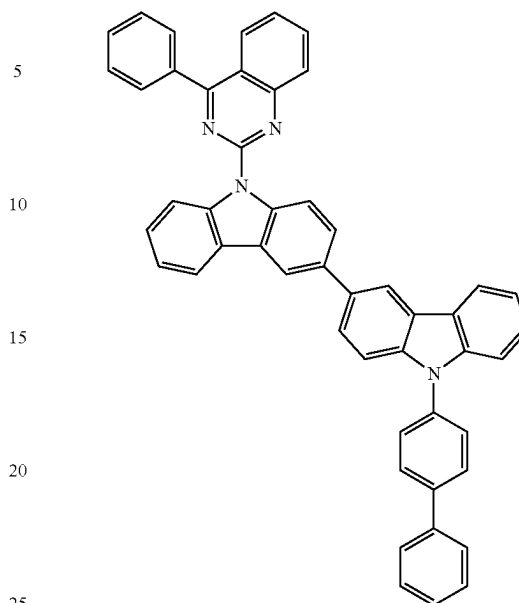
Formula 21



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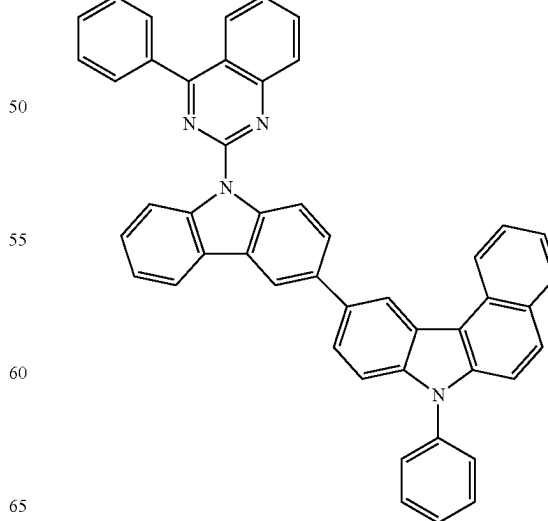
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L2



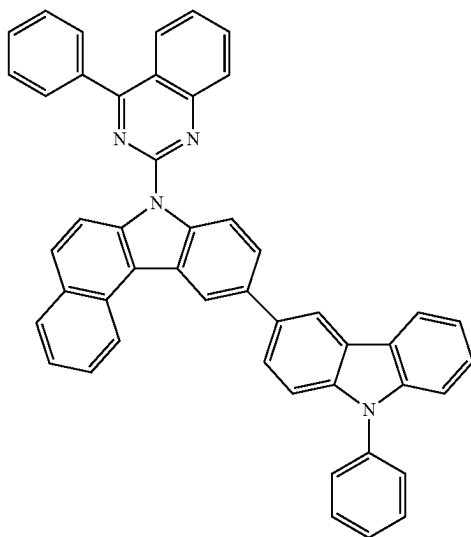
L1

L3



55

-continued

**56**

-continued

L4

5

10

15

20

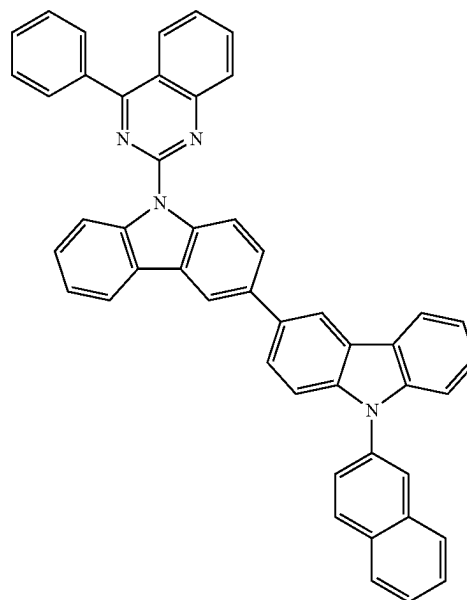
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L6



L5 45

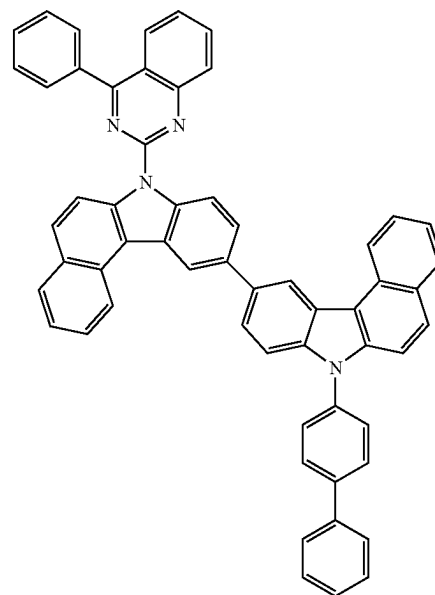
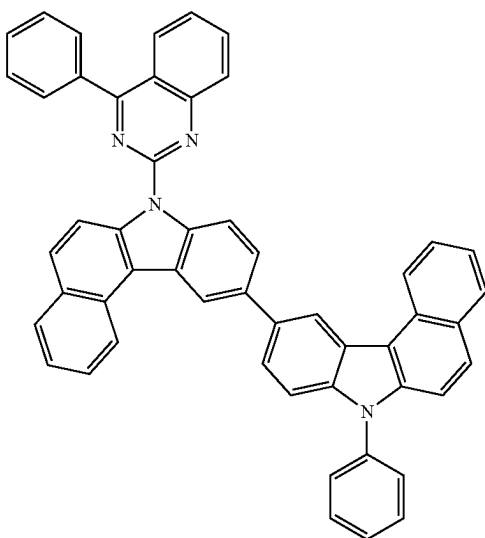
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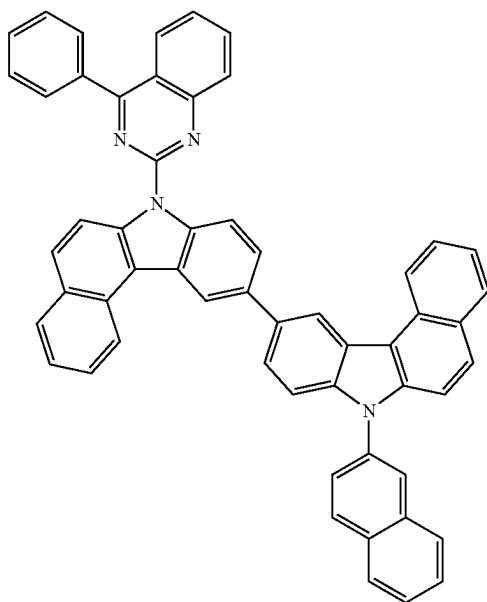
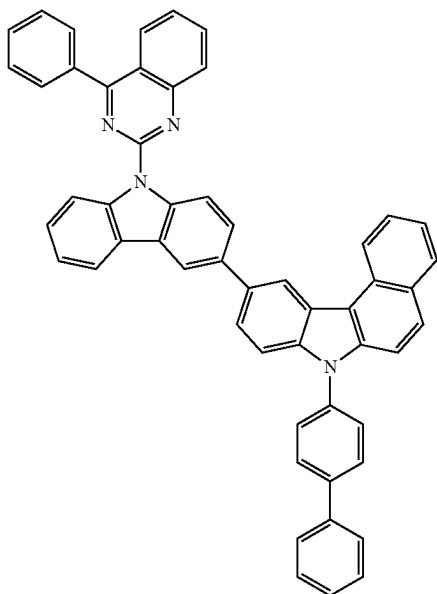
65

L7



57

-continued

**58**

-continued

L8

L10

5

10

15

20

25

30

35

40

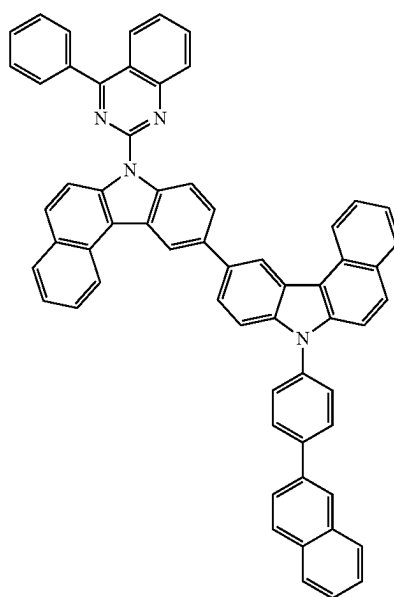
L9

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L11

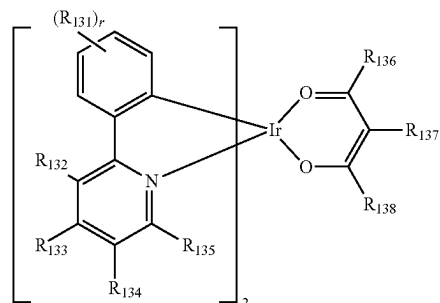


In one exemplary aspect, the P-type red host and the N-type red host in the lower middle EML **940A** may be mixed with a weight ratio of, but is not limited to, about 1:9 to about 9:1, for example, about 2:8 to about 8:2 or about 7:3 to about 3:7.

The red dopant **944a** may include at least one of red phosphorescent material, red fluorescent material and red delayed fluorescent material. In one exemplary aspect, the red phosphorescent material may have, but is not limited to, the following structure of Formula 22:

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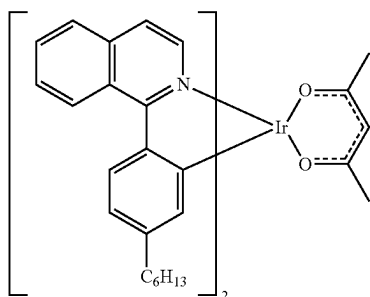
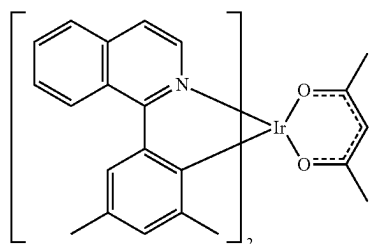
Formula 22



wherein R_{131} is protium, deuterium, a halogen atom, a C_1 - C_6 alkyl group, a C_3 - C_6 cyclo alkyl group, a C_6 - C_{10} aryl group or a C_3 - C_{10} hetero aryl group, when r is more than 2, R_{131} may be identical to or different from each other; each of R_{132} to R_{135} is independently protium, deuterium, a halogen atom, a C_1 - C_6 alkyl group, a C_3 - C_6 cyclo alkyl group, a C_6 - C_{10} aryl group or a C_3 - C_{10} hetero aryl group, or adjacent two of R_{132} to R_{135} form a C_6 - C_{10} aryl ring, wherein at least one of adjacent two of R_{132} , R_{133} , R_{134} and R_{135} may form a C_6 - C_{10} aryl ring; each of R_{136} , R_{137} and R_{138} is independently protium, deuterium or a C_1 - C_6 alkyl group; r is a number of substituents and is an integer of 0 to 4.

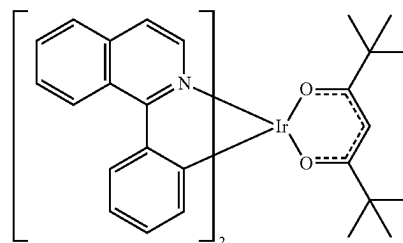
In Formula 22, R_{131} may comprise protium, deuterium and a C_1 - C_6 alkyl group, each of R_{132} , R_{133} , R_{134} and R_{135} may comprise independently protium, deuterium, a C_1 - C_6 alkyl group, or at least one of adjacent two of R_{132} , R_{133} , R_{134} and R_{135} may form a phenyl ring. As an example, the red dopant **944a** may be selected from, but is not limited to, the following red phosphorescent material having the structure of Formula 23:

Formula 23

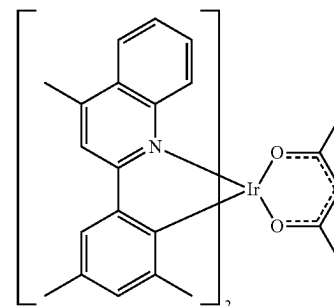


60

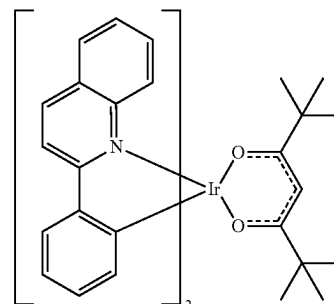
-continued



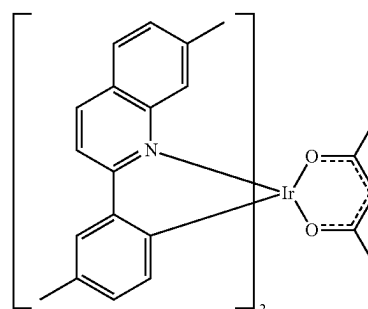
M3



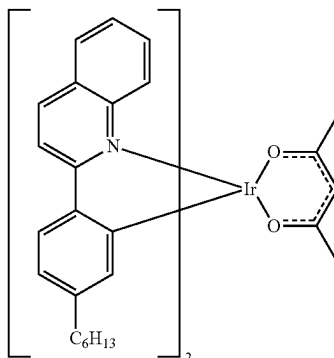
M4



M5



M6



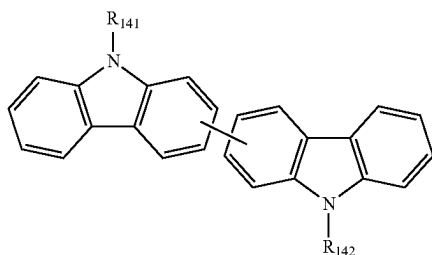
M7

61

The upper middle EML **940B** may be a green emitting material layer. In this case, the upper middle EML **940B** may include green host (fourth host) **942b** and green dopant (fourth dopant) **944b**. In one exemplary aspect, the green host **942b** may include a P-type green host (hole-type green host) and an N-type green host (electron-type green host).

As an example, the P-type green host may include, but is not limited to, a biscarbazole-based organic compound having the following structure of Formula 24:

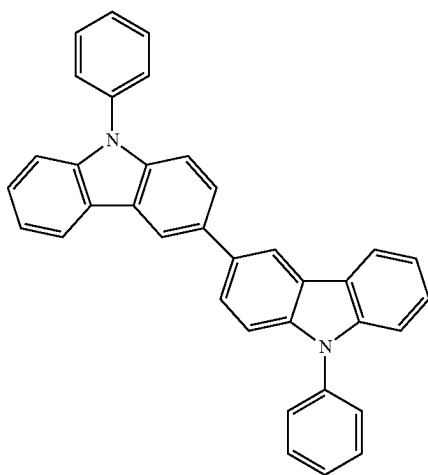
Formula 24



wherein each of R_{141} and R_{142} is independently a C_6 - C_{30} aryl group, wherein the C_6 - C_{30} aryl group may be unsubstituted or further substituted with other C_6 - C_{10} aryl group.

In Formula 24, each of R_{141} and R_{142} may comprise independently a phenyl group and a naphthyl group each of which may be independently unsubstituted or substituted with other phenyl group and/or other naphthyl group. As an example, the P-type green host may be selected from, but is not limited to, the following biscarbazole-based organic compounds having the structure of Formula 25:

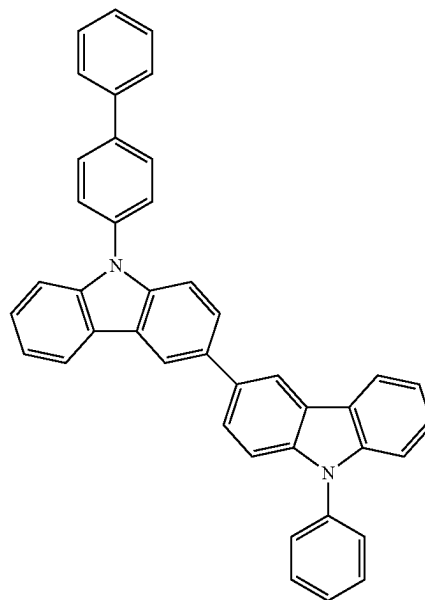
Formula 25



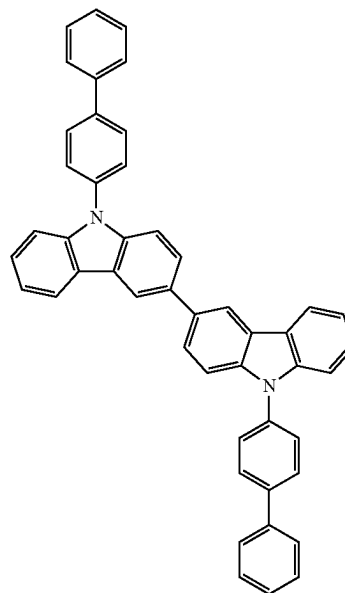
62

-continued

P2

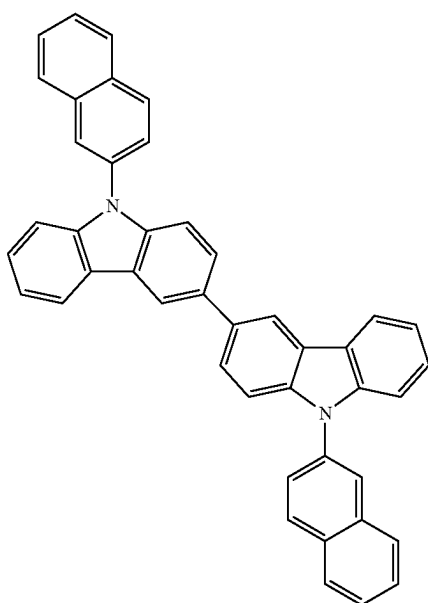
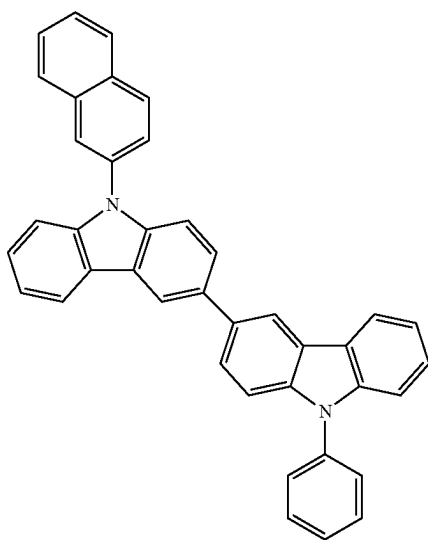


P3



63

-continued

**64**

-continued

P4

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P5

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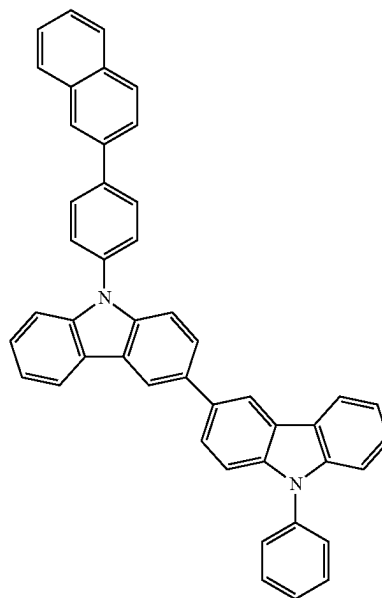
50

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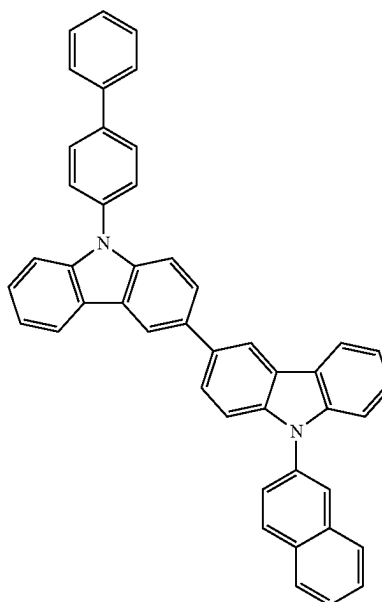
60

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P6

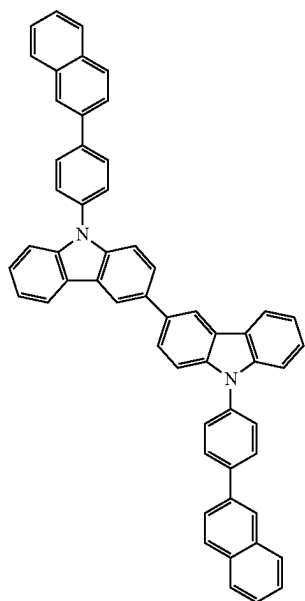
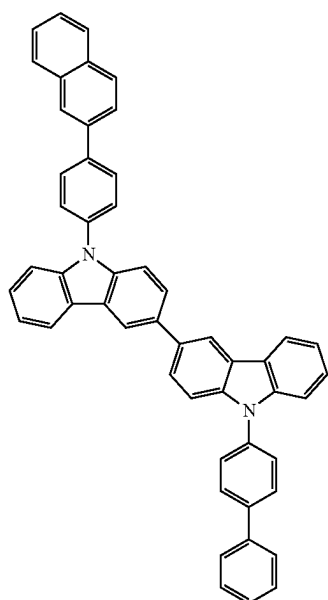


P7



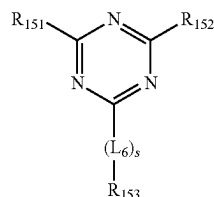
65

-continued



The N-type green host may include, but is not limited to, a triazine-based organic compound having the following structure of Formula 26:

Formula 26



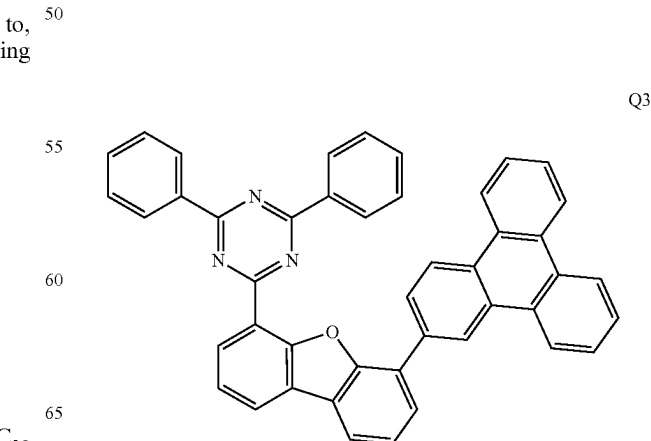
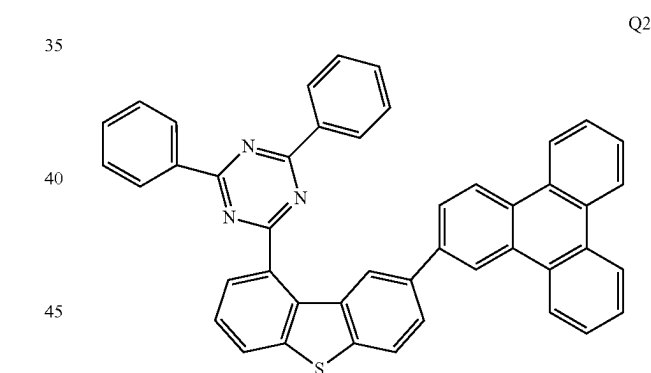
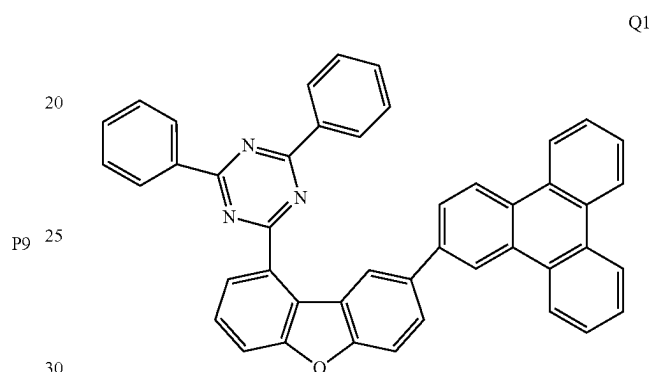
wherein each of R_{151} and R_{152} is independently a C_6 - C_{30} aryl group; R_{153} is a C_{10} - C_{20} fused hetero aryl group which

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may be unsubstituted or further substituted with C_{10} - C_{20} fused aryl group; L_6 is a C_6 - C_{30} arylene group; s is 0 or 1.

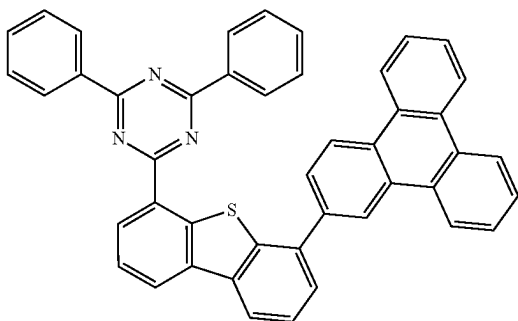
In Formula 26, each of R_{151} and R_{152} may comprise independently a phenyl group, R_{153} may comprise a dibenzofuran group and a dibenzothiophenyl group each of which may be independently unsubstituted or substituted with a triphenylenyl group and/or a phenanthrenyl group, and L_6 may comprise a phenylene group. As an example, the N-type green host may be selected from, but is not limited to, the following triazine-based organic compound having the structure of Formula 27:

Formula 27



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-continued



Q4

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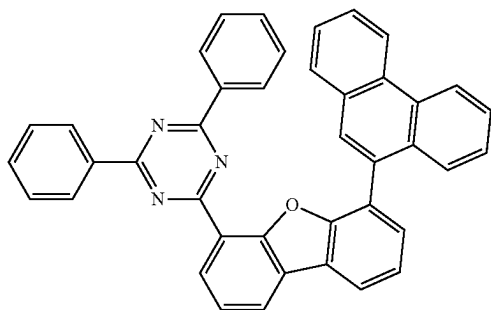
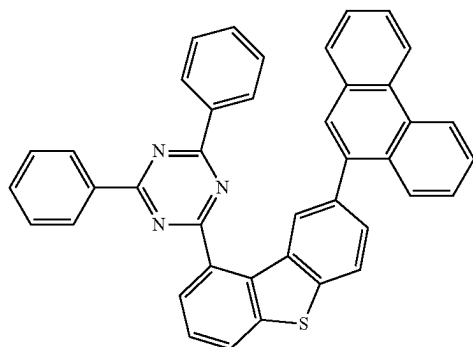
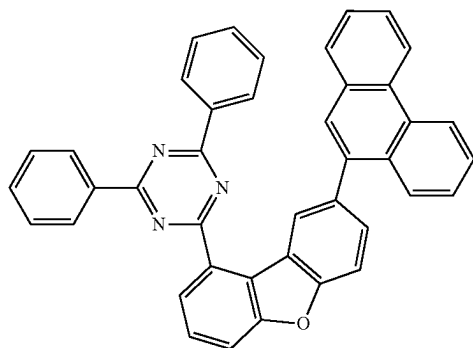
15

Q5

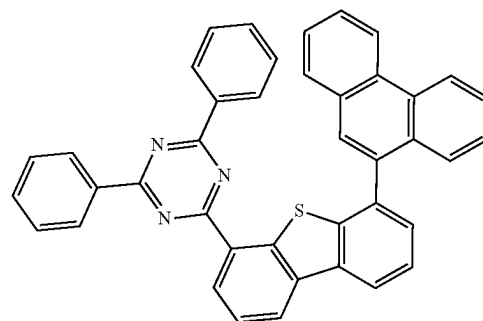
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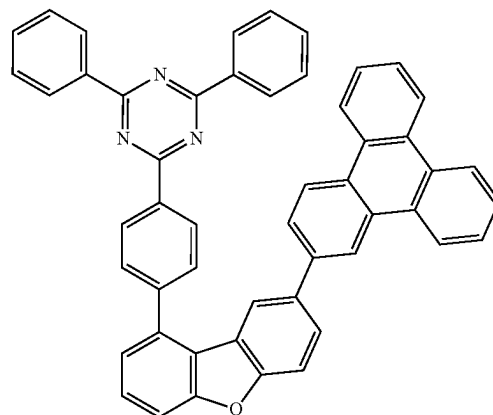
**68**

-continued



Q8

15



Q9

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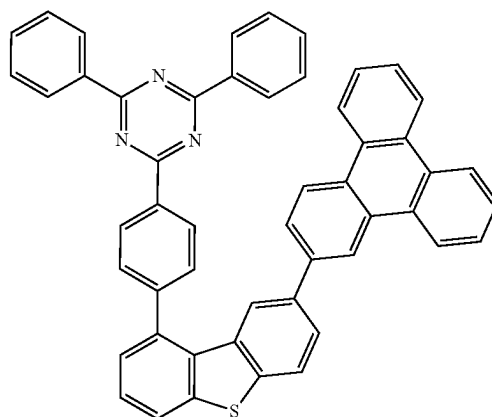
30

Q6

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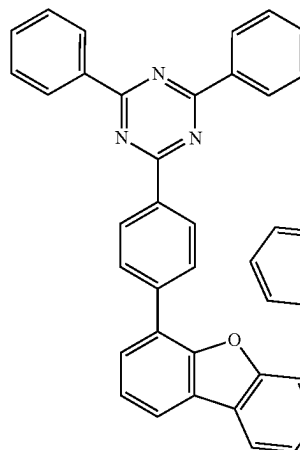
40

45



Q10

50



Q11

Q7

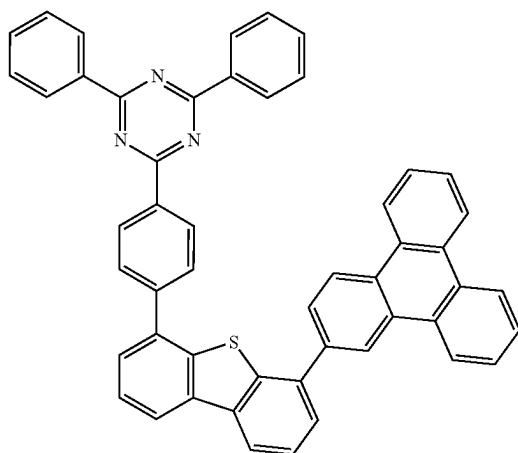
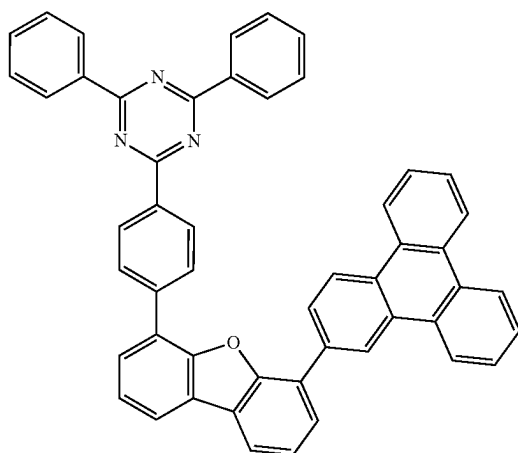
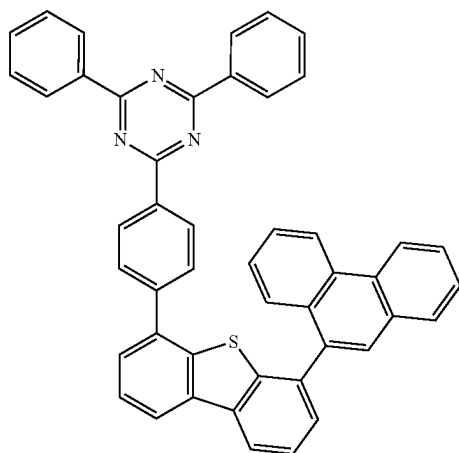
55

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69

-continued

**70**

-continued

Q12

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Q13

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Q14

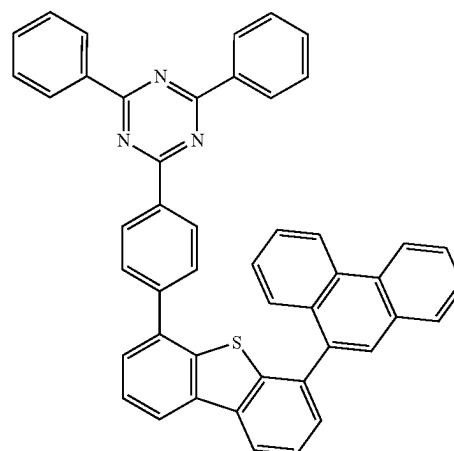
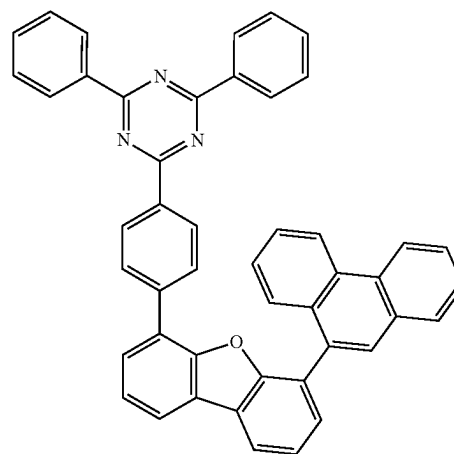
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Q15

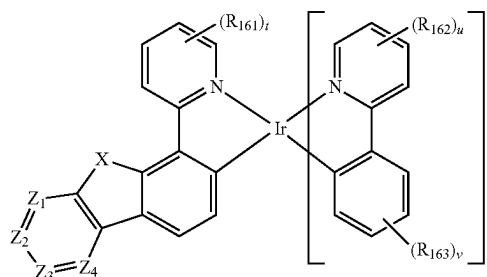
Q16



In one exemplary aspect, the P-type green host and the N-type green host in the upper middle EML **940B** may be mixed with a weight ratio of, but is not limited to, about 1:9 to about 9:1, for example, about 2:8 to about 8:2 or about 7:3 to about 3:7.

The green dopant **944b** may comprise at least one of green phosphorescent material, green fluorescent material and green delayed fluorescent material. In one exemplary aspect, the green phosphorescent material may have, but is not limited to, the following structure of Formula 28:

Formula 28



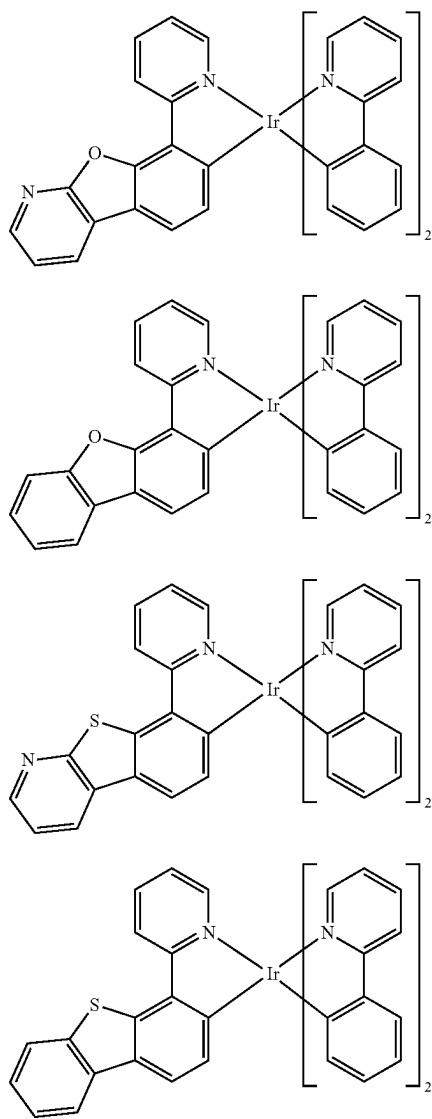
wherein each of R_{161} , R_{161} and R_{163} is independently pro-
tium, deuterium, a halogen atom, a C_1 - C_{10} alkyl group, a
 C_3 - C_6 cyclo alkyl group, a C_6 - C_{10} aryl group or a C_3 - C_{10}

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hetero aryl group; each of t, u and v is a number of substituents and is independently an integer of 0 to 4; X is O or S; each of Z₁ to Z₄ is independently N or CR₁₆₄, wherein R₁₆₄ is protium, deuterium, a halogen atom, a C₁-C₁₀ alkyl group, a C₃-C₆ cyclo alkyl group, a C₆-C₁₀ aryl group or a C₃-C₁₀ hetero aryl group.

In Formula 28, each of R₁₆₁, R₁₆₁ and R₁₆₃ may comprise independently protium, deuterium and a C₁-C₁₀ alkyl group, and at least one, for example at least two or at least three, of Z₁ to Z₄ may be CR₁₆₄, and R₁₆₄ may comprise protium, deuterium and a C₁-C₁₀ alkyl group. As an example, the green dopant **944b** may be selected from, the following green phosphorescent materials having the structure of Formula 29:

Formula 29



The contents of the green dopant **944b** in the upper middle EML **940B** may be, but is not limited to, about 1 wt % to about 10 wt %, for example, about 1 wt % to about 5 wt %. The upper middle EML **940B** may have a thickness of, but is not limited to, about 30 Å to about 600 Å, for example, about 50 Å to about 400 Å.

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The EML **1740** disposed adjacently to the first electrode **610** includes the first host **742** having excellent hole injection property and the EML **2840** disposed adjacently to the second electrode **620** includes the second host **842** having excellent electron injection property. Accordingly, the OLED **D2** can lower its driving voltage and improve its luminous efficiency and luminous lifespan. The white light emitted from the OLED **D2** in the red pixel region RP, the green pixel region GP and the blue pixel region BP transmits the color filter layer **580** (FIG. 4) in each pixel region, so that the organic light emitting display device **500** (FIG. 4) can implement full-color images.

In FIG. 5, the OLED **D2** has three emitting parts to form a tandem structure. Alternatively, an OLED may further comprise at least one emitting part and at least one charge generation layer between the emitting parts.

In addition, an organic light emitting device in accordance with the present disclosure may include a color conversion layer. FIG. 6 is a schematic cross-sectional view illustrating an organic light emitting display device as an organic light emitting device in still another exemplary aspect of the present disclosure.

As illustrated in FIG. 6, the organic light emitting display device **1000** comprises a first substrate **1002** that defines each of a red pixel region RP, a green pixel region GP and a blue pixel region BP, a second substrate **1004** facing the first substrate **1002**, a thin film transistor Tr over the first substrate **1002**, an organic light emitting diode (OLED) **D** disposed between the first and second substrates **1002** and **1004** and emitting blue (B) light and a color conversion layer **1080** disposed between the OLED **D** and the second substrate **1004**. Although not shown in FIG. 6, a color filter layer may be disposed between the second substrate **1004** and the respective color conversion layer **1080**.

The thin film transistor Tr is disposed over the first substrate **1002** correspondingly to each of the red pixel region RP, the green pixel region GP and the blue pixel region BP. A passivation layer **1060**, which has a drain contact hole exposing one electrode, for example a drain electrode, constituting the thin film transistor Tr, is formed with covering the thin film transistor Tr over the whole first substrate **1002**.

The OLED **D**, which includes a first electrode **1110**, an emissive layer **1130** and the second electrode **1120**, is disposed over the passivation layer **1060**. The first electrode **1110** may be connected to the drain electrode of the thin film transistor Tr through the drain contact hole. In addition, a bank layer **1064** covering edges of the first electrode **1110** is formed at the boundary between the red pixel region RP, the green pixel region GP and the blue pixel region BP. In this case, the OLED **D** may have a structure of FIG. 2 and can emit blue (B) light. The OLED **D** is disposed in each of the red pixel region RP, the green pixel region GP and the blue pixel region BP to provide blue (B) light.

The color conversion layer **1080** may include a first color conversion layer **1082** corresponding to the red pixel region RP and a second color conversion layer **1084** corresponding to the green pixel region GP. As an example, the color conversion layer **1080** may include an inorganic luminescent material such as quantum dot (QD).

The blue (B) light emitted from the OLED **D** in the red pixel region RP is converted into red (R) light by the first color conversion layer **1082** and the blue (B) light emitted from the OLED **D** in the green pixel region GP is converted into green (G) light by the second color conversion layer **1084**. Accordingly, the organic light emitting display device **1000** can implement a color image.

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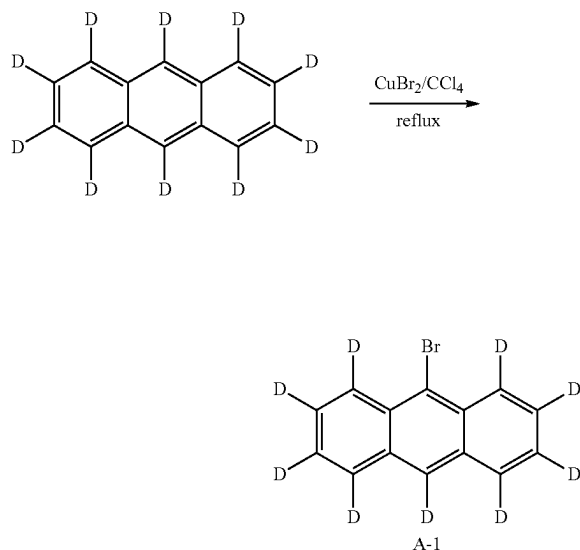
In addition, when the light emitted from the OLED D is displayed through the first substrate **1002**, the color conversion layer **1080** may be disposed between the OLED D and the first substrate **1002**.

Synthesis Example 1

Synthesis of Host 1-2

(1) Synthesis of Intermediate A-1

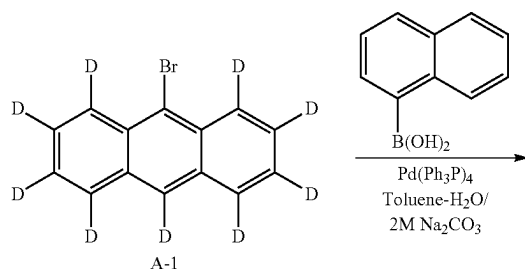
Reaction Formula 1-1



Anhydrous cupric bromide (45 g, 0.202 mol) was added into CCl_4 solution dissolving anthracene-D10 (18.8 g, 0.10 mol). The reactants were stirred for 12 hours with heating under nitrogen atmosphere. After the reaction was complete, white CuBr(I) was filtered off, the filtrate was passed through 35 nm Alumina column to be purified. The solvent in the purified reactant solution was removed under vacuum condition to obtain a mixture including the Intermediate A-1 (9-bromoanthracene-D9). The mixture includes the Intermediate A-1 as major component, and the starting material (anthracene-D10) and the dibromo-byproduct as minor component. The mixture without additional purification was used as the starting material in the reaction Formula 1-2.

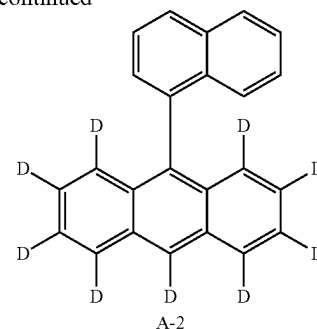
(2) Synthesis of Intermediate A-2

Reaction Formula 1-2



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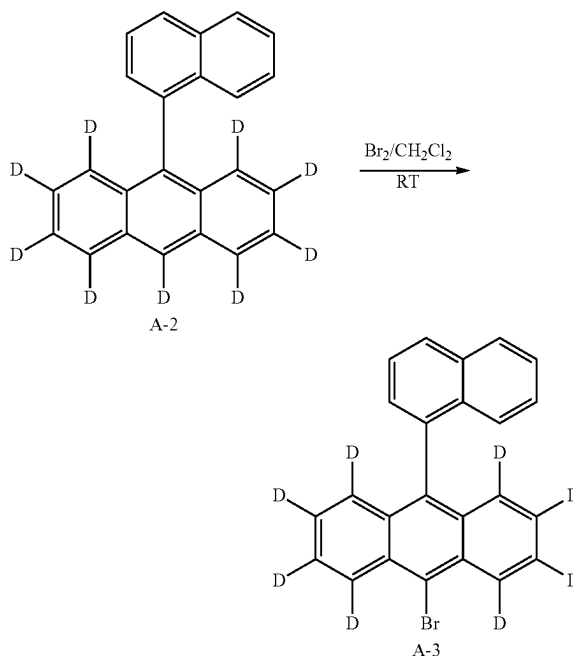
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The intermediate A-1 (2.66 g, 0.01 mol) and naphthalene-1-boronic acid (1.72 g, 0.01 mol) was added into a rounded-bottom flask, and then toluene (30 ml) was added into the flask to form a mixed solution. Under a nitrogen atmosphere, the mixed solution was stirred with addition of Na_2CO_3 aqueous solution dissolving Na_2CO_3 (2.12 g) in distilled water (10 ml). $\text{Pd(PPh}_3)_4$ (tetrakis(triphenylphosphine)palladium(0), 0.25 g, 0.025 mmol) as catalyst was further added into the mixed solution and stirred. After the reaction was complete, the reaction solution was added into methanol solution to precipitate a product, and the precipitated product was filtered. In the reduce-pressure filter, the precipitated product was washed sequentially using water, hydrogen chloride aqueous solution (10% concentration), water and methanol. The precipitated product was purified to give an intermediate A-2 of white powder (2.6 g, yield: 83%).

(3) Synthesis of Intermediate A-3

Reaction Formula 1-3



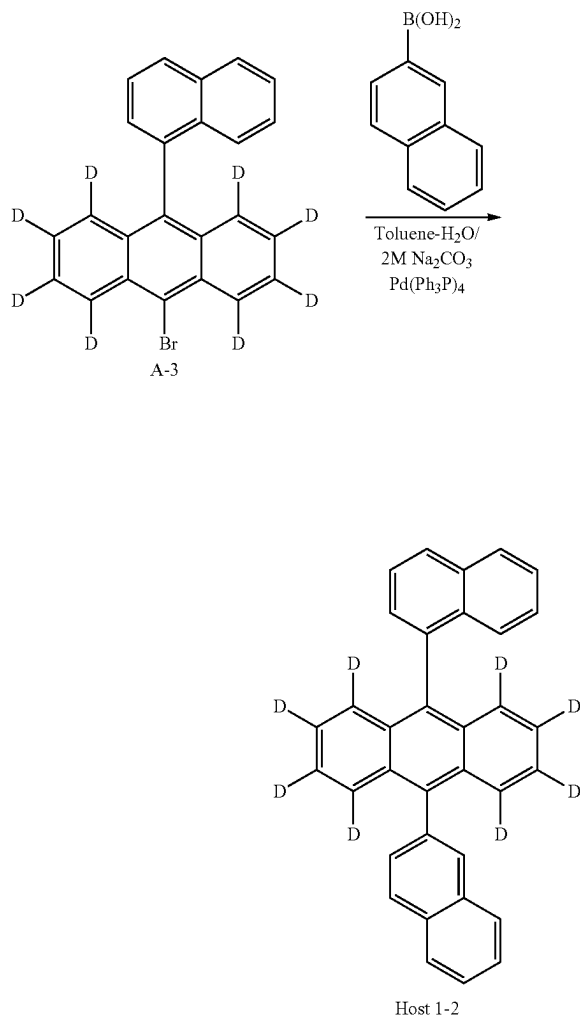
After dissolving the intermediate A-2 (2.8 g, 8.75 mmol) into dichloromethane (50 ml), Br_2 (1.4 g, 8.75 mmol) was added into the solution and then the solution was stirred at room temperature (RT). After the reaction was complete, 2M of $\text{Na}_2\text{S}_2\text{O}_3$ aqueous solution (10 ml) was added into the

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reactant and stirred. The organic layer was separated and washed with $\text{Na}_2\text{S}_2\text{O}_3$ aqueous solution (10% concentration, 10 ml) and distilled water. The organic layer was separated again, and water in the organic layer was removed by MgSO_4 . After the organic solution was concentrated, excessive methanol was added to obtain a product. The product was filtered to give the intermediate A-3 (3.3 g)

(4) Synthesis of Host 1-2

Reaction Formula 1-4



The intermediate A-3 (1.96 g, 0.05 mol) and naphthalene-2-boronic acid (1.02 g, 0.06 mol) were added and dissolved into toluene (30 ml). The mixed solution was stirred under a nitrogen atmosphere. Na_2CO_3 aqueous solution (1 ml) dissolving Na_2CO_3 (1.90 g) into distilled water (8 ml) was added into the mixed solution and then $\text{Pd(PPh}_3)_4$ (0.125 g, 0.0125 mmol) was further added. The mixed solution was heated and reacted under a nitrogen atmosphere. After the reaction was complete, the organic layer was separated, and methanol was added into the organic layer to precipitate a white solid mixture. The white solid mixture was purified by silica-gel column chromatography using chloroform and hexane (volume ratio=1:3) as an eluent to give the Host 1-2 (2.30 g).

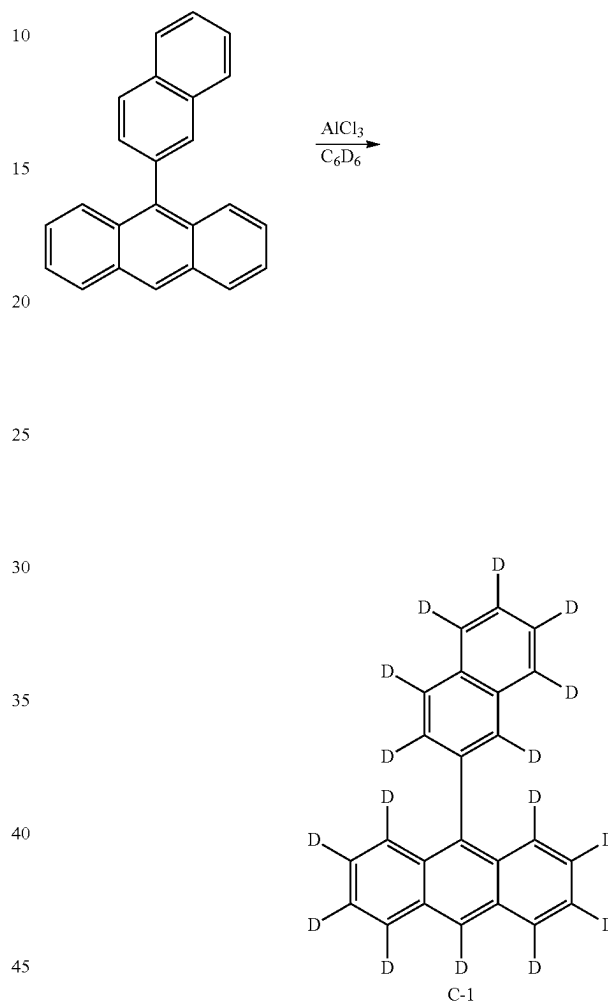
76

Synthesis Example 2

Synthesis of Host 1-3

(1) Synthesis of Intermediate B-1

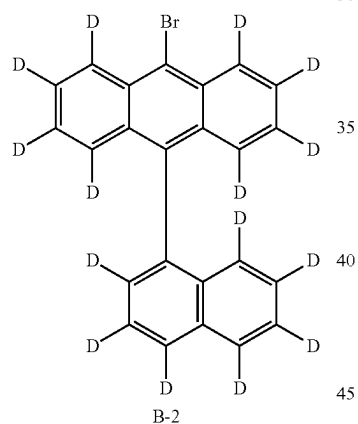
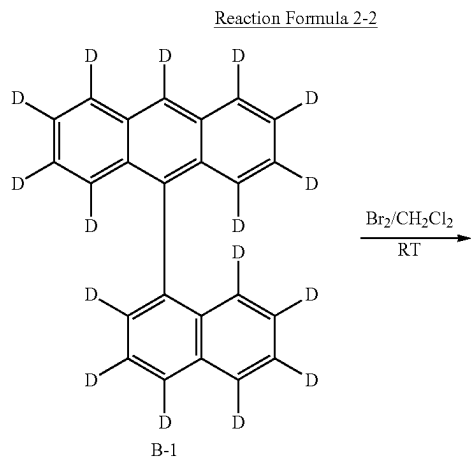
Reaction Formula 2-1



AlCl_3 (0.391 g, 4 mmol) was added into a benzene- D_6 solution (C_6D_6 , 100 ml) dissolving 10-(naphthalen-1-yl)anthracene (3.05 g, 10 mmol). After the mixed solution was stirred at RT for 6 hours, D_2O (50 ml) was added into the mixed solution. Organic layer was separated from the aqueous layer, and then the aqueous layer was washed with dichloromethane (CH_2Cl_2). After the organic layer was separated, MgSO_4 was added with stirring to dry the organic layer, and then, only the organic solvent was separated through filtering. The separated organic solvent was removed by rotation evaporation to obtain a crude product. The crude product was purified with column chromatography to give the Intermediate B-1 (2.88 g, 9 mmol).

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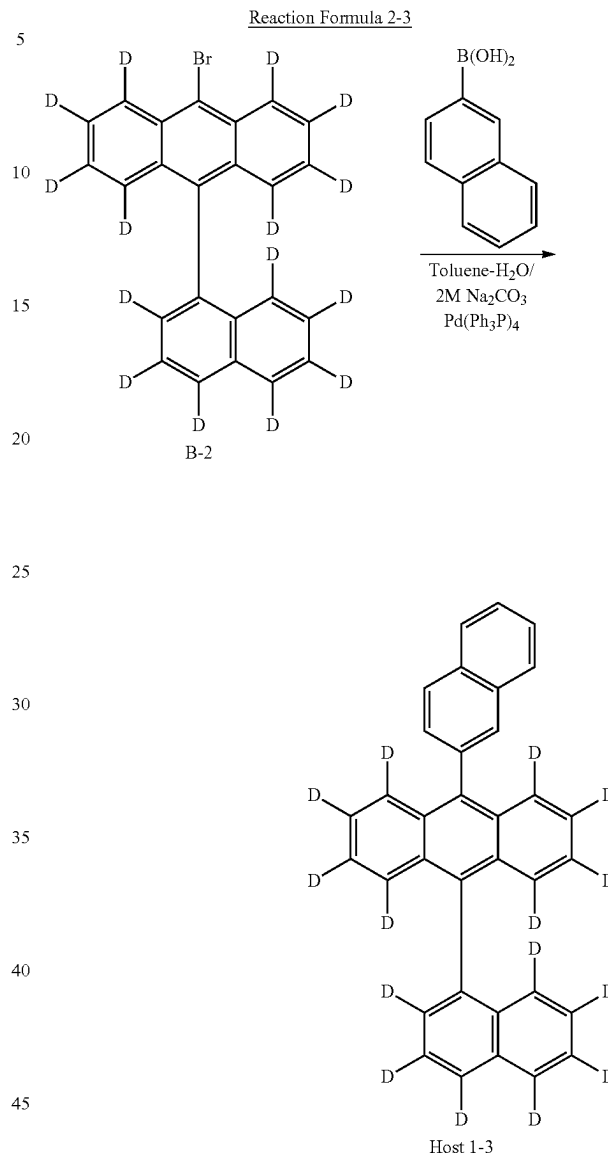
(2) Synthesis of Intermediate B-2



After the Intermediate B-1 (2.88 g, 9 mmol) was dissolved in dichloromethane (50 ml), Br₂ (1.45 g, 9 mmol) was added into the solution and then the solution was stirred. After the reaction was complete, 2M of Na₂S₂O₃ aqueous solution (10 ml) was added into the reactant and stirred. The organic layer was separated and washed with Na₂S₂O₃ aqueous solution (10% concentration, 10 ml) and distilled water. The organic layer was separated again, and water in the organic layer was removed by MgSO₄. After the organic solution was concentrated, excessive methanol was added to obtain a product. The product was filtered to give the intermediate B-2 (2.8 g).

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(3) Synthesis of Host 1-3



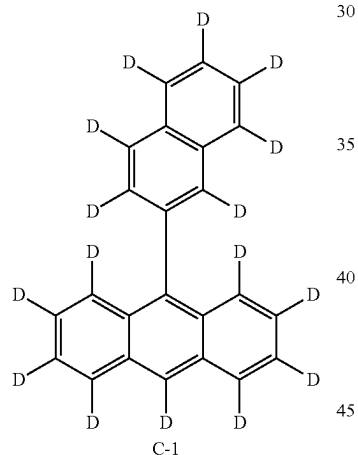
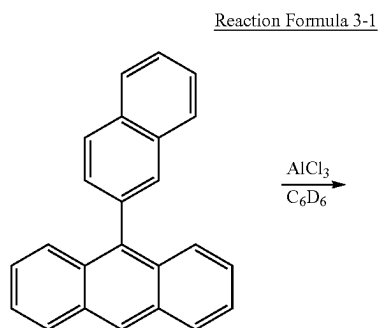
The intermediate B-2 (2.8 g, 7 mmol) and naphthalene-2-boronic acid (1.38 g, 8 mmol) was added into a rounded-bottom flask, and then toluene (30 ml) was added into the flask to form a mixed solution. Under a nitrogen atmosphere, the mixed solution was stirred with addition of Na₂CO₃ aqueous solution dissolving Na₂CO₃ (2.12 g) in distilled water (10 ml). Pd(PPh₃)₄ (0.25 g, 0.025 mmol) as catalyst was further added into the mixed solution and stirred. After the reaction was completed, the reaction solution was added into methanol solution to precipitate a product, and the precipitated product was filtered. In the reduce-pressure filter, the precipitated product was washed sequentially using water, hydrogen chloride aqueous solution (10% concentration), water and methanol. The precipitated product was purified to give Host 1-3 (2.65 g).

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Synthesis Example 3

Synthesis of Intermediate 1-4

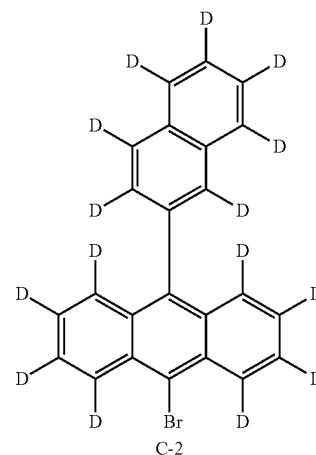
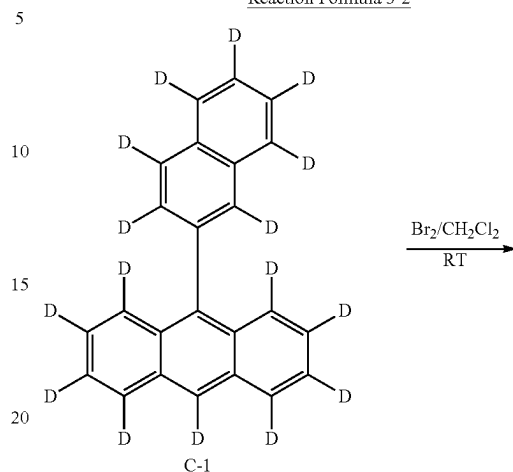
(1) Synthesis of Intermediate C-1



80

(2) Synthesis of Intermediate C-2

Reaction Formula 3-2

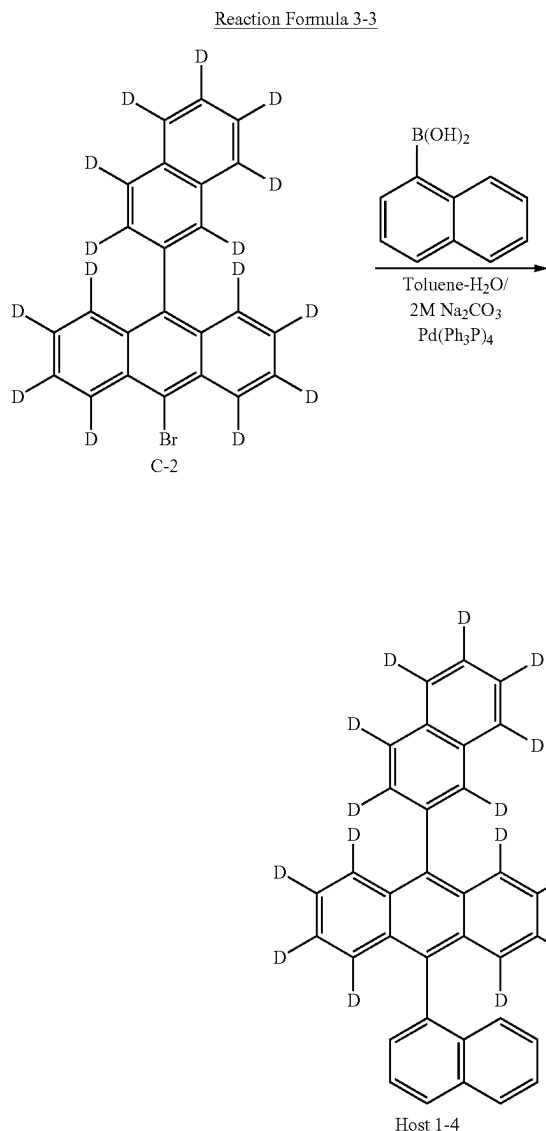


AlCl_3 (0.391 g, 4 mmol) was added into a benzene- D_6 solution (C_6D_6 , 100 ml) dissolving 10-(naphthalen-2-yl)anthracene (3.05 g, 10 mmol). After the mixed solution was stirred at RT for 6 hours, D_2O (50 ml) was added into the mixed solution. Organic layer was separated from the aqueous layer, and then the aqueous layer was washed with dichloromethane (CH_2Cl_2). After the organic layer was separated, MgSO_4 was added with stirring to dry the organic layer, and then, only the organic solvent was separated through filtering. The separated organic solvent was removed by rotation evaporation to obtain a crude product. The crude product was purified with column chromatography to give the Intermediate C-1 (2.88 g, 9 mmol).

After the Intermediate C-1 (2.88 g, 9 mmol) was dissolved in dichloromethane (50 ml), Br_2 (1.45 g, 9 mmol) was added into the solution and then the solution was stirred. After the reaction was complete, 2M of $\text{Na}_2\text{S}_2\text{O}_3$ aqueous solution (10 ml) was added into the reactant and stirred. The organic layer was separated and washed with $\text{Na}_2\text{S}_2\text{O}_3$ aqueous solution (10% concentration, 10 ml) and distilled water. The organic layer was separated again, and water in the organic layer was removed by MgSO_4 . After the organic solution was concentrated, excessive methanol was added to obtain a product. The product was filtered to give the intermediate C-2 (2.8 g).

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(3) Synthesis of Host 1-4

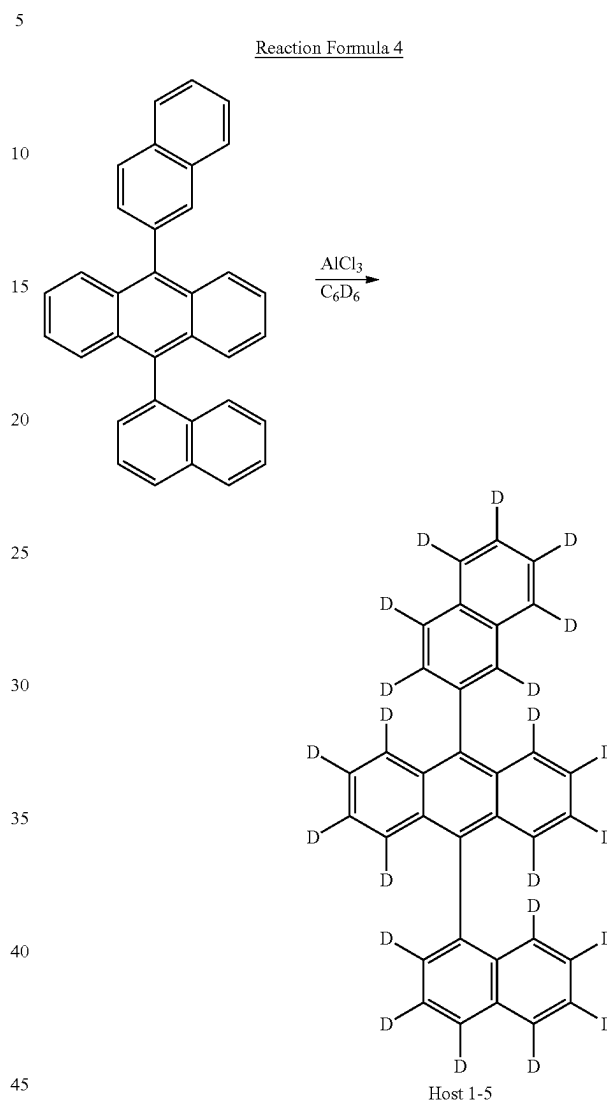


The intermediate C-2 (2.8 g, 7 mmol) and naphthalene-1-boronic acid (1.38 g, 8 mmol) was added into a rounded-bottom flask, and then toluene (30 ml) was added into the flask to form a mixed solution. Under a nitrogen atmosphere, the mixed solution was stirred with addition of Na_2CO_3 aqueous solution dissolving Na_2CO_3 (2.12 g) in distilled water (10 ml). $\text{Pd}(\text{PPh}_3)_4$ (0.25 g, 0.025 mmol) as catalyst was further added into the mixed solution and stirred. After the reaction was complete, the reaction solution was added into methanol solution to precipitate a product, and the precipitated product was filtered. In the reduce-pressure filter, the precipitated product was washed sequentially using water, hydrogen chloride aqueous solution (10% concentration), water and methanol. The precipitated product was purified to give Host 1-4 (2.65 g).

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Synthesis Example 4

Synthesis of Host 1-5



AlCl_3 (0.391 g, 4 mmol) was added into a benzene- D_6 solution (C_6D_6 , 100 ml) dissolving α,β -ADN (9-(1-naphthyl)-10-(2-naphthyl)anthracene, 4.3 g, 10 mmol). After the mixed solution was stirred at RT for 6 hours, D_2O (50 ml) was added into the mixed solution. Organic layer was separated from the aqueous layer, and then the aqueous layer was washed with dichloromethane (CH_2Cl_2). After the organic layer was separated, MgSO_4 was added with stirring to dry the organic layer, and then, only the organic solvent was separated through filtering. The separated organic solvent was removed by rotation evaporation to obtain a crude product. The crude product was purified with column chromatography to give Host 1-5 (4 g).

Example 1 (EX.1)

Fabrication of Organic Light Emitting Diode (OLED)

An organic light emitting diode having a tandem structure with two emitting parts was fabricated by applying Host 1-5 as the first host in the EML1 and Host 1-3 as the second host

in the EML2. A glass substrate onto which ITO was coated as a thin film was washed with UV-treated ozone, and then transferred to a vacuum chamber for depositing emissive layer. Subsequently, an emissive layer and a cathode were deposited by evaporation from a heating boat under about 10^{-7} Torr as the following order.

HIL (E3 in Formula 8 (97 wt %), I1 (3 wt %) in Formula 9, 100 Å); HTL1 (E3 in Formula 8, 300 Å), EML1 (Host 1-5 (98 wt %), Dopant 1 (2 wt %), 250 Å), ETL1 (F1 in Formula 11, 150 Å), n-CGL (H1 in Formula 15 (95 wt %), Li (2 wt %), 120 Å), p-CGL (E3 in Formula 8 (85 wt %), I1 in Formula 9 (3 wt %), 75 Å), HTL2 (E3 in Formula 8, 300 Å), EML2 (Host 1-3 (98 wt %), Dopant 1 (2 wt %), 250 Å), ETL2 (F1 in Formula 11 and G1 in Formula 13 (1:1 by weight, 50 wt %), Liq (50 wt %), 350 Å), EIL (LiF (50 wt %), Yb (50 wt %), 20 Å), and a cathode (Ag:Mg=10:1 by weight, 175 Å).

And then, the OLED was transferred from the deposition chamber to a dry box for film formation, followed by encapsulation using UV-curable epoxy and moisture getter.

Example 2 (EX. 2)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-4 as the second host in the EML2 instead of Host 1-3 was used.

Example 3 (EX. 3)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example, 1, except that Host 1-2 as the first host in the EML1 instead of Host 1-5 and Host 1-4 as the second host in the EML2 instead of Host 1-3 were used.

Example 4 (EX. 4)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 3, except that Host 1-5 as the second host in the EML2 instead of Host 1-4 was used.

Comparative Example 1 (Ref. 1)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-1 as the first host in the EML1 instead of Host 1-5 and Host 1-1 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 2 (Ref. 2)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-1 as the first host in the EML1 instead of Host 1-5 and Host 1-5 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 3 (Ref. 3)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-2 as the first host in the EML1 instead of Host 1-5 and Host 1-2 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 4 (Ref. 4)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-3 as the first host in the EML1 instead of Host 1-5 was used.

Comparative Example 5 (Ref. 5)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-4 as the first host in the EML1 instead of Host 1-5 and Host 1-4 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 6 (Ref. 6)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-5 as the second host in the EML2 instead of Host 1-3 was used.

Comparative Example 7 (Ref. 7)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-3 as the first host in the EML1 instead of Host 1-5 and Host 1-5 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 8 (Ref. 8)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Host 1-4 as the first host in the EML1 instead of Host 1-5 and Host 1-5 as the second host in the EML2 instead of Host 1-3 were used.

Experimental Example 1

Measurements of Luminous Properties of OLEDs

Each of the OLEDs fabricated in Examples 1 to 4 and Comparative Examples 1 to 8 was connected to an external power source and then luminous properties for all the OLEDs were evaluated using a constant current source (KEITHLEY) and a photometer PR650 at room temperature. In particular, driving voltage (V), current efficiency (Cd/A) and color coordinates at a current density of 10 mA/cm² and time period (T₉₅) at which the luminance was reduced to 95% from initial luminance at 3000 nit, at RT and at a current density of 22.5 mA/cm². The measurement results are indicated in the following Table 1.

TABLE 1

Luminous Properties of OLED						
Sample	Host ^a	V	cd/A	(T ₉₅ , hr)	(CIE x, y)	
Ref. 1	1-1/1-1	7.08	13.73	30	0.143	0.041
Ref. 2	1-1/1-5	7.23	13.04	36	0.143	0.041
Ref. 3	1-2/1-2	7.18	13.04	40	0.143	0.041
Ref. 4	1-3/1-3	7.21	13.72	44	0.143	0.041
Ref. 5	1-4/1-4	7.19	13.55	44	0.143	0.041
Ref. 6	1-5/1-5	7.25	12.91	45	0.143	0.041
Ref. 7	1-3/1-5	7.23	13.04	43	0.143	0.041
Ref. 8	1-4/1-5	7.23	13.04	43	0.143	0.041
Ex. 1	1-5/1-3	6.98	14.42	45	0.143	0.041

TABLE 1-continued

Luminous Properties of OLED					
Sample	Host ^a	V	cd/A	(T ₉₅ , hr)	(CIE x, y)
Ex. 2	1-5/1-4	6.98	14.55	45	0.143 0.041
Ex. 3	1-2/1-4	7.14	14.55	45	0.142 0.042
Ex. 4	1-2/1-5	7.00	14.21	45	0.142 0.042

^aEML1/EML2

As indicated in Table 1, compared to the OLEDs fabricated in Comparative Examples 1 and 3-5 where the EML1 and the EML2 includes the same host of Host 1-1 (anthracene-based compound in which none of the carbon atoms in the aromatic rings is substituted with deuterium, 0% deuteration rate), Host 1-2, Host 1-3 or Host 1-4 (anthracene-based compound in which a part of carbon atoms in the aromatic rings is substituted with deuterium), the OLEDs fabricated in Examples 1-4 where the EML1 includes Host 1-5 (anthracene-based compound in which all the carbon atoms in the aromatic rings are substituted with deuterium, 100% of deuteration rate) or Host 1-2 (anthracene-based compound in which the carbon atoms in only the anthracene core are substituted with deuterium, about 36% of deuteration rate) as the first host, and the EML2 includes Host 1-3, Host 1-4 (anthracene-based compound having about 68% of deuteration rate) or Host 1-5 as the second host lowered their driving voltage by up to 2.8% and improved their current efficiency and luminous lifespan by up to 11.6% and 50%, respectively.

In addition, compared to the OLEDs fabricated in Comparative Examples 2 and 6-8 where the EML1 includes Host 1-1 (0% of deuteration rate), Host 1-5 (100% of deuteration rate), Host 1-3 or Host 1-4 (about 68% of deuteration rate) as the first host and the EML2 includes Host 1-5 (100% of deuteration rate) as the second host, the OLEDs fabricated in Examples lowered their driving voltage by up to 3.7% and improved their current efficiency and luminous lifespan by up to 11.6% and 25%, respectively.

Example 5 (EX. 5)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 1, except that Dopant 2 as the dopant in the EML1 and the EML2 instead of Dopant 1 was used.

Example 6 (EX. 6)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-4 as the second host in the EML2 instead of Host 1-3 was used.

Comparative Example 9 (Ref. 9)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-1 as the first host in the EML1 instead of Host 1-5 and Host 1-1 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 10 (Ref. 10)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-1 as the first host in the EML1 instead of Host 1-5 and Host 1-5 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 11 (Ref. 11)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-2 as the first host in the EML1 instead of Host 1-5 and Host 1-2 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 12 (Ref. 12)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-3 as the first host in the EML1 instead of Host 1-5 was used.

Comparative Example 13 (Ref. 13)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-4 as the first host in the EML1 instead of Host 1-5 and Host 1-4 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 14 (Ref. 14)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-5 as the second host in the EML2 instead of Host 1-3 was used.

Comparative Example 15 (Ref. 15)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-3 as the first host in the EML1 instead of Host 1-5 and Host 1-5 as the second host in the EML2 instead of Host 1-3 were used.

Comparative Example 16 (Ref. 16)

Fabrication of OLED

An OLED was fabricated using the same procedure and the same material as Example 5, except that Host 1-4 as the first host in the EML1 instead of Host 1-5 and Host 1-5 as the second host in the EML2 instead of Host 1-3 were used.

Experimental Example 2

Measurement of Luminous Properties of OLEDs

Luminous properties for each of the OLEDs fabricated in Examples 5 to 6 and Comparative Examples 9 to 16 were measured as Experimental Example 1. The measurement results are indicated in the following Table 2.

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TABLE 2

Luminous Properties of OLED						
Sample	Host ^a	V	cd/A	(T ₉₅ , hr)	(CIE x, y)	
Ref. 9	1-1/1-1	7.08	14.01	25	0.14	0.039
Ref. 10	1-1/1-5	7.23	13.31	30	0.14	0.039
Ref. 11	1-2/1-2	7.18	13.31	33	0.14	0.039
Ref. 12	1-3/1-3	7.21	13.23	36	0.14	0.039
Ref. 13	1-4/1-4	7.11	13.35	37	0.14	0.039
Ref. 14	1-5/1-5	7.25	13.17	37	0.14	0.039
Ref. 15	1-3/1-5	7.23	13.31	36	0.14	0.039
Ref. 16	1-4/1-5	7.23	13.31	36	0.14	0.039
Ex. 5	1-5/1-3	6.98	14.71	37	0.14	0.039
Ex. 6	1-5/1-4	6.98	14.85	37	0.14	0.039

^aEML1/EML2

As indicated in Table 2, compared to the OLEDs fabricated in Comparative Examples 9 and 11-13 where the EML1 and the EML2 includes the same host of Host 1-1 (anthracene-based compound in which none of the carbon atoms in the aromatic rings is substituted with deuterium, 0% deuteration rate), Host 1-2, Host 1-3 or Host 1-4 (anthracene-based compound in which a part of carbon atoms in the aromatic rings is substituted with deuterium), the OLEDs fabricated in Examples 5-6 where the EML1 includes Host 1-5 (anthracene-based compound in which all the carbon atoms in the aromatic rings are substituted with deuterium, 100% of deuteration rate) as the first host, and the EML2 includes Host 1-3 or Host 1-4 (anthracene-based compound having about 68% of deuteration rate) as the second host lowered their driving voltage by up to 2.8% and improved their current efficiency and luminous lifespan by up to 11.6% and 48%, respectively.

In addition, compared to the OLEDs fabricated in Comparative Examples 10 and 14-16 where the EML1 includes Host 1-1 (0% of deuteration rate), Host 1-5 (100% of deuteration rate), Host 1-3 or Host 1-4 (about 68% of deuteration rate) as the first host and the EML2 includes Host 1-5 (100% of deuteration rate) as the second host, the OLEDs fabricated in Examples lowered their driving voltage by up to 3.7% and improved their current efficiency and luminous lifespan by up to 11.6% and 23.3%, respectively.

It will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the scope of the invention. Thus, it is intended that the present disclosure cover the modifications and variations of the present disclosure provided they come within the scope of the appended claims.

The various embodiments described above can be combined to provide further embodiments. All of the U.S. patents, U.S. patent application publications, U.S. patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to

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include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

The invention claimed is:

1. An organic light emitting diode, comprising:

a first electrode;

a second electrode facing the first electrode;

a first emitting part disposed between the first and second electrodes and comprising a first emitting material layer;

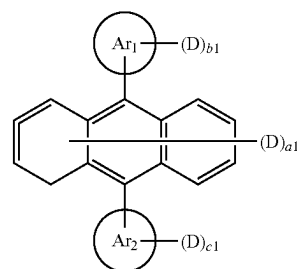
a second emitting part disposed between the first emitting part and the second electrode and comprising a second emitting material layer; and

a first charge generation layer disposed between the first and second emitting parts,

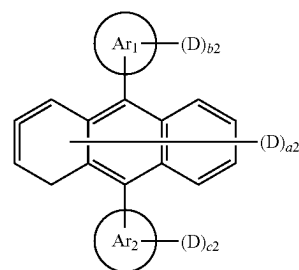
wherein the first emitting material layer includes a first host having the following structure of Formula 1; and

wherein the second emitting material layer includes a second host having the following structure of Formula 3:

Formula 1



Formula 3



wherein:

each of Ar₁ and Ar₂ is independently a C₆-C₂₀ aryl group;

D represents deuterium;

each of a₁ and a₂ is independently an integer of 0 to 8;

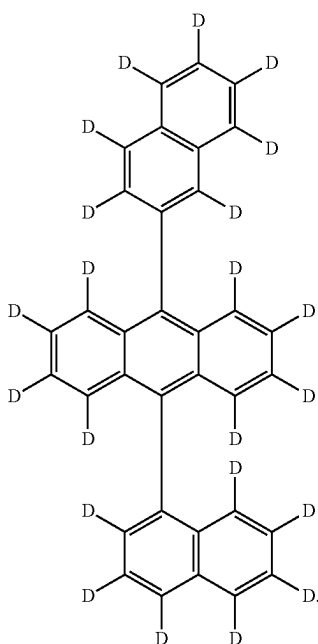
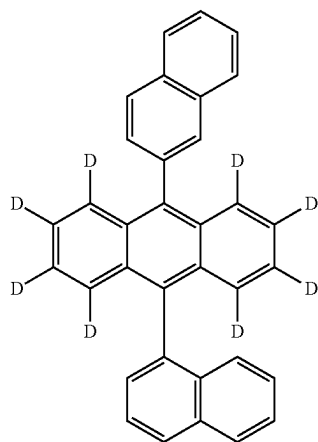
each of b₁, b₂, c₁ and c₂ is independently a number of deuterium substituted carbon atoms in Ar₁ and Ar₂ not linked to the anthracene core; and

a sum of a₁, b₁ and c₁ is different from a sum of a₂, b₂ and c₂.

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2. The organic light emitting diode of claim 1, wherein each of Ar1 and Ar2 is selected from the group consisting of a phenyl group, a naphthyl group and an anthracenyl group.

3. The organic light emitting diode of claim 1, wherein the first host is selected from one of the following organic compounds:

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4. The organic light emitting diode of claim 1, wherein the second host is selected from one of the following organic compounds:

Host 1-2

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Host 1-5

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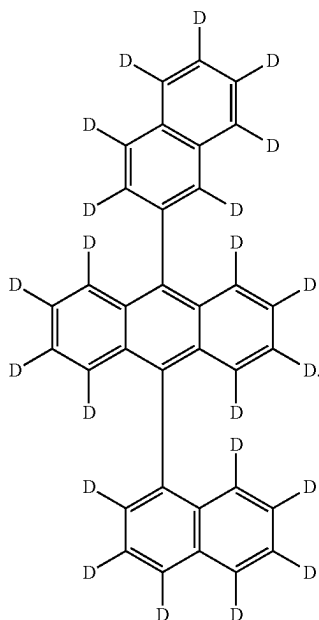
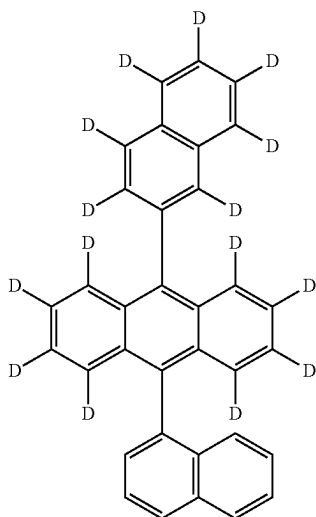
Host 1-1

Host 1-2

Host 1-3

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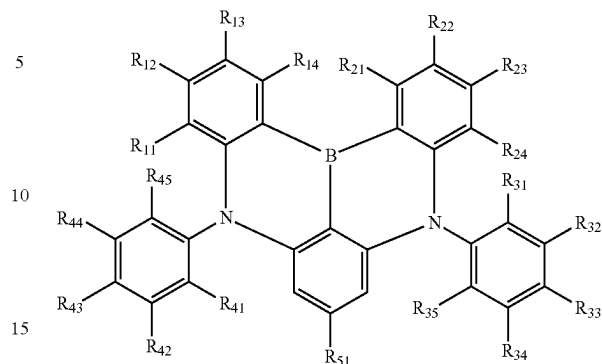
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5. The organic light emitting diode of claim 1, wherein the first emitting material layer includes a first dopant and the second emitting material layer includes a second dopant, and wherein each of the first dopant and the second dopant includes independently a boron-based compound having the following structure of Formula 5:

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Host 1-4 Formula 5

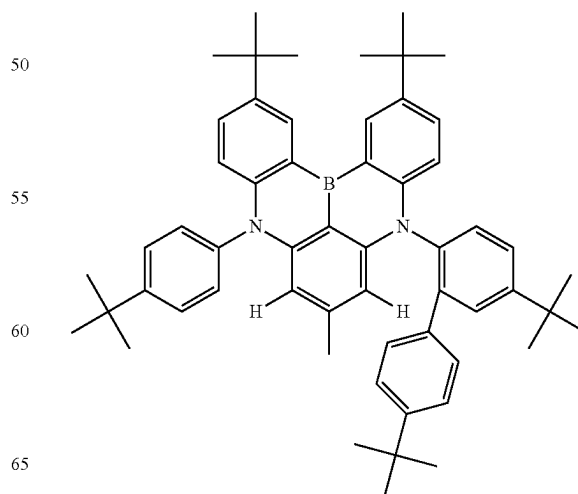


wherein:

each of R_{11} , R_{12} , R_{13} , R_{14} , R_{21} , R_{22} , R_{23} , R_{24} , R_{31} , R_{32} , R_{33} , R_{34} , R_{35} , R_{41} , R_{42} , R_{43} , R_{44} and R_{45} is independently selected from the group consisting of hydrogen, deuterium, a C_1 - C_{10} alkyl group, a C_6 - C_{30} aryl group, a C_6 - C_{30} aryl amino group and a C_5 - C_{30} hetero aryl group, or adjacent two of R_{31} , R_{32} , R_{33} , R_{34} , and R_{35} together with the carbon atoms to which they are attached or adjacent two of R_{41} , R_{42} , R_{43} , R_{44} and R_{45} together with the carbon atoms to which they are attached forms independently an unsubstituted or substituted C_6 - C_{10} aromatic ring or an unsubstituted or substituted C_5 - C_{10} hetero aromatic ring; and R_{51} is selected from the group consisting of hydrogen, deuterium, a C_1 - C_{10} alkyl group, a C_3 - C_{15} cyclo alkyl group, a C_6 - C_{30} aryl group, a C_5 - C_{30} hetero aryl group and a C_6 - C_{30} aryl amino group, wherein each of the C_6 - C_{30} aryl group and the C_5 - C_{30} hetero aryl group is optionally substituted with deuterium or a C_1 - C_{10} alkyl group, and wherein the aryl ring of the C_6 - C_{30} aryl amino group is optionally substituted with deuterium or at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{20} aryl group.

6. The organic light emitting diode of claim 5, wherein each of the first dopant and the second dopant is independently selected from one of the following compounds:

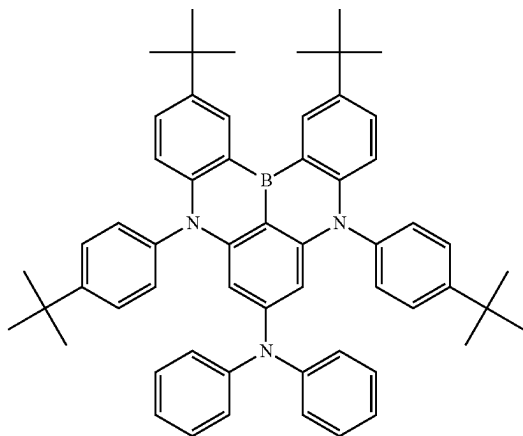
Dopant 1



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Dopant 2



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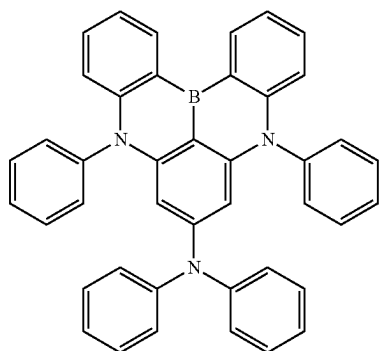
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Dopant 3



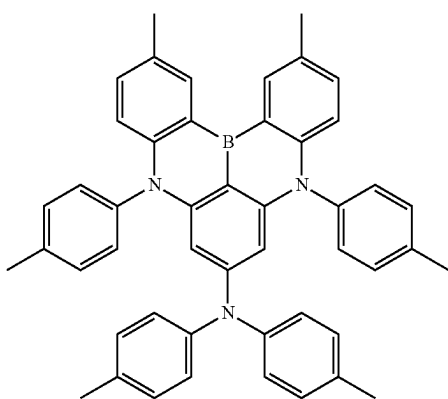
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Dopant 4



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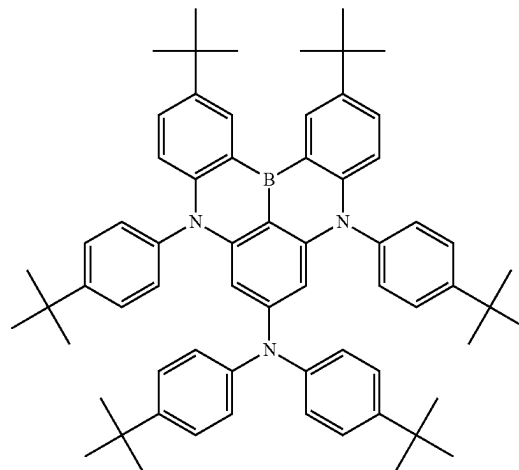
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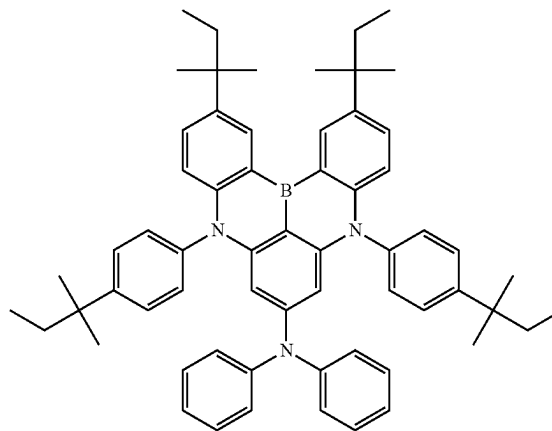
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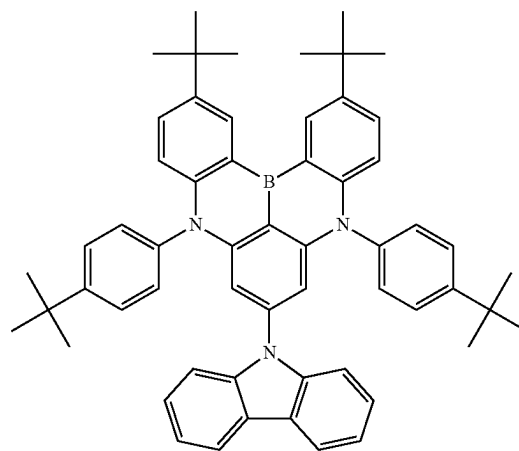
Dopant 5



Dopant 6



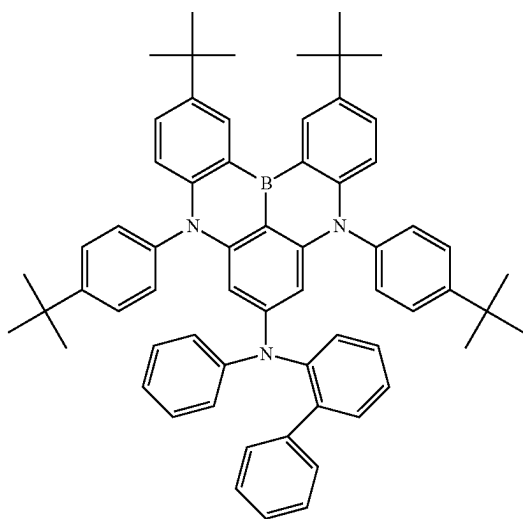
Dopant 7



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Dopant 8



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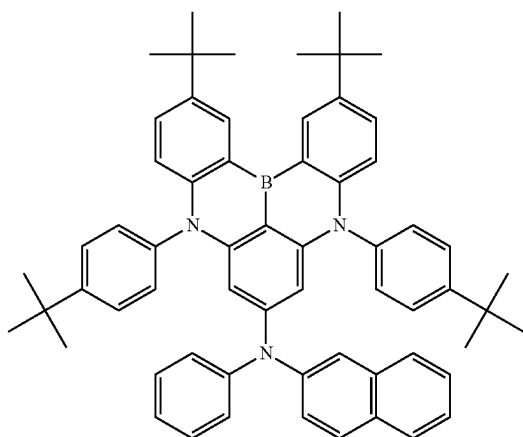
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Dopant 9

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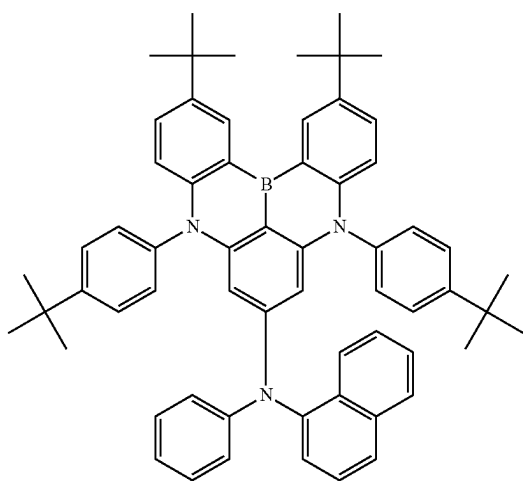
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Dopant 10

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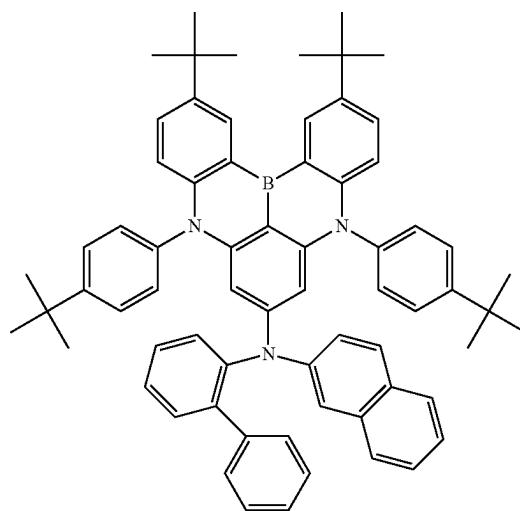
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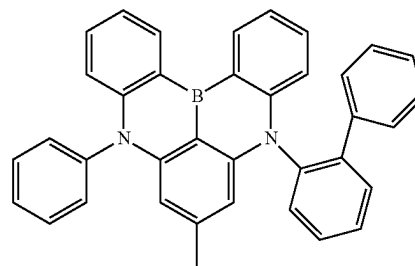
96

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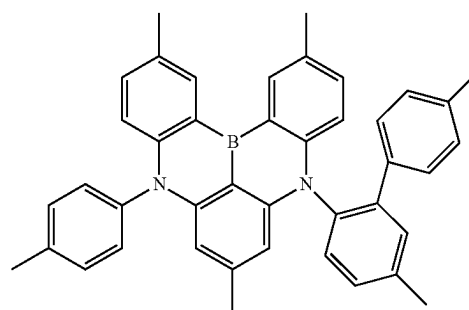
Dopant 11



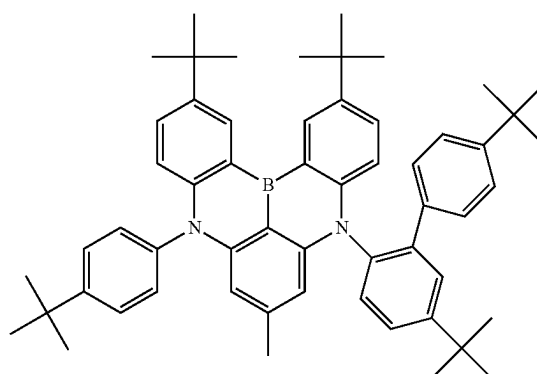
Dopant 12



Dopant 13



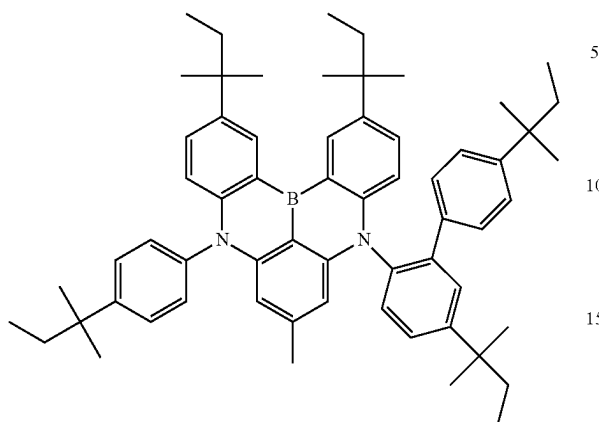
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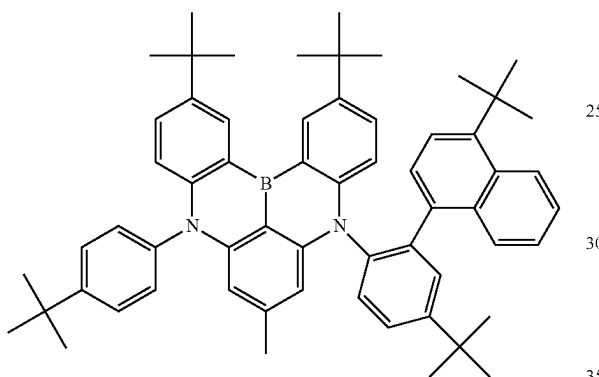
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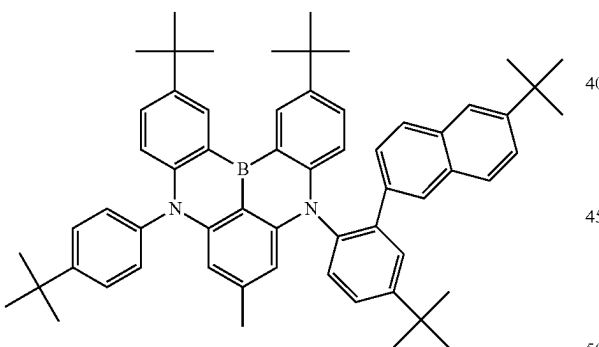
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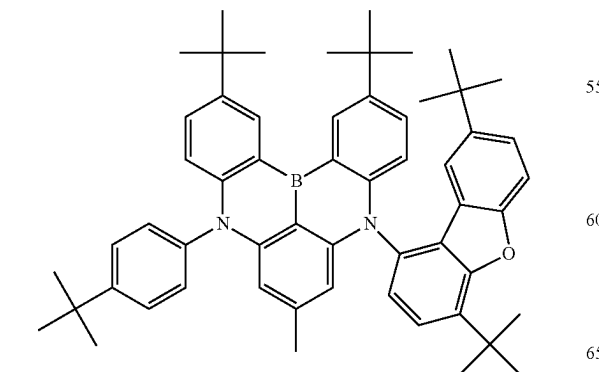
Dopant 16



Dopant 17



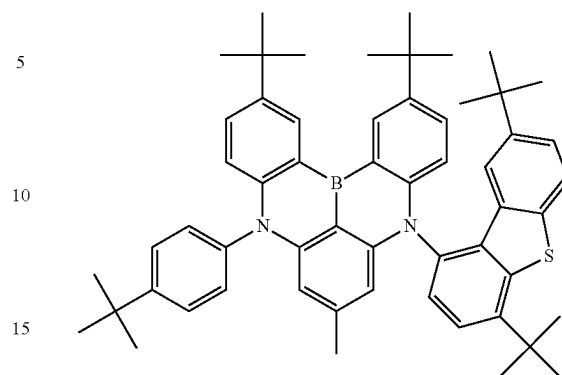
Dopant 18



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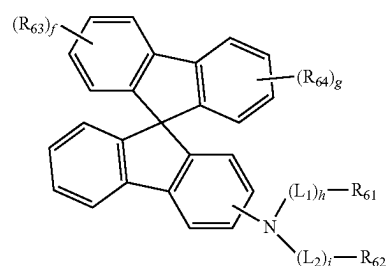
Dopant 19



7. The organic light emitting diode of claim 1, wherein the first emitting part further comprises at least one of a first hole transport layer disposed between the first electrode and the first emitting material layer and a first electron transport layer disposed between the first emitting material layer and the first charge generation layer, and the second emitting part further comprises at least one of a second hole transport layer disposed between the first charge generation layer and the second emitting material layer and a second electron transport layer disposed between the second emitting material layer and the second electrode.

8. The organic light emitting diode of claim 7, wherein each of the first and second hole transport layer independently comprises a spirofluorene-based organic compound having the following structure of Formula 7:

Formula 7



wherein:

each of R_{61} and R_{62} is independently a C_6 - C_{30} aryl group or a C_3 - C_{30} hetero aryl group, wherein each of the C_6 - C_{30} aryl group and the C_3 - C_{30} hetero aryl group is optionally substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{30} aryl group;

each of R_{63} and R_{64} is independently hydrogen, deuterium or a C_1 - C_{20} alkyl group;

each of f and g is a number of substituents and is independently an integer of 0 to 4;

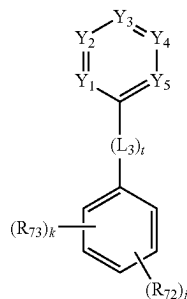
each of L_1 and L_2 is independently a C_6 - C_{30} arylene group, wherein the C_6 - C_{30} arylene group is optionally substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{30} aryl group; and

each of h and i is independently an integer of 0 or 1.

9. The organic light emitting diode of claim 7, wherein the first electron transport layer includes an azine-based organic compound having the following structure of Formula 10:

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Formula 10



wherein:

each of Y_1 , Y_2 , Y_3 , Y_4 and Y_5 is independently CR_{71} or nitrogen, provided that one to three of Y_1 , Y_2 , Y_3 , Y_4 and Y_5 is nitrogen;

R_{71} is hydrogen or a C_6-C_{30} aryl group;

L_3 is a C_6-C_{30} arylene group;

R_{72} is a C_6-C_{30} aryl group or a C_5-C_{30} hetero aryl group, wherein each of the C_6-C_{30} aryl group and the C_5-C_{30} hetero aryl group is optionally substituted further with a C_6-C_{30} aryl group or a C_3-C_{30} hetero aryl group;

R_{73} is hydrogen or adjacent two of R_{73} form a C_6-C_{20} aromatic ring when k is 2 or more;

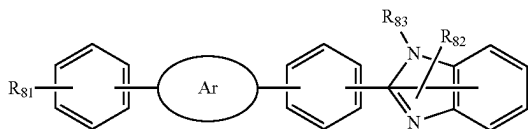
j is 1 or 2;

k is an integer of 0 to 4; and

l is 0 or 1.

10. The organic light emitting diode of claim 9, wherein the second electron transport layer comprises a benzimidazole-based organic compound having the following structure of Formula 12:

Formula 12



wherein:

Ar is a $C_{10}-C_{30}$ arylene group;

R_{81} is a C_6-C_{30} aryl group or a C_5-C_{30} hetero aryl group, wherein each of the C_6-C_{30} aryl group and the C_5-C_{30} hetero aryl group is optionally substituted with a C_1-C_{10} alkyl group; and

each of R_{82} and R_{83} is independently hydrogen, a C_1-C_{10} alkyl group or a C_6-C_{30} aryl group.

11. The organic light emitting diode of claim 1, further comprising a third emitting part disposed between the first charge generation layer and the second emitting part and including a third emitting material layer, and a second charge generation layer disposed between the second emitting part and the third emitting part.

12. The organic light emitting diode of claim 11, wherein:

the first emitting part further comprises at least one of a first hole transport layer disposed between the first electrode and the first emitting material layer and a first electron transport layer disposed between the first emitting material layer and the first charge generation layer; the second emitting part further comprises at least one of a second hole transport layer disposed between the

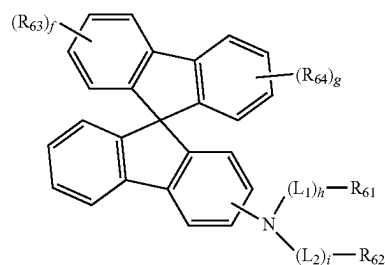
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second charge generation layer and the second emitting material layer and a second electron transport layer disposed between the second emitting material layer and the second electrode; and

the third emitting part further comprises at least one of a third hole transport layer disposed between the first charge generation layer and the third emitting material layer and a third electron transport layer disposed between the third emitting material layer and the second charge generation layer.

13. The organic light emitting diode of claim 12, wherein each of the first hole transport layer, the second hole transport layer and the third hole transport layer comprises independently a spirofluorene-based organic compound having the following structure of Formula 7:

Formula 7



wherein:

each of R_{61} and R_{62} is independently a C_6-C_{30} aryl group or a C_3-C_{30} hetero aryl group, wherein each of the C_6-C_{30} aryl group and the C_3-C_{30} hetero aryl group is optionally substituted with at least one of a C_1-C_{10} alkyl group and a C_6-C_{30} aryl group;

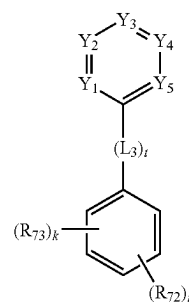
each of R_{63} and R_{64} is independently hydrogen, deuterium, or a C_1-C_{20} alkyl group; each of f and g is a number of substituents and is independently an integer of 0 to 4;

each of L_1 and L_2 is independently a C_6-C_{30} arylene group, wherein the C_6-C_{30} arylene group is optionally substituted with at least one of a C_1-C_{10} alkyl group and a C_6-C_{30} aryl group; and

each of h and i is an integer of 0 or 1.

14. The organic light emitting diode of claim 12, wherein the first electron transport layer includes an azine-based organic compound having the following structure of Formula 10:

Formula 10



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wherein:

each of Y_1 , Y_2 , Y_3 , Y_4 and Y_5 is independently CR_{71} or nitrogen, provided that one to three of Y_1 , Y_2 , Y_3 , Y_4 and Y_5 is nitrogen;

R_{71} is hydrogen or a C_6 - C_{30} aryl group;

L_3 is a C_6 - C_{30} arylene group;

R_{72} is a C_6 - C_{30} aryl group or a C_5 - C_{30} hetero aryl group, wherein each of the C_6 - C_{30} aryl group and the C_5 - C_{30} hetero aryl group is optionally substituted further with a C_6 - C_{30} aryl group or a C_3 - C_{30} hetero aryl group;

R_{73} is hydrogen or adjacent two of R_{73} form a C_6 - C_{20} aromatic ring when k is 2 or more;

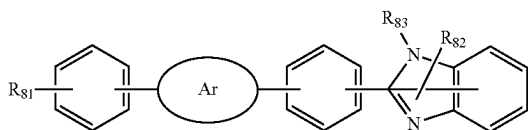
j is 1 or 2;

k is an integer of 0 to 4; and

l is 0 or 1.

15. The organic light emitting diode of claim 14, wherein each of the second electron transport layer and the third electron transport layer comprises independently a benzimidazole-based organic compound having the following structure of Formula 12:

Formula 12



wherein:

Ar is a C_{10} - C_{30} arylene group;

R_{81} is a C_6 - C_{30} aryl group or a C_5 - C_{30} hetero aryl group, wherein each of the C_6 - C_{30} aryl group and the C_5 - C_{30} hetero aryl group is optionally substituted with a C_1 - C_{10} alkyl group; and

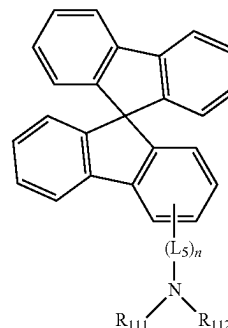
each of R_{82} and R_{83} is independently hydrogen, a C_1 - C_{10} alkyl group or a C_6 - C_{30} aryl group.

16. The organic light emitting diode of claim 12, wherein the first emitting part further comprises a first electron blocking layer disposed between the first hole transport layer and the first emitting material layer, and the second emitting part further comprises a second electron blocking layer disposed between the second hole transport layer and the second emitting material layer.

17. The organic light emitting diode of claim 16, wherein each of the first electron blocking layer and the second electron blocking layer comprises independently a spirofluorene-based organic compound having the following structure of Formula 18:

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Formula 18



wherein:

L_5 is a C_6 - C_{30} arylene group;

n is 0 or 1; and

each of R_{111} and R_{112} is independently a C_6 - C_{30} aryl group or a C_5 - C_{30} hetero aryl group,

wherein each of the C_6 - C_{30} aryl group and the C_5 - C_{30} hetero aryl group is optionally substituted with at least one of a C_1 - C_{10} alkyl group and a C_6 - C_{20} aryl group.

18. The organic light emitting diode of claim 12, wherein the third emitting material layer comprises a lower emitting material layer disposed between the third hole transport layer and the third electron transport layer and an upper emitting material layer disposed between the lower emitting material layer and the third electron transport layer, and wherein one of the lower emitting material layer and the upper emitting material layer includes a red emitting material layer and the other of the lower emitting material layer and the upper emitting material layer includes a green emitting material layer.

19. An organic light emitting device, comprising:

a substrate; and

an organic light emitting diode of claim 1 over the substrate.

20. The organic light emitting device of claim 19, wherein the substrate defines a red pixel region, a green pixel region and a blue pixel region, wherein the organic light emitting diode is located correspondingly to the red pixel region, the green pixel region and the blue pixel region, and the organic light emitting device further comprises a color filter layer disposed between the substrate and the organic light emitting diode or over the organic light emitting diode correspondingly to the red pixel region, the green pixel region and the blue pixel region.

21. The organic light emitting device of claim 19, wherein the substrate defines a red pixel region, a green pixel region and a blue pixel region, wherein the organic light emitting diode is located correspondingly to the red pixel region, the green pixel region and the blue pixel region, and the organic light emitting device further comprises a color conversion layer disposed between the substrate and the organic light emitting diode or over the organic light emitting diode correspondingly to the red pixel region and the green pixel region.

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